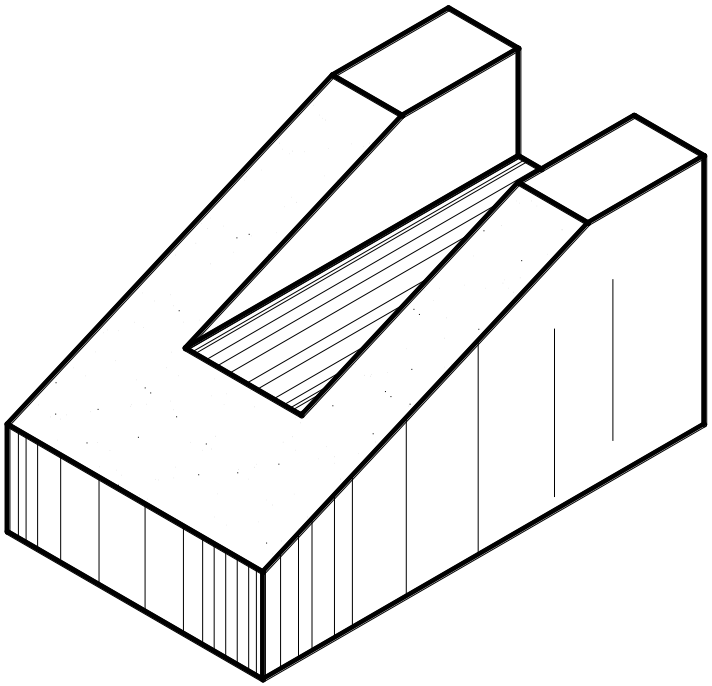
§5-1　组合体的基本知识

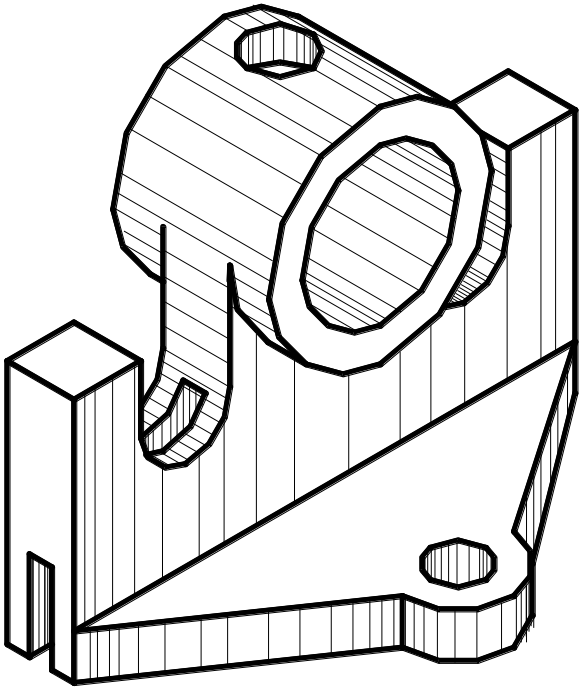
组合体的组合形式及其表面的相对位置

1、组合体的组合形式

组合体是由基本形体组合而成。其基本组合形式可分为切割和叠加两类。



切割形成的组合体

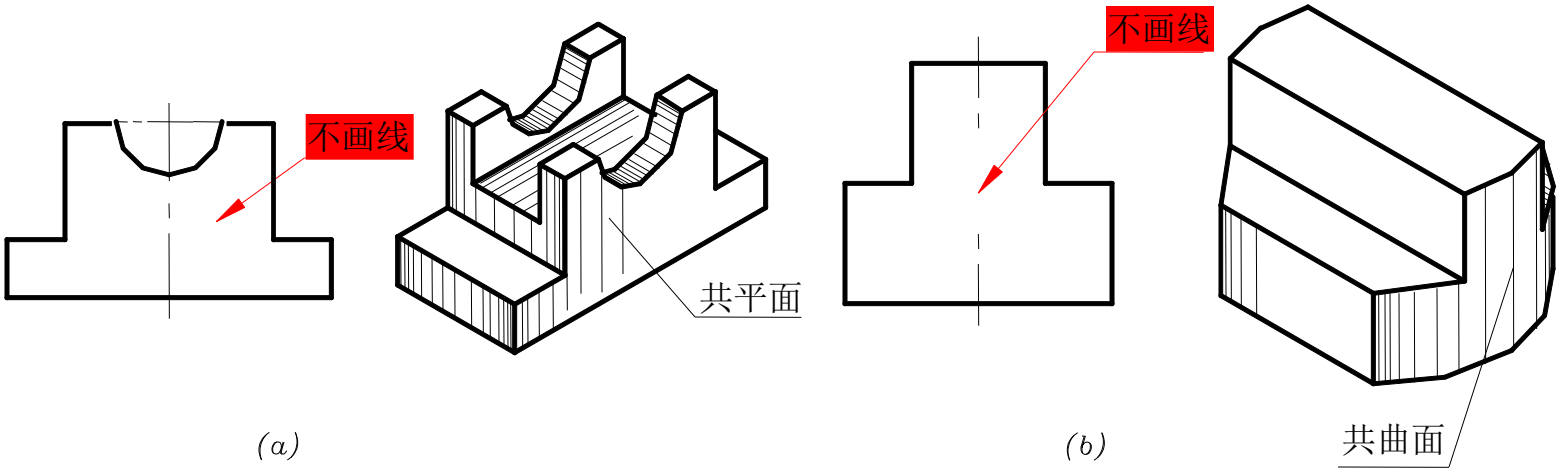


叠加形成的组合体

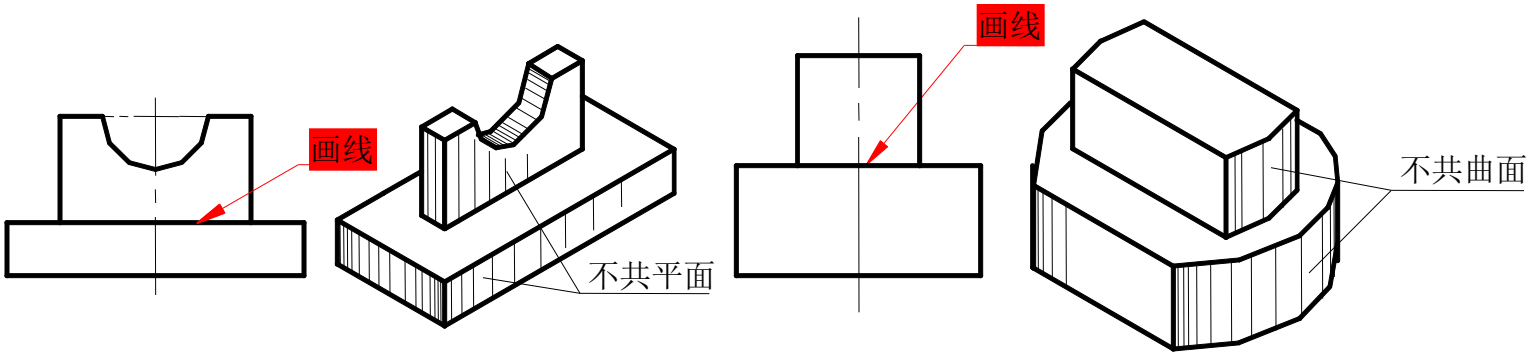
§5-1　组合体的基本知识

2、组合体表面间的相对位置

(1)共面



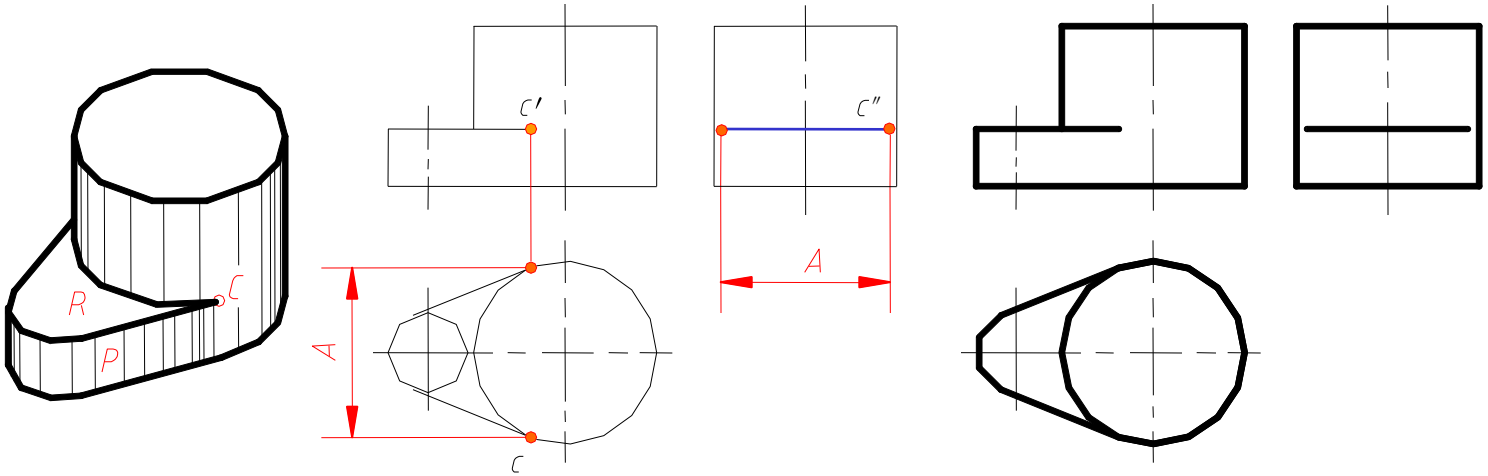
(a)　　(b)



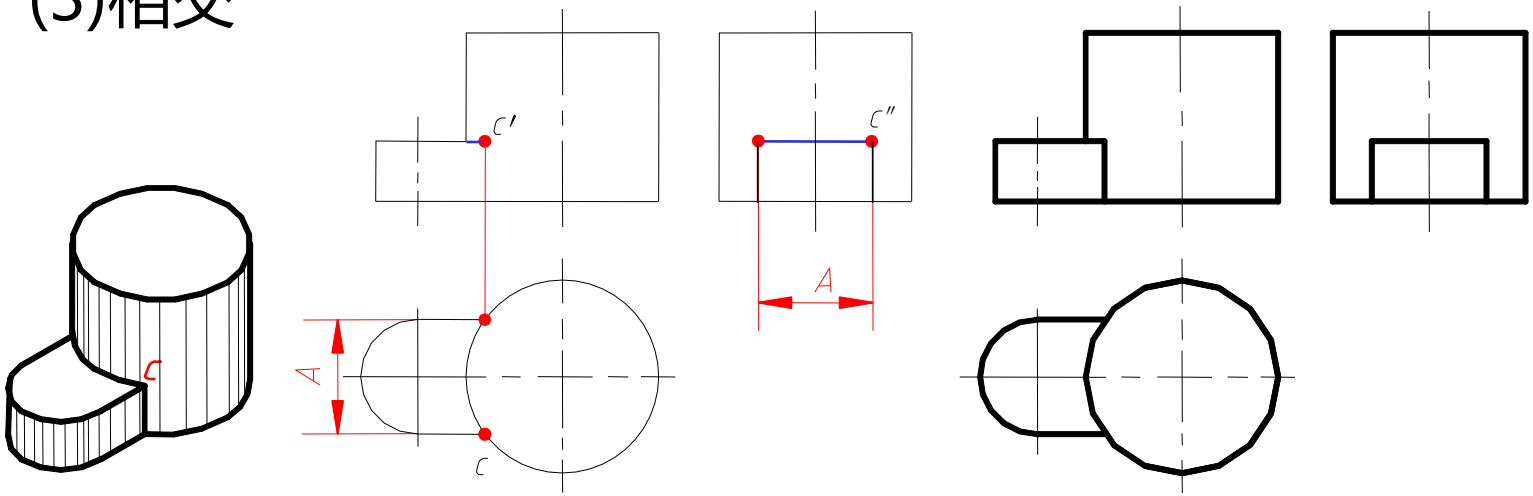
(c)　　(d)

§5-1　组合体的基本知识

(2)相切



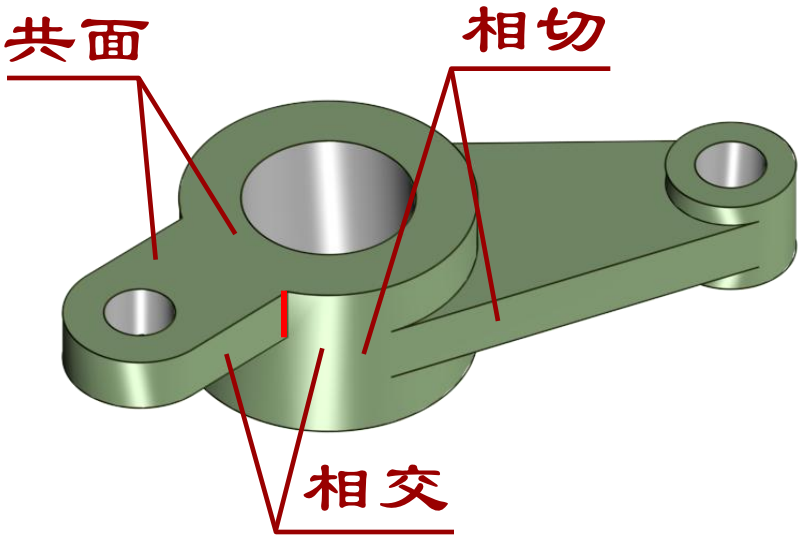
(3)相交



§5-1　组合体的基本知识

相邻形体表面之间的关系有几种？

- **共面**：共面处无分界线。
- **相交**：有表面交线。
- **相切**：相切处无分界线。



§5-2 组合体三视图的画法

步骤：

1、进行形体分析。

2、确定主视图。

3、选比例、定图幅。

4、布置图面，画作图基准线。

5、逐个画出各形体的三视图。

6、检查、描深，完成作图。

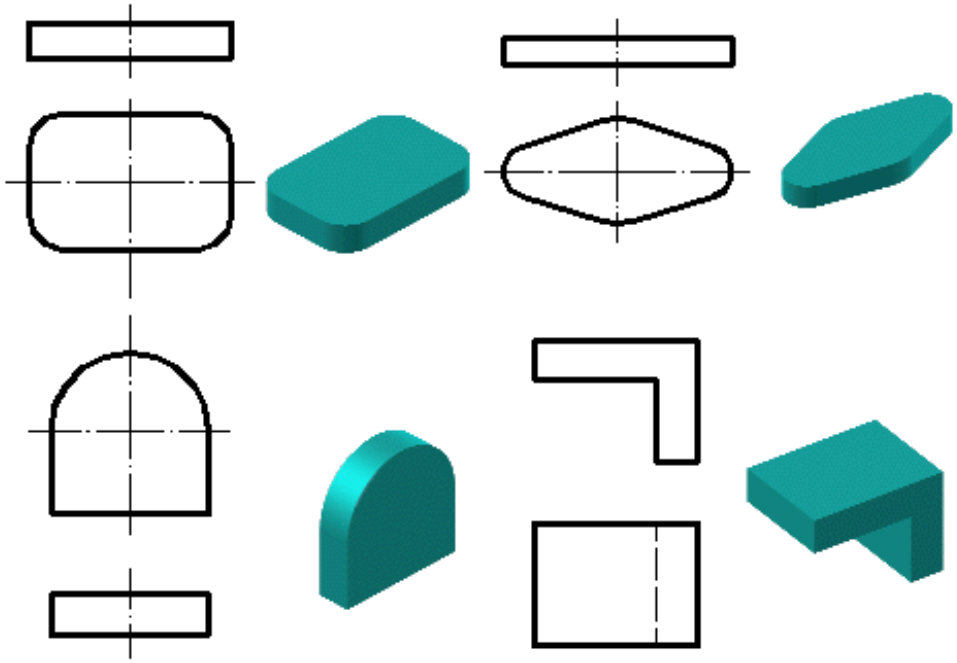
§5-2 组合体三视图的画法

1. 形体分析

叠加还是挖切？相对位置？相邻表面之间的关系？

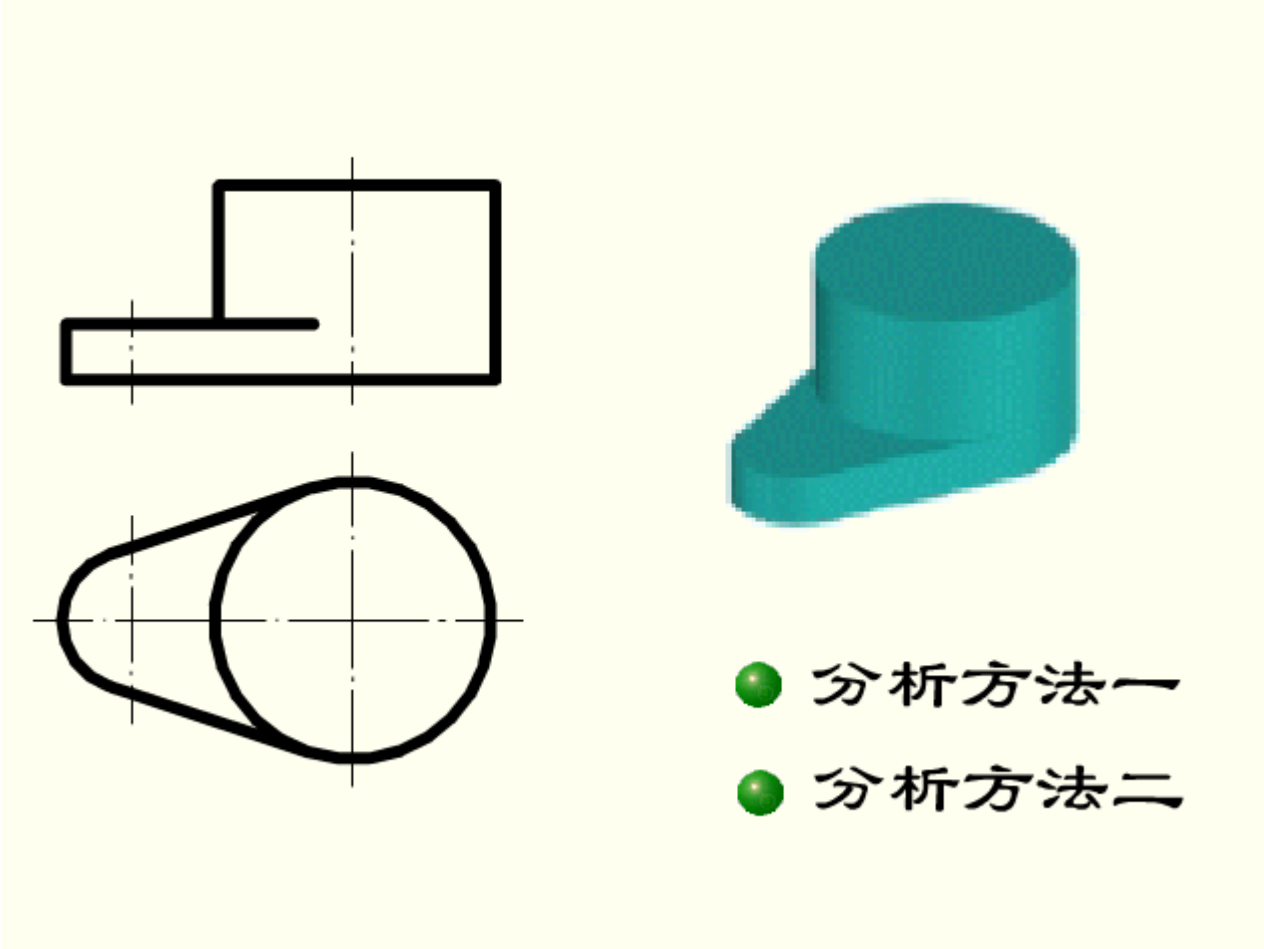
注意点：

（1）对于一些常见的简单的组合体，一般可看成为一个立体，不必再作更细的分析。这种立体通常称为**简单形体**。



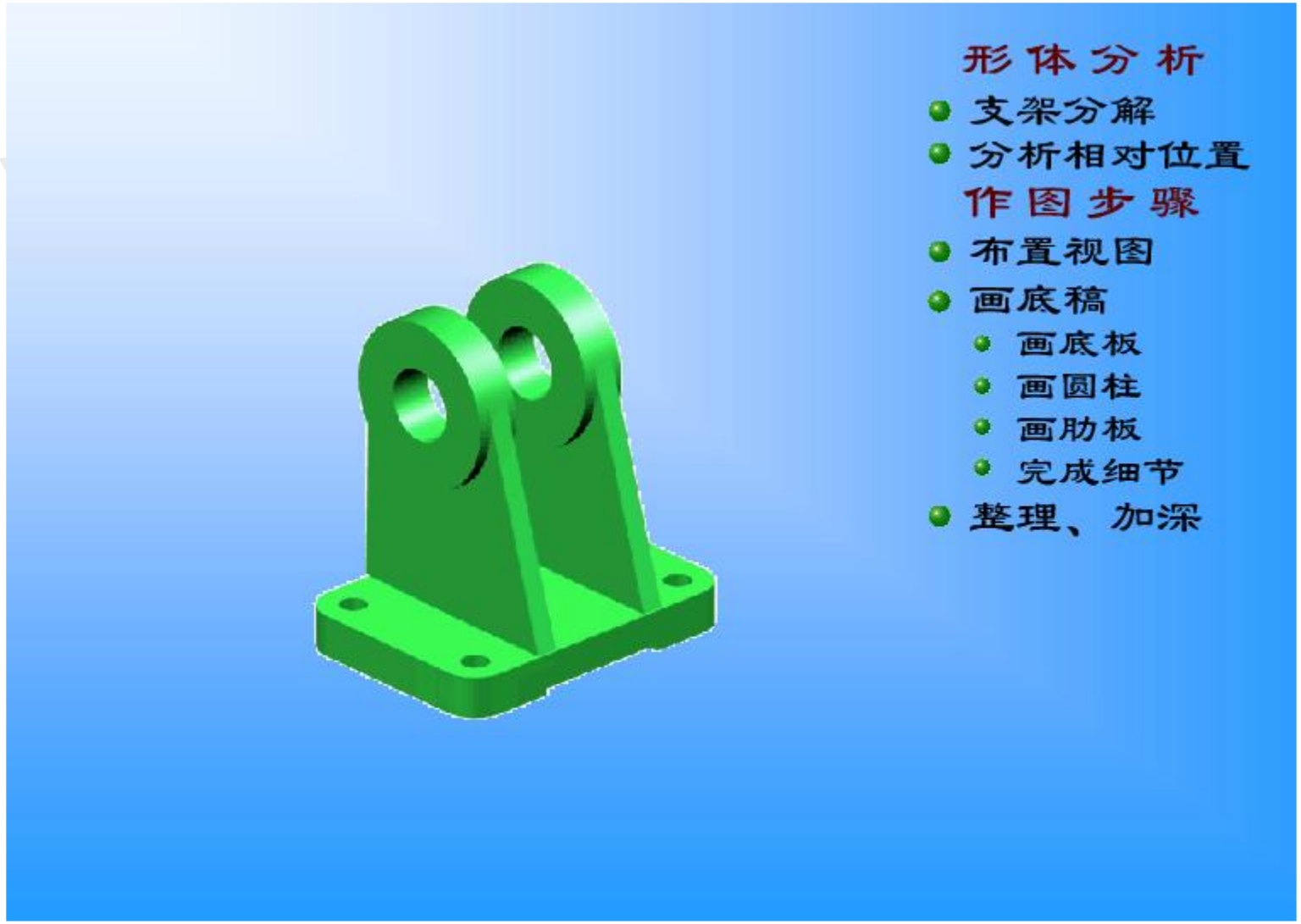
§5-2 组合体三视图的画法

（2）在许多情况下，立体的分解往往不是唯一的。



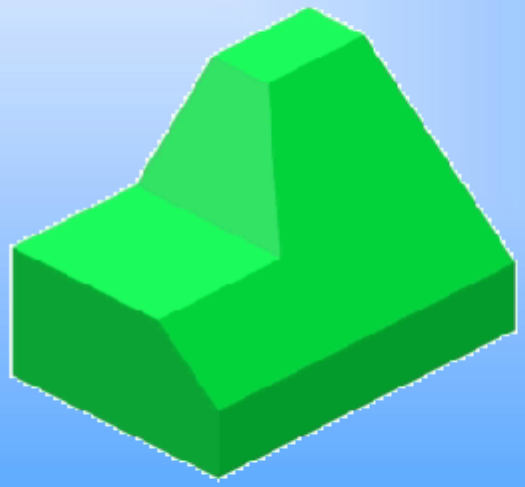
§5-2 组合体三视图的画法

例1. 支架的画法



§5-2 组合体三视图的画法

例2. 垫块的画法



- 形体分析
- 作图步骤
- 画底稿
  - 画长方体
  - 画前上方切角
  - 画左上方切角
- 整理、加深

§5-2 组合体三视图的画法

2. 确定主视图

原则：

（1）最能反映组合体的形体特征。

（2）最能反映相邻表面间相对位置关系。

（3）组合体三视图中虚线最少。

§5-2 组合体三视图的画法

形体分析

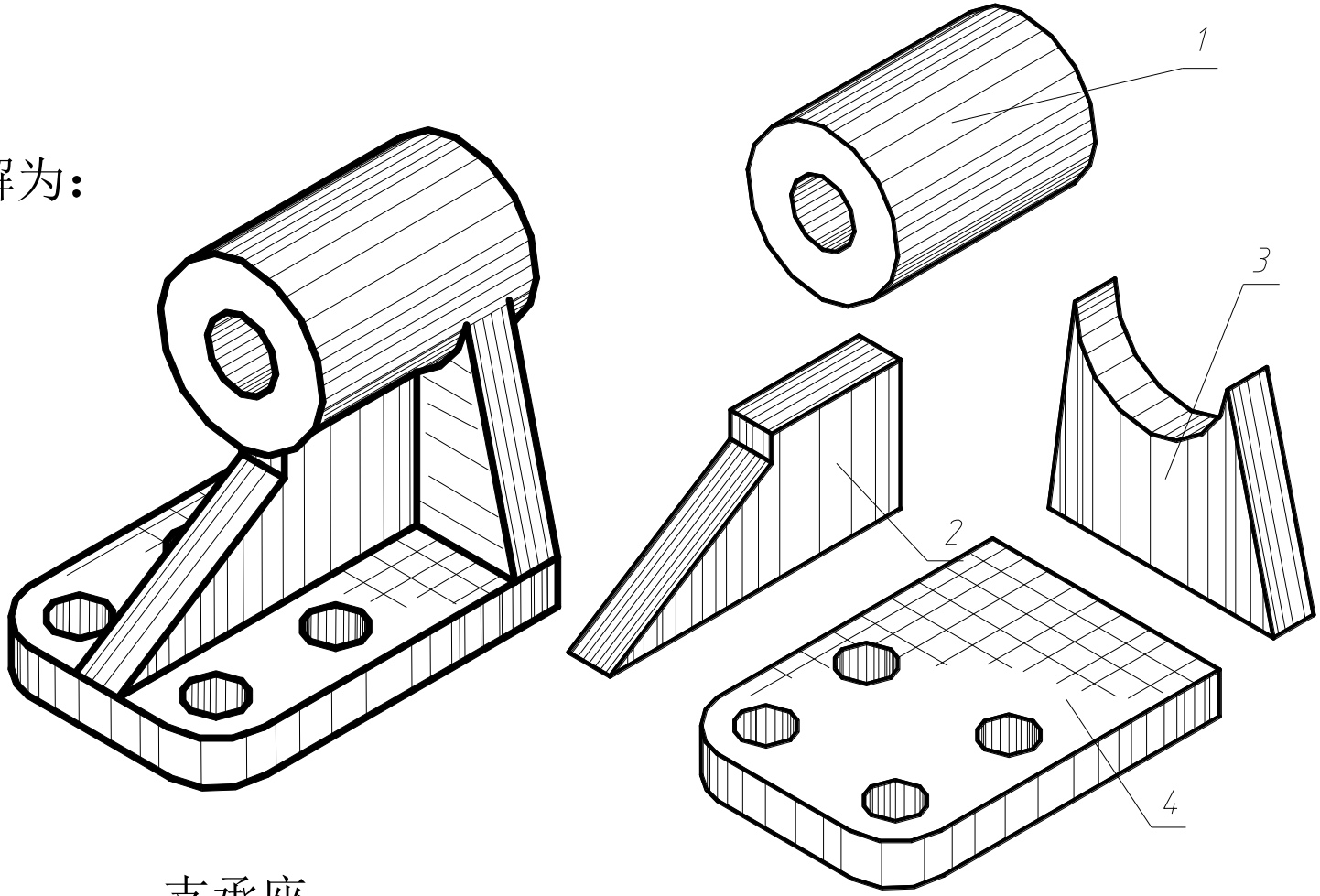
将支承座分解为：

1、圆柱

2、肋板

3、支承板

4、底板



支承座

§5-2 组合体三视图的画法

3. 选比例、定图幅

比例(一)

种　类	比　　例		
原值比例	1:1		
放大比例	5:1 $5\times10^n:1$	2:1 $2\times10^n:1$	$1\times10^n:1$
缩小比例	1:2 $1:2\times10^n$	1:5 $1:5\times10^n$	1:10 $1:1\times10^n$

注：n为正整数。

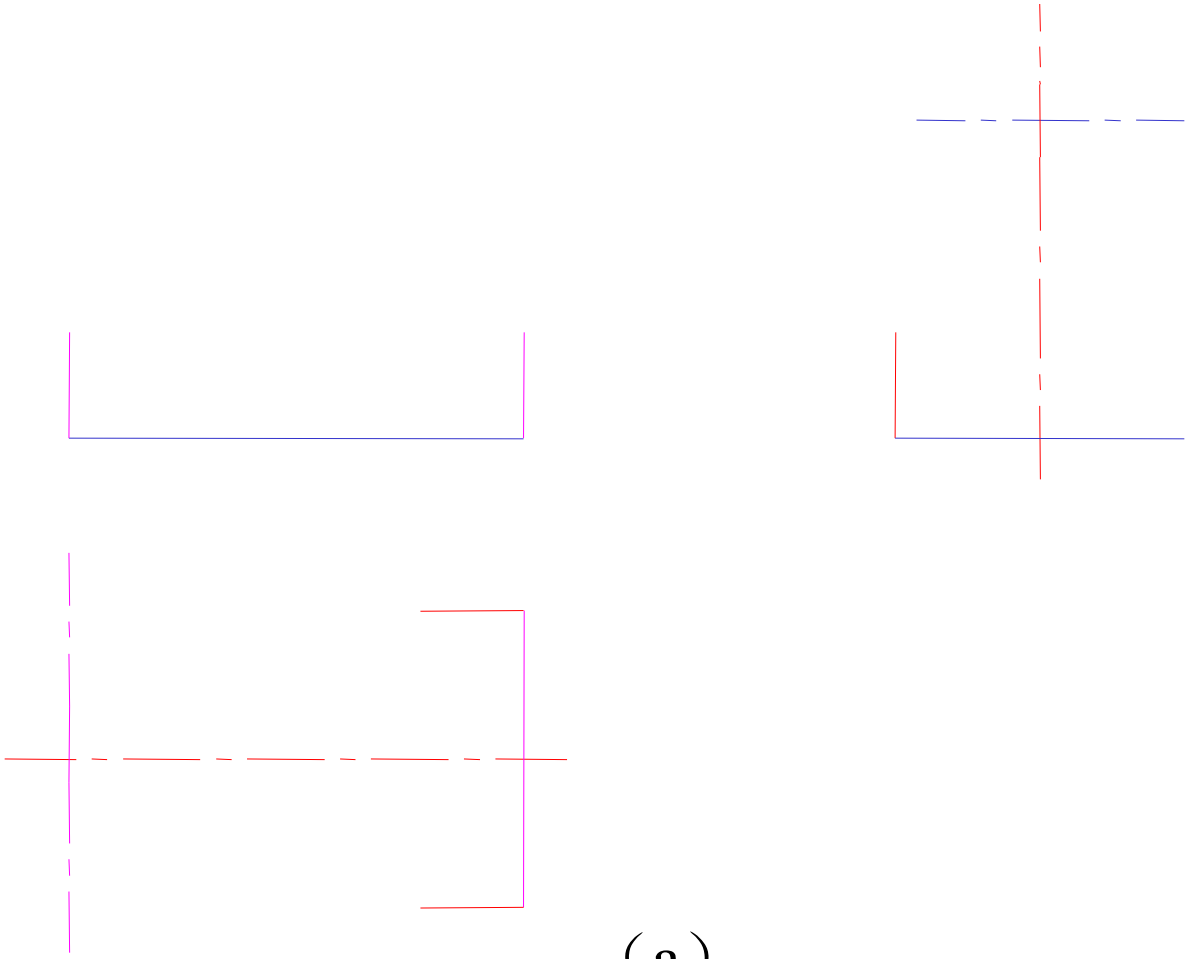
比例(二)

种　类	比　　例				
放大比例	4:1 $4\times10^n:1$		2.5:1 $2.5\times10^n:1$		
缩小比例	1:1.5 $1:1.5\times10^n$	1:2.5 $1:2.5\times10^n$	1:3 $1:3\times10^n$	1:4 $1:4\times10^n$	1:6 $1:6\times10^n$

注：n为正整数。

§5-2 组合体三视图的画法

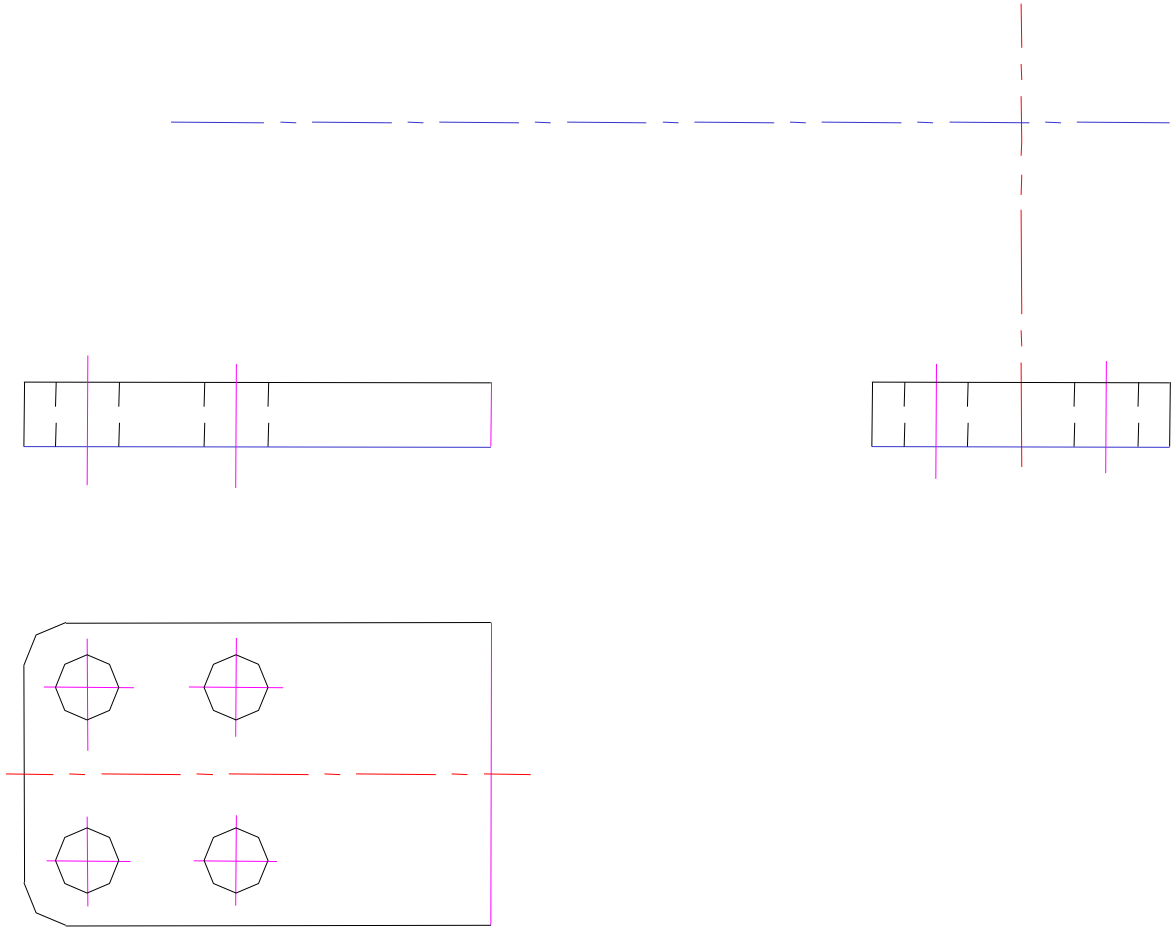
4. 布置图面，画作图基准线。



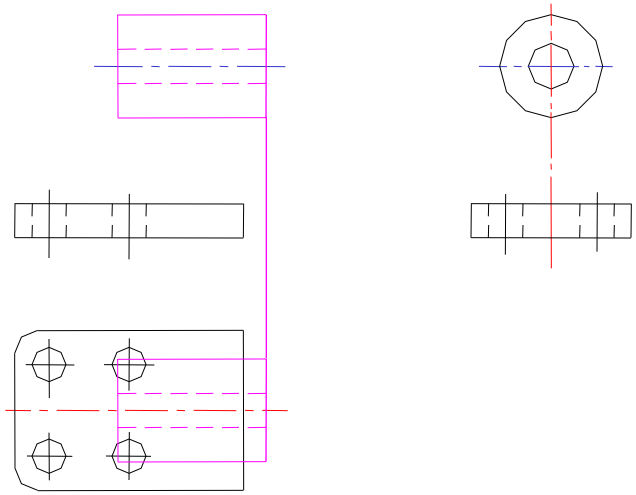
（a）

§5-2 组合体三视图的画法

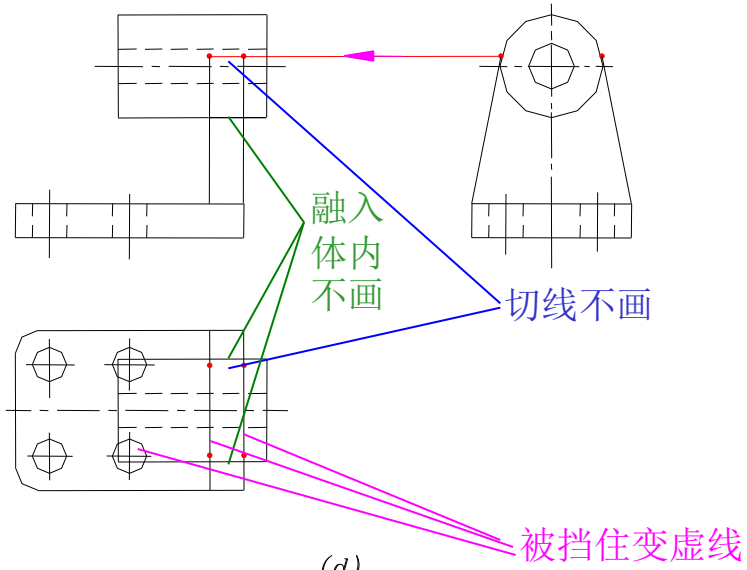
5. 逐一绘制各基本形体。



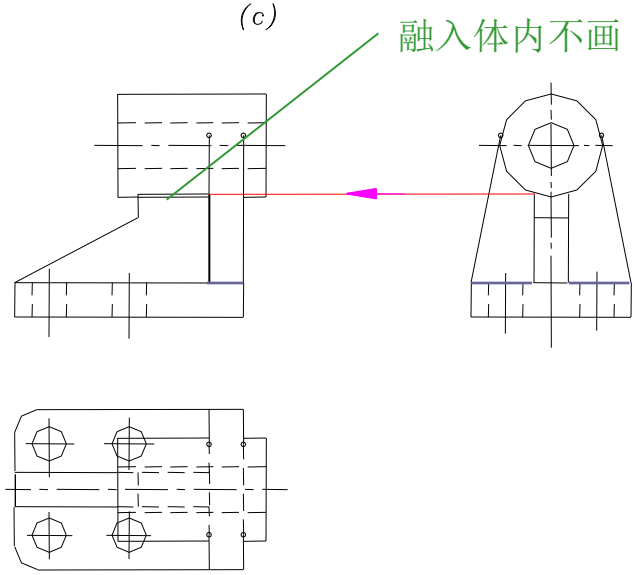
(b)



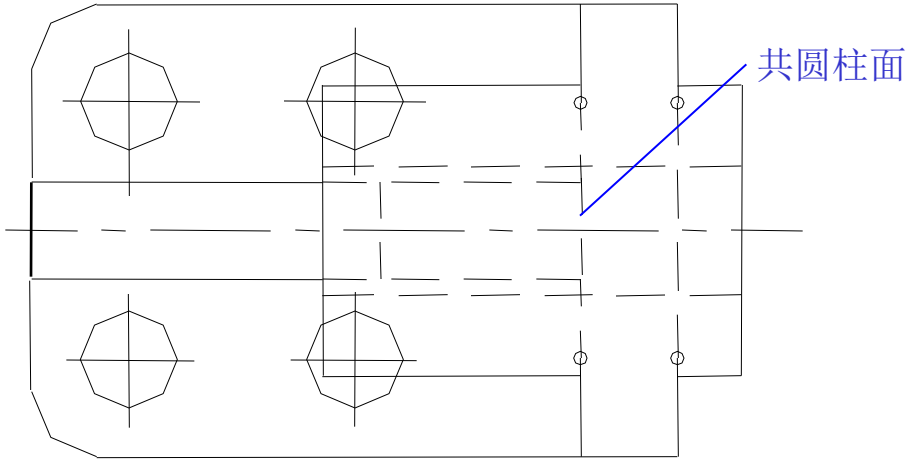
(c)



(d)

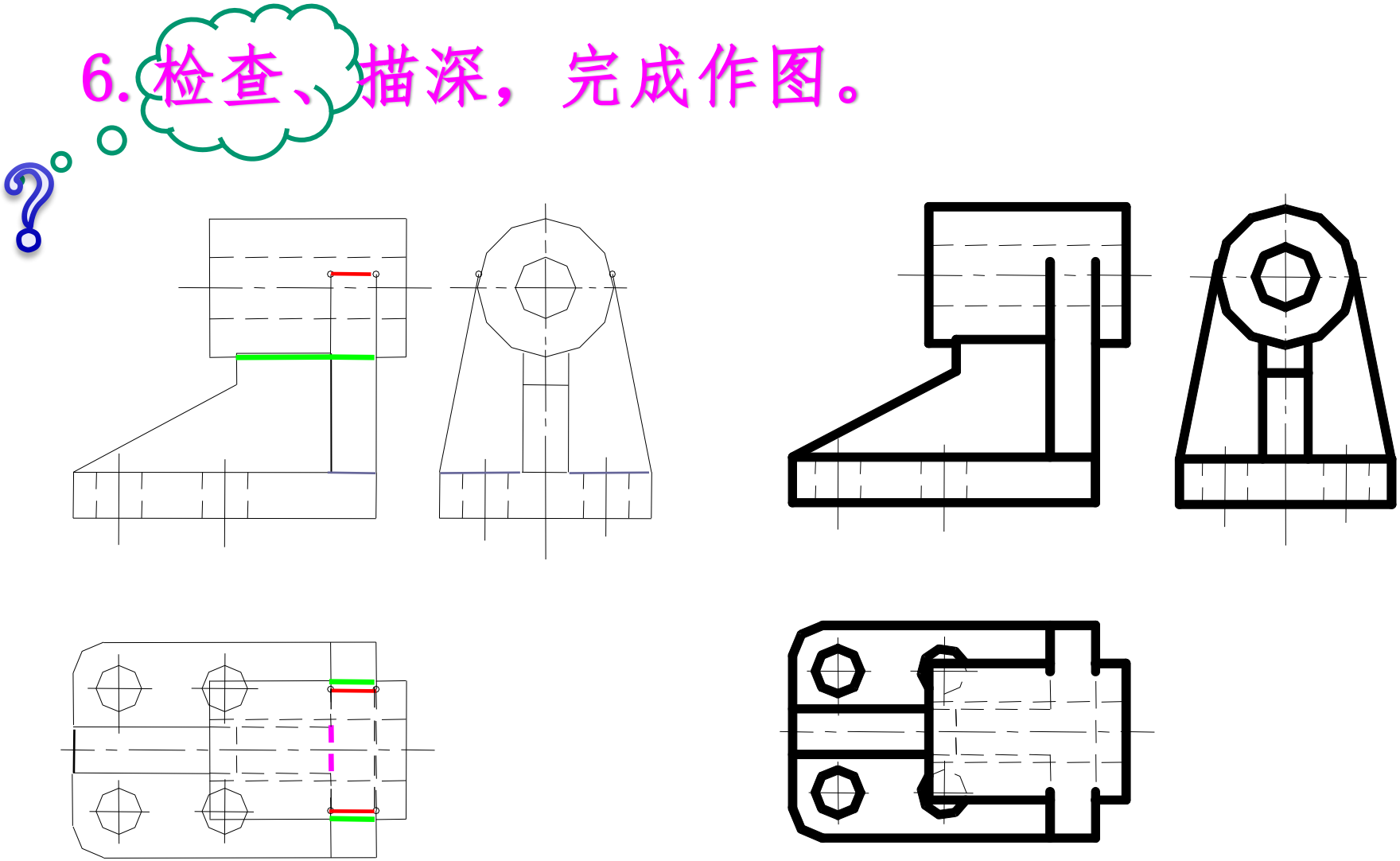


(e)



§5-2 组合体三视图的画法

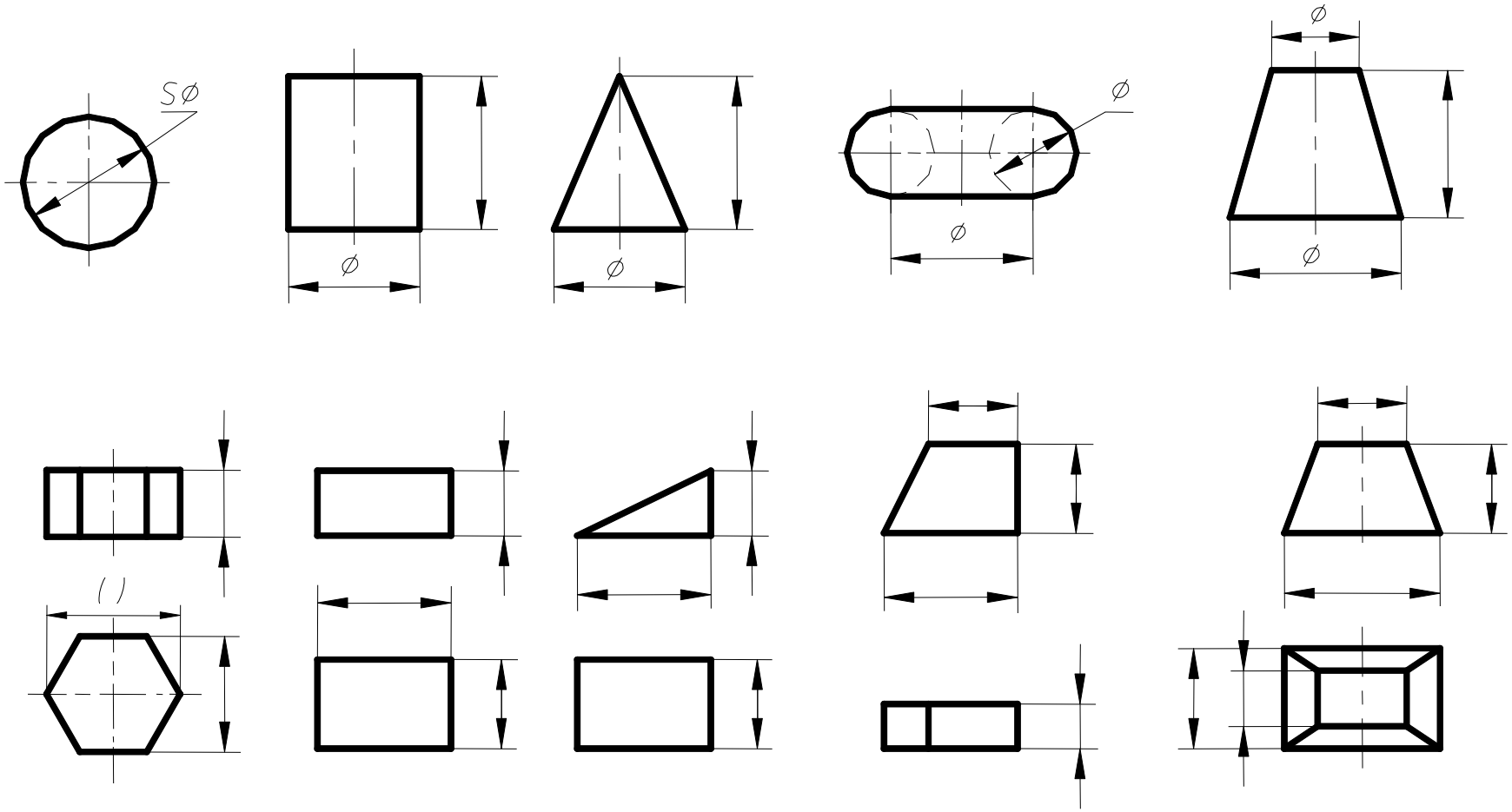
6. 检查、描深，完成作图。



§5-3 组合体的尺寸注法

标注尺寸总的要求是：正确、完整、清晰、合理

1、定形尺寸：定形尺寸是确定基本形体大小的尺寸，如圆柱、圆锥、六棱柱、六面体等所标注的尺寸。



§5-3 组合体的尺寸注法

2、定位尺寸：

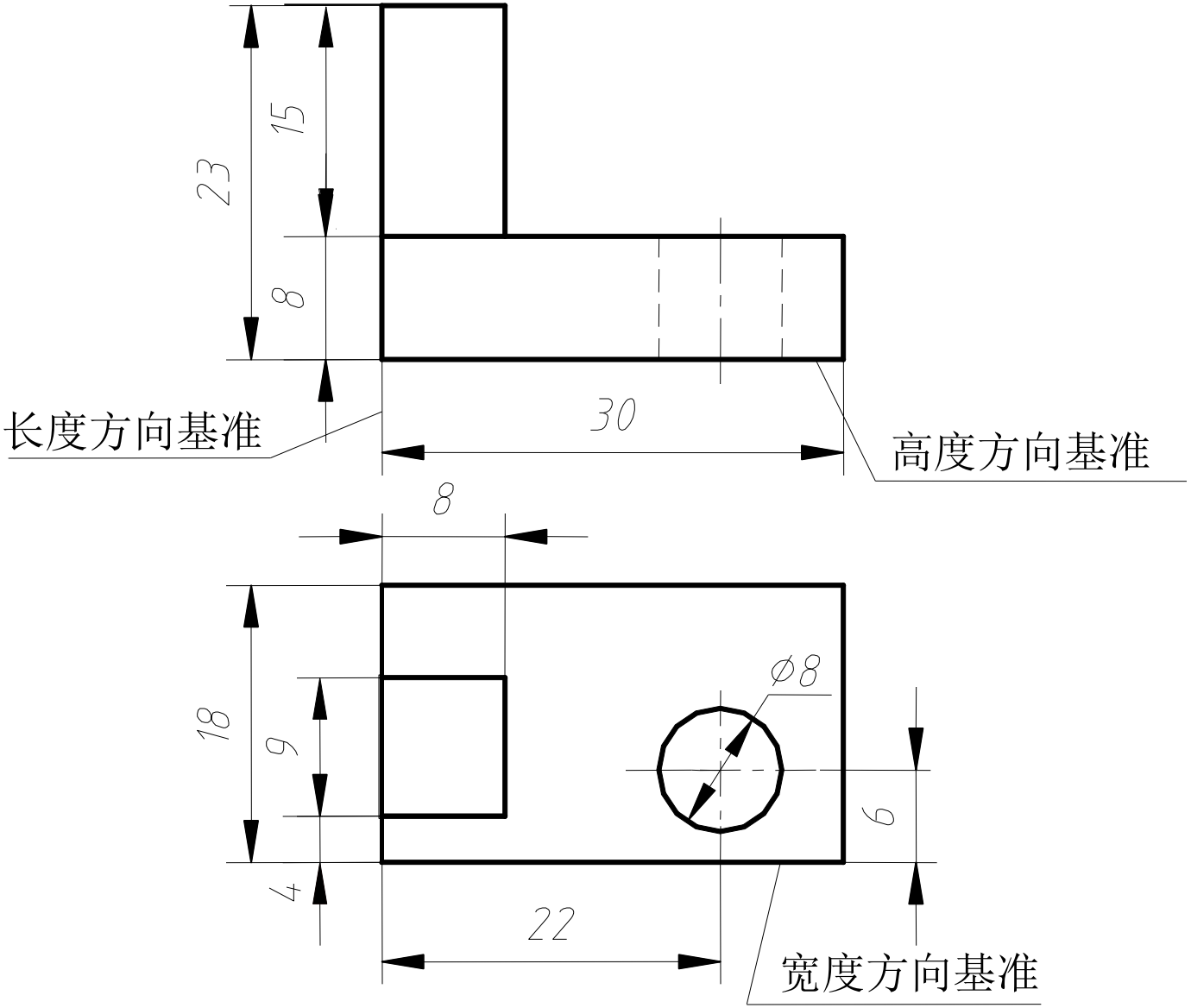
从尺寸、基准出发，定各基本形体的相对位置尺寸。

基准：

标 注尺寸的起点。

3、总体尺寸：

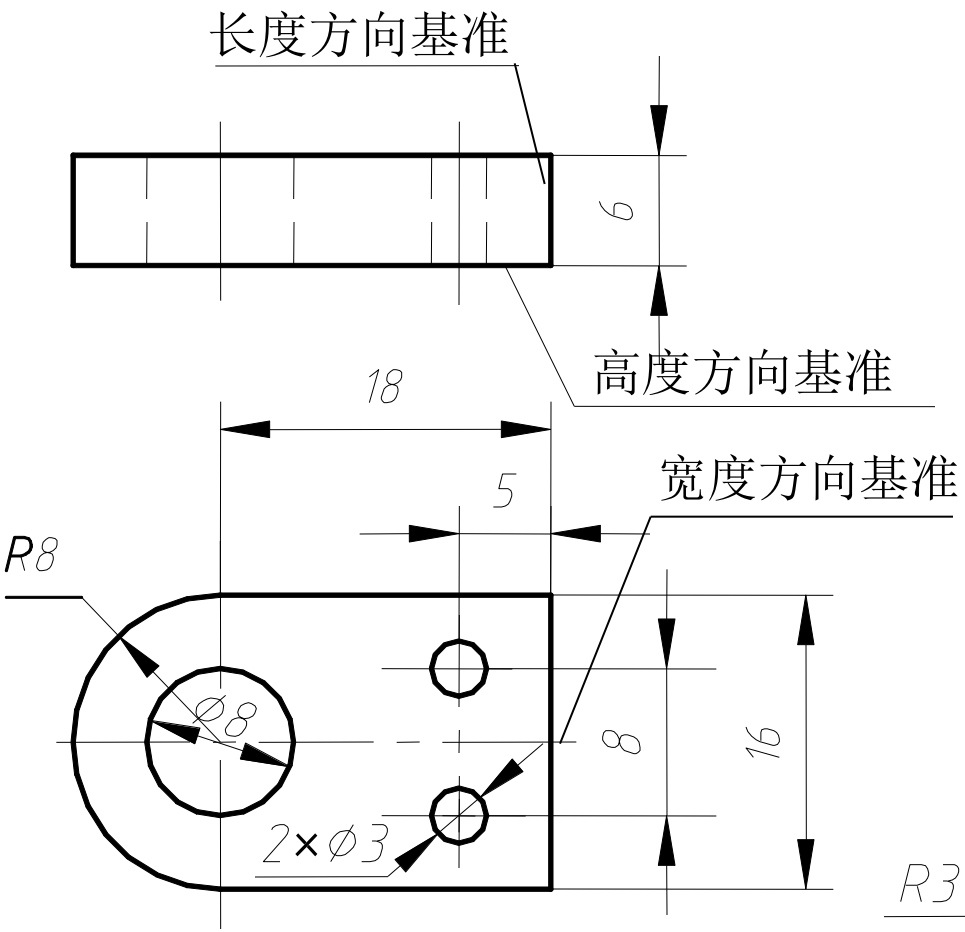
组合体的总高，总长和总宽。

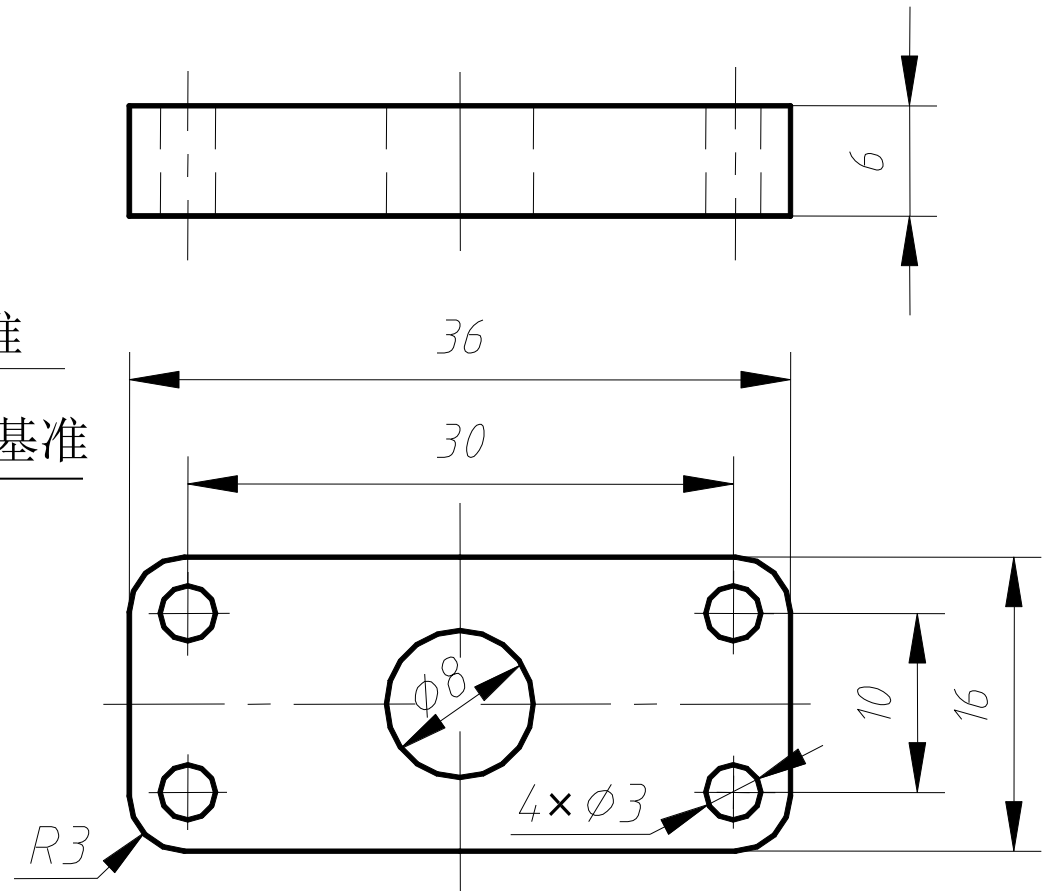


§5-3 组合体的尺寸注法

标注定形尺寸、定位尺寸
又标注总体尺寸

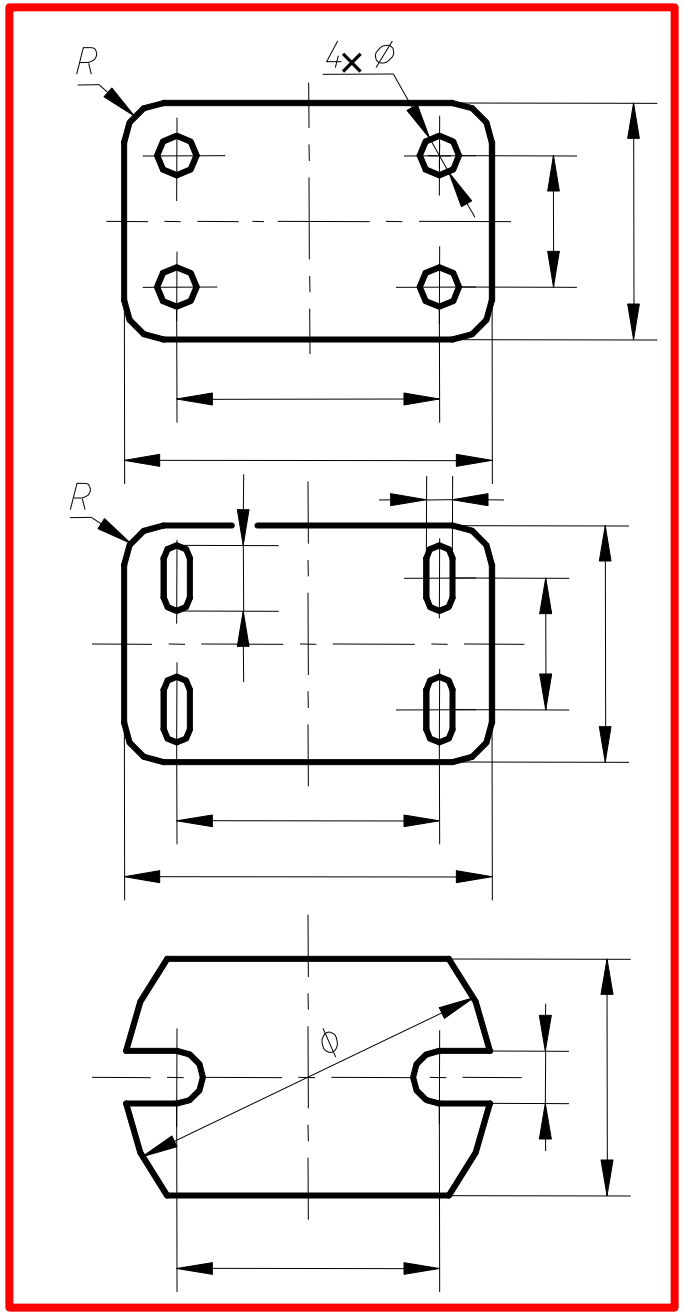
总体尺寸通过计算给定

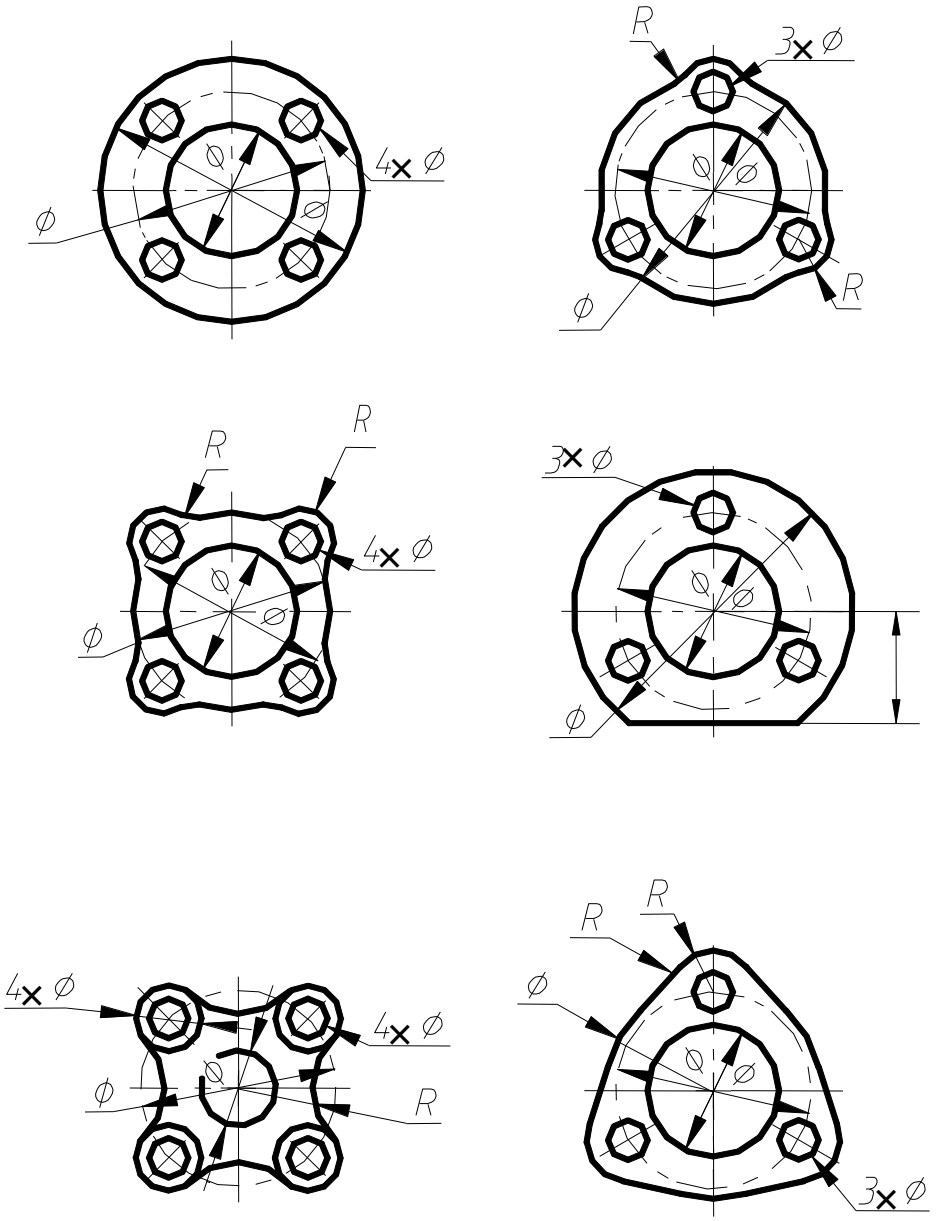




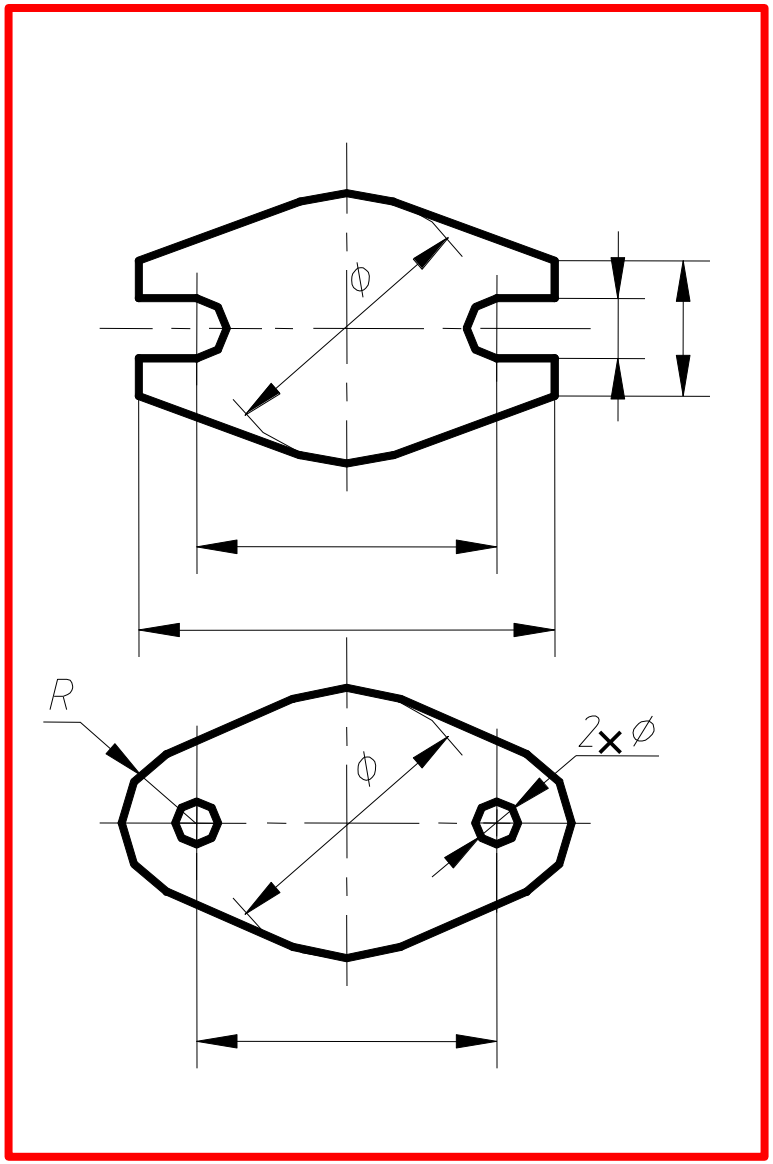
§5-3 组合体的尺寸注法

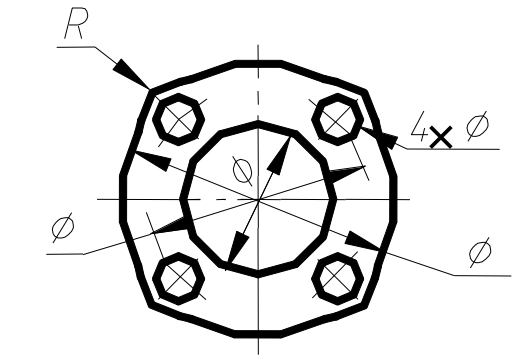
4、常见结构形状的底板端板尺寸注法

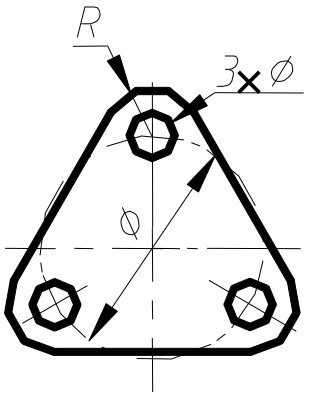


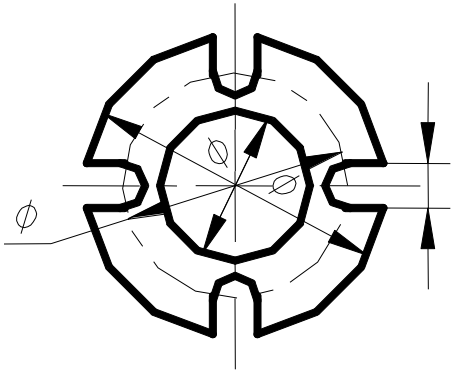


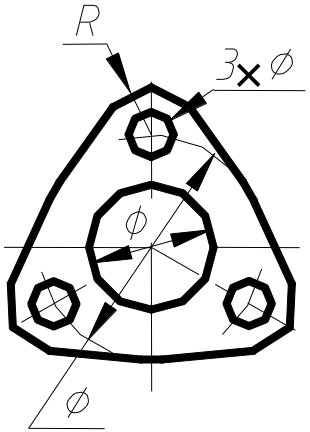
§5-3 组合体的尺寸注法







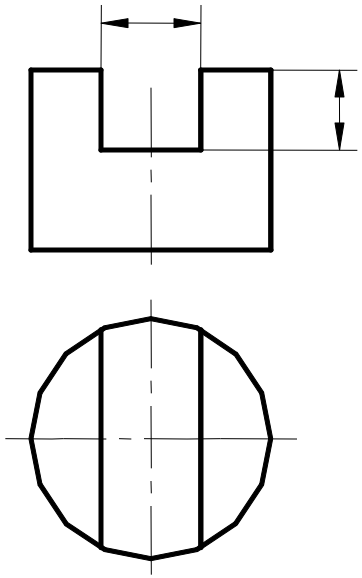




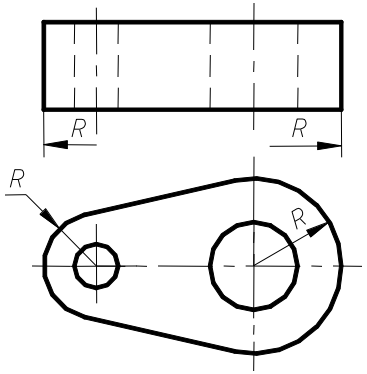
§5-3 组合体的尺寸注法

5、标注尺寸注意事项:

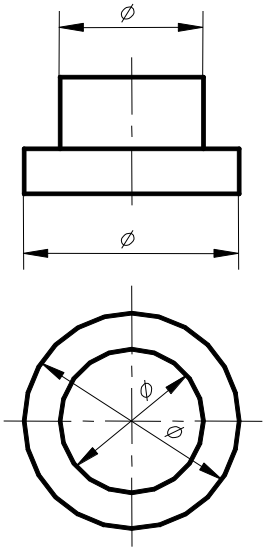
(1)尺寸应标注在反映形体特征明显的视图上，相关尺寸应集中标注。



(2)小于或等于半圆的圆弧半径尺寸注在反映圆弧实形的视图上。

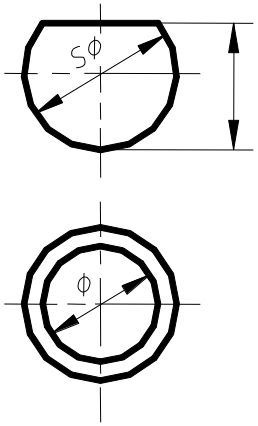


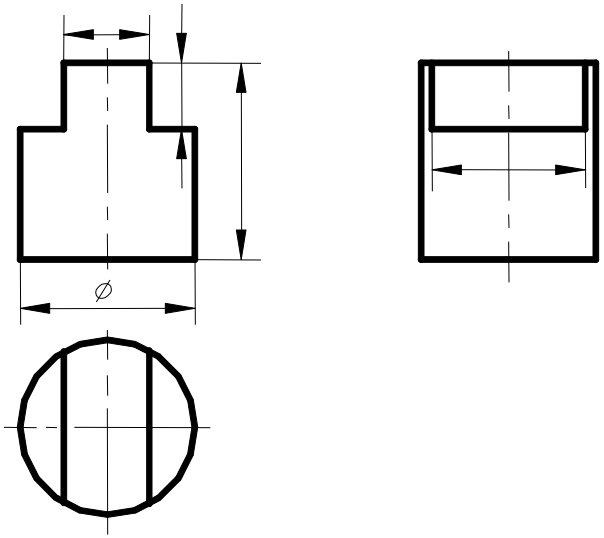
(3)直径尺寸最好注在非圆视图上。

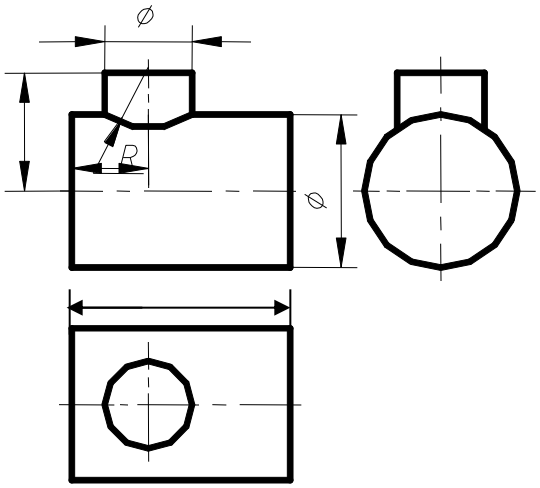


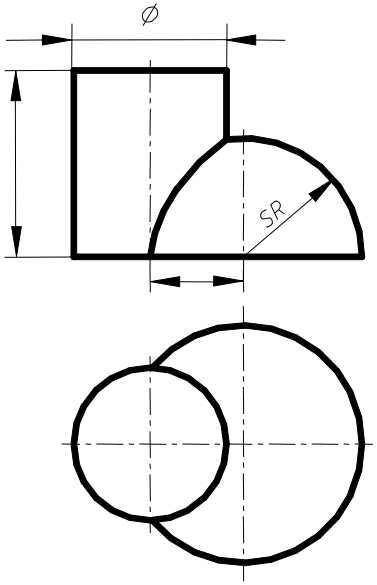
§5-3 组合体的尺寸注法

(4)立体表面交线一般不应标注尺寸。







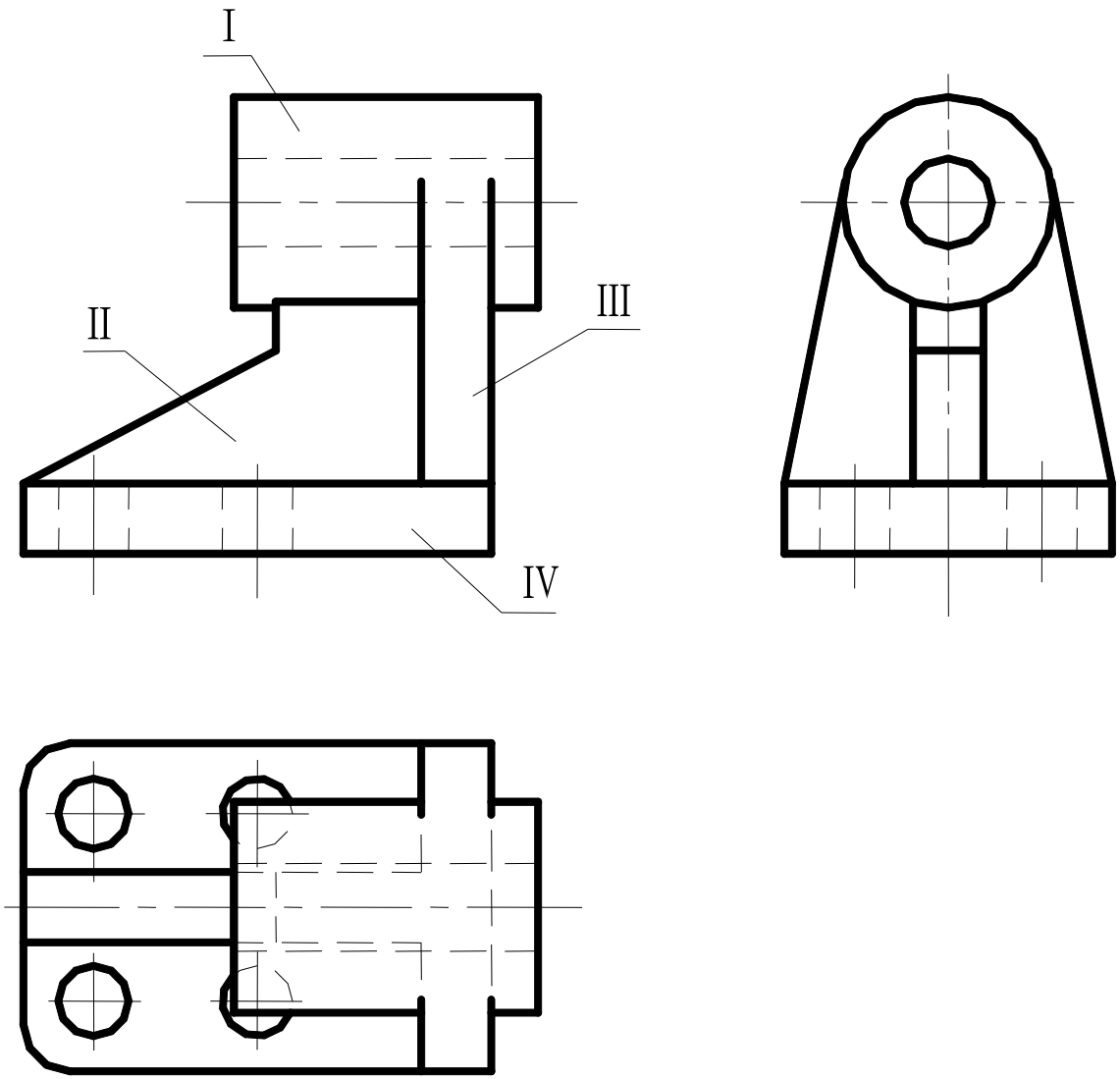


§5-3 组合体的尺寸注法

6、举例：

标注支承座尺寸。

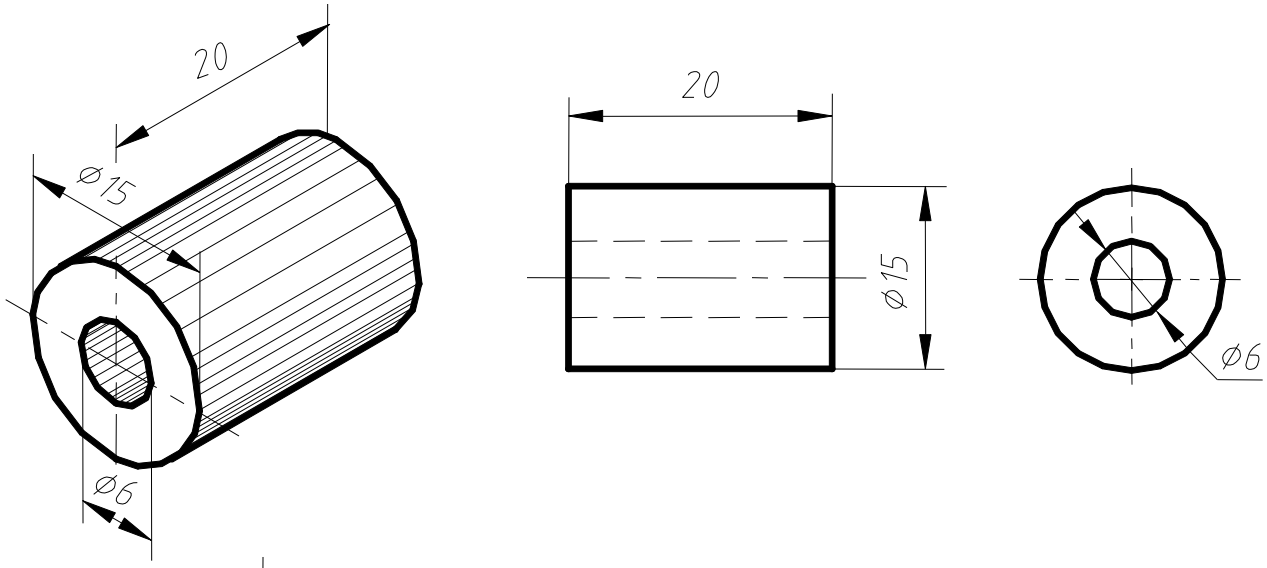
(1)形体分析。



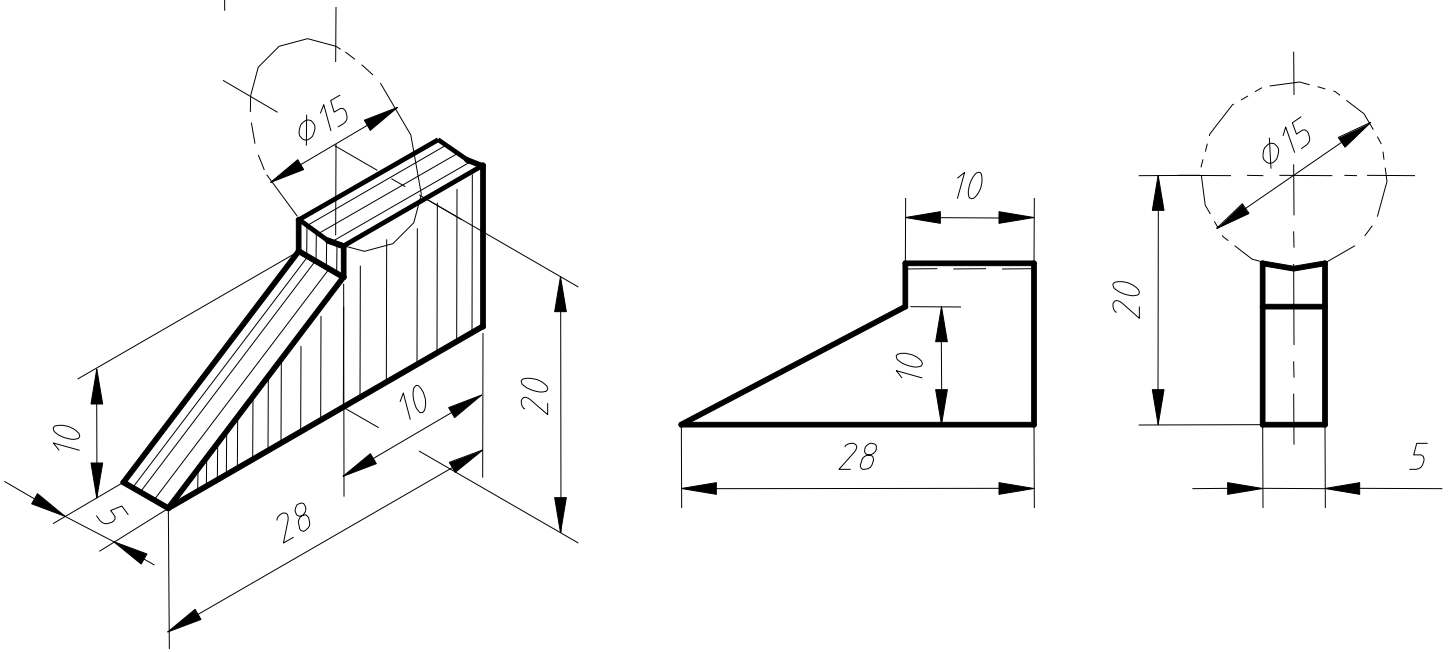
§5-3 组合体的尺寸注法

(2)标注各基本形体定形尺寸

圆柱定形尺寸

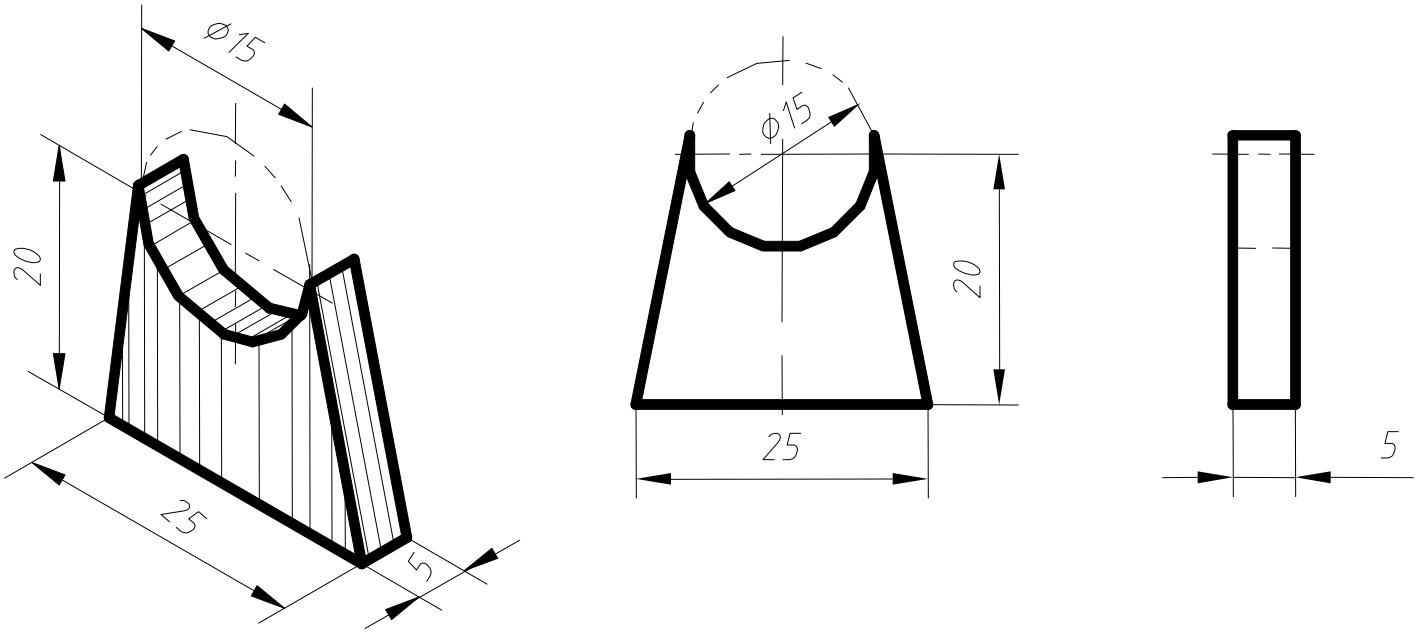


肋板定形尺寸

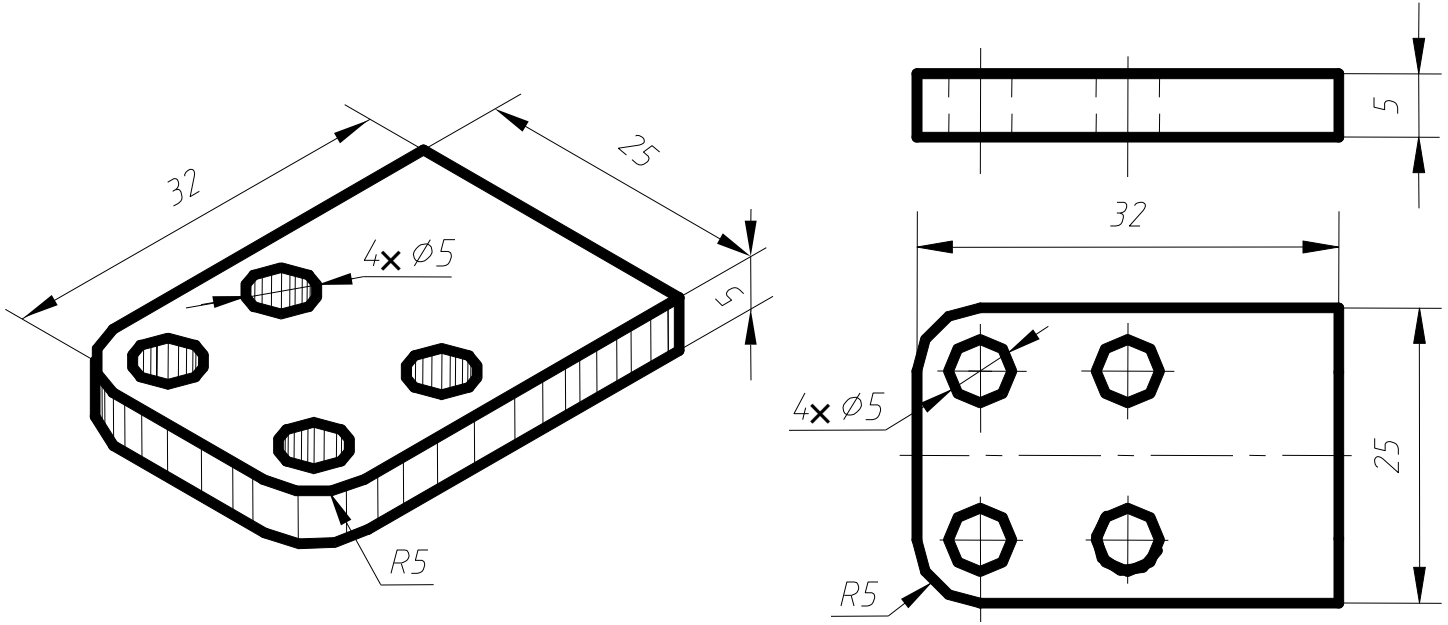


§5-3 组合体的尺寸注法

支承板定形尺寸

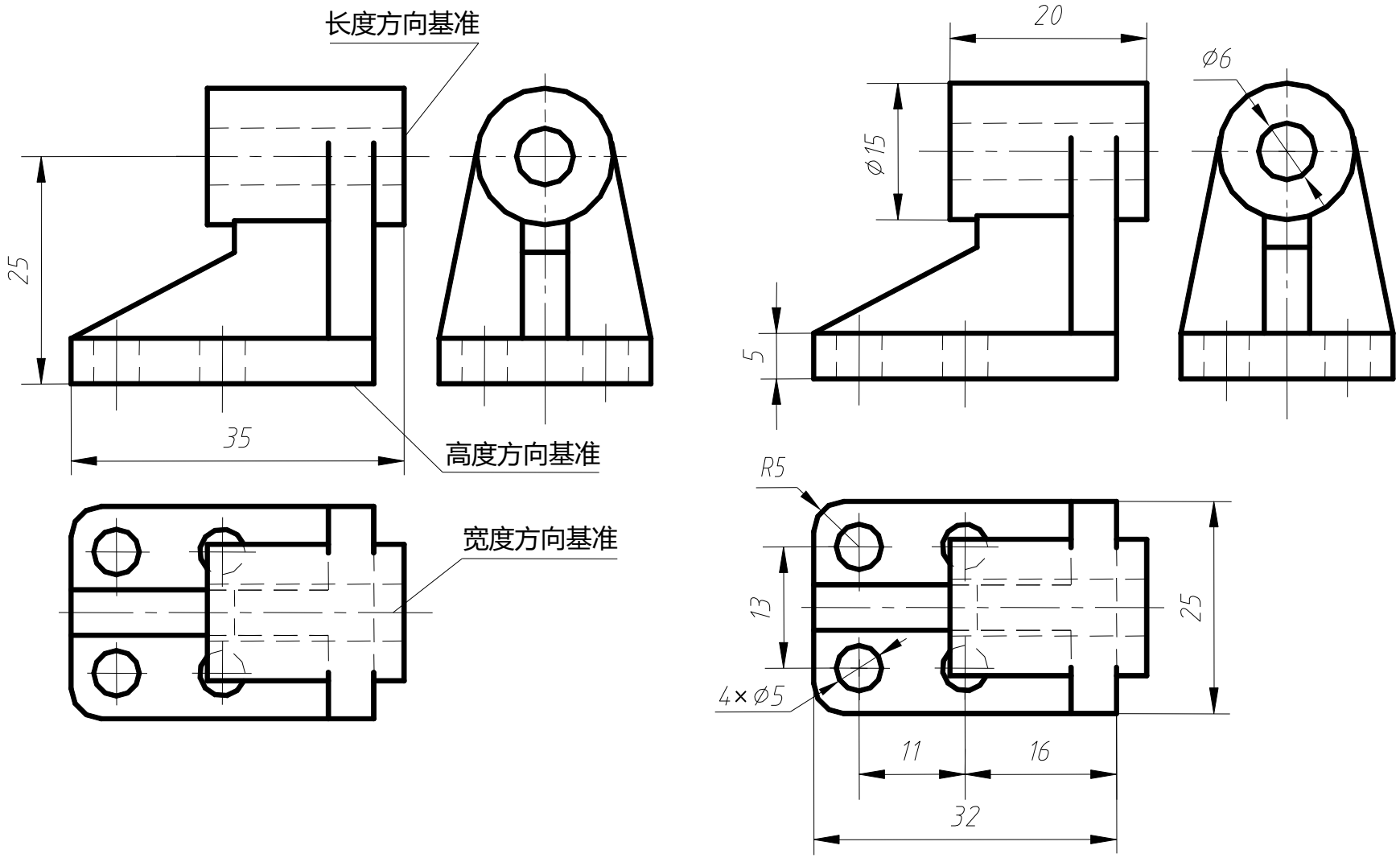


底板定形尺寸

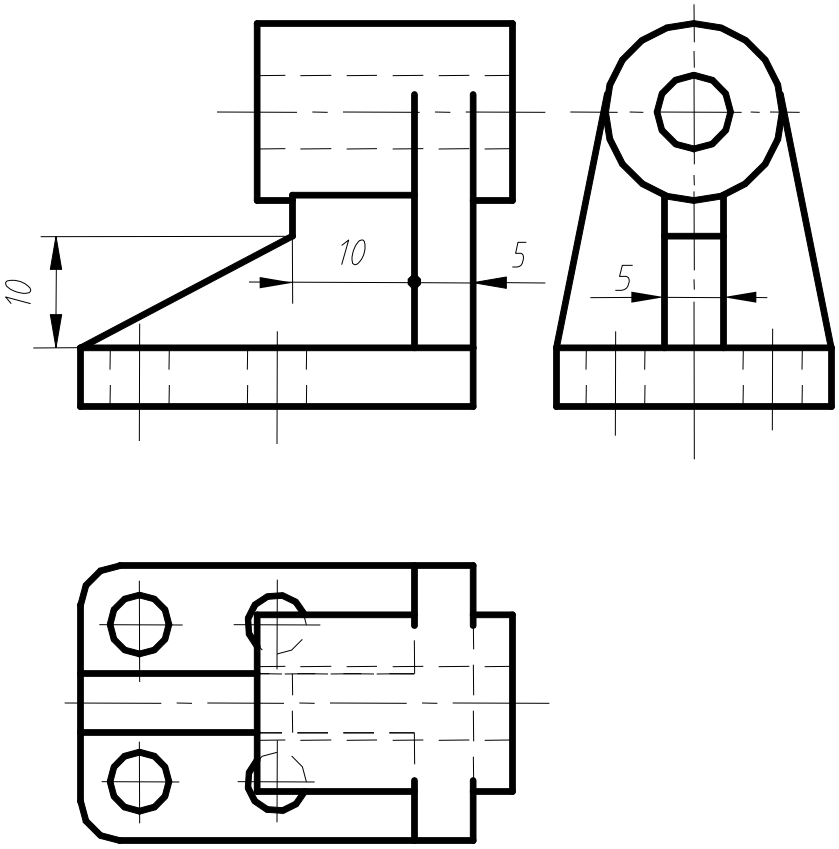


§5-3 组合体的尺寸注法

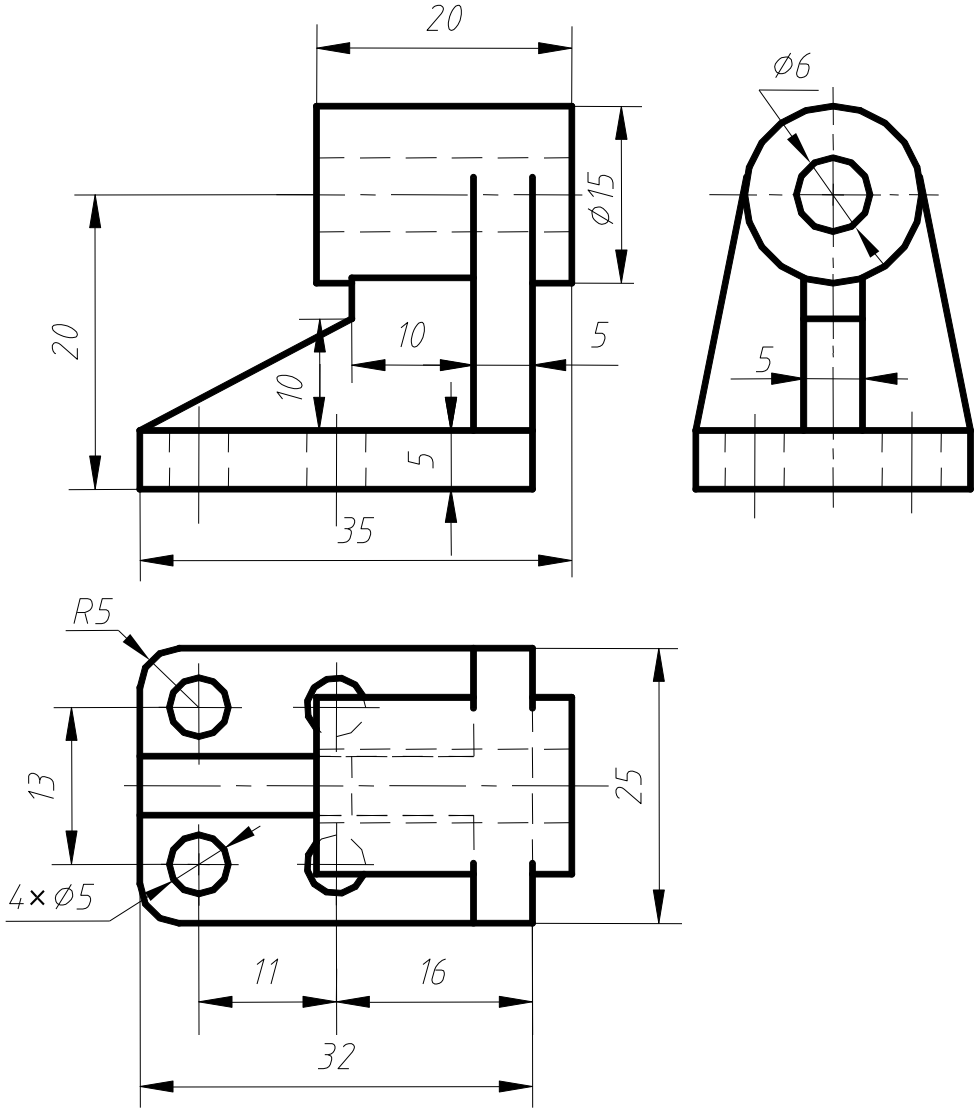
(3)支承座尺寸标注顺序



§5-3 组合体的尺寸注法



(c)

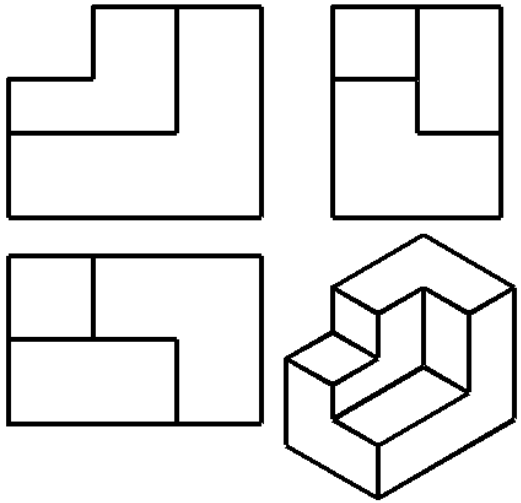


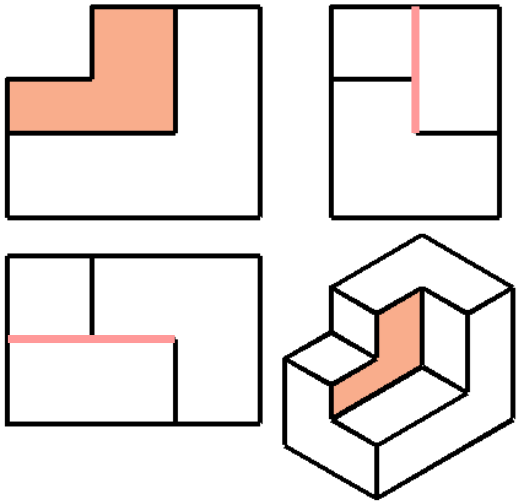
(d)

§5-4 读组合体视图

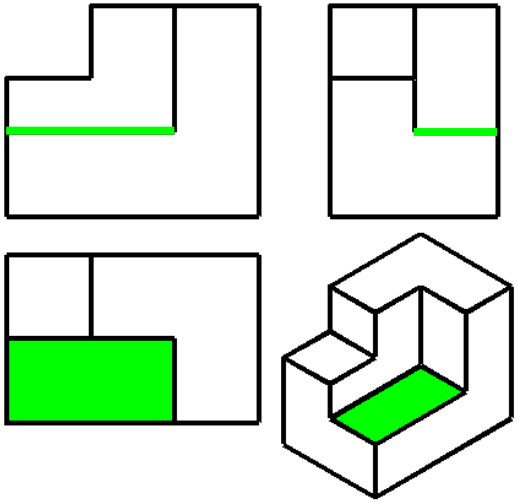
1、平行面的投影具有实形性和积聚性

题目

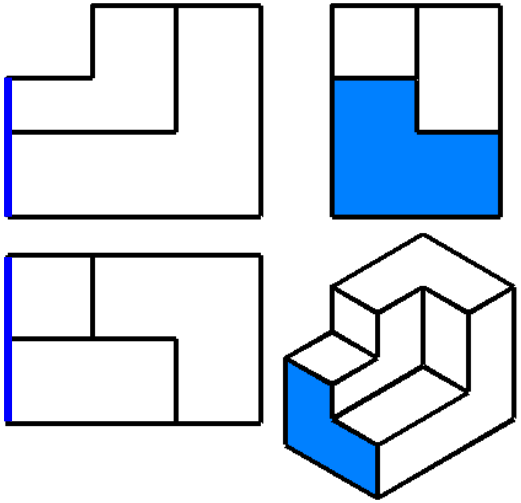




正平面的投影



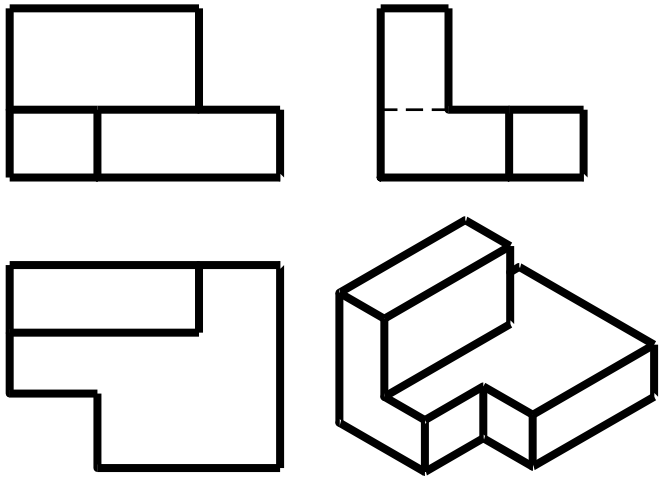
水平面的投影



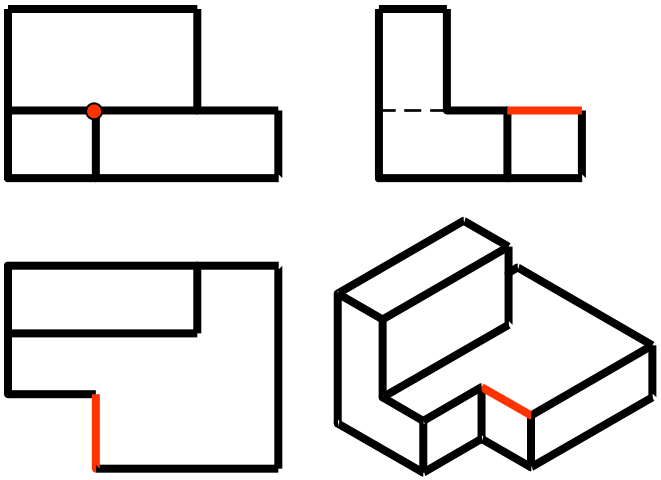
侧平面的投影

§5-4 读组合体视图

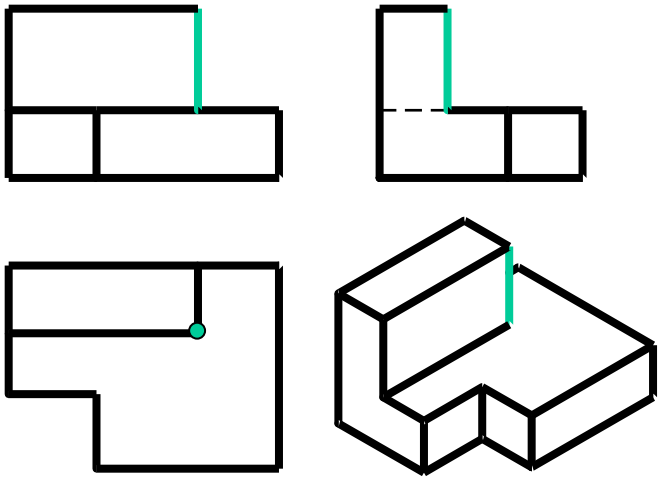
2、垂直线的投影具有实长性和积聚性



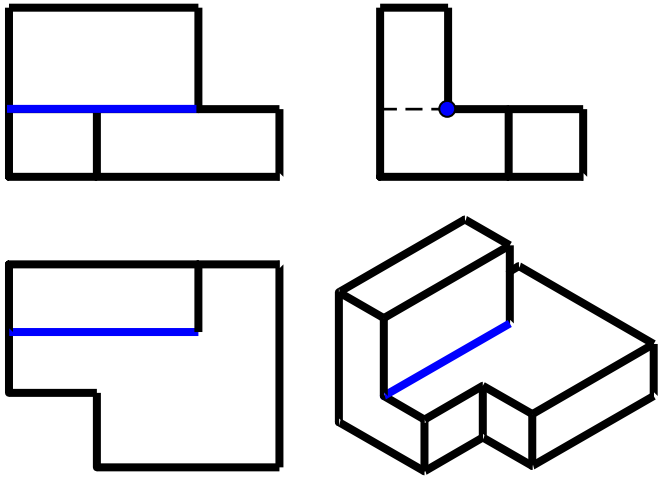
题目



正垂线的投影



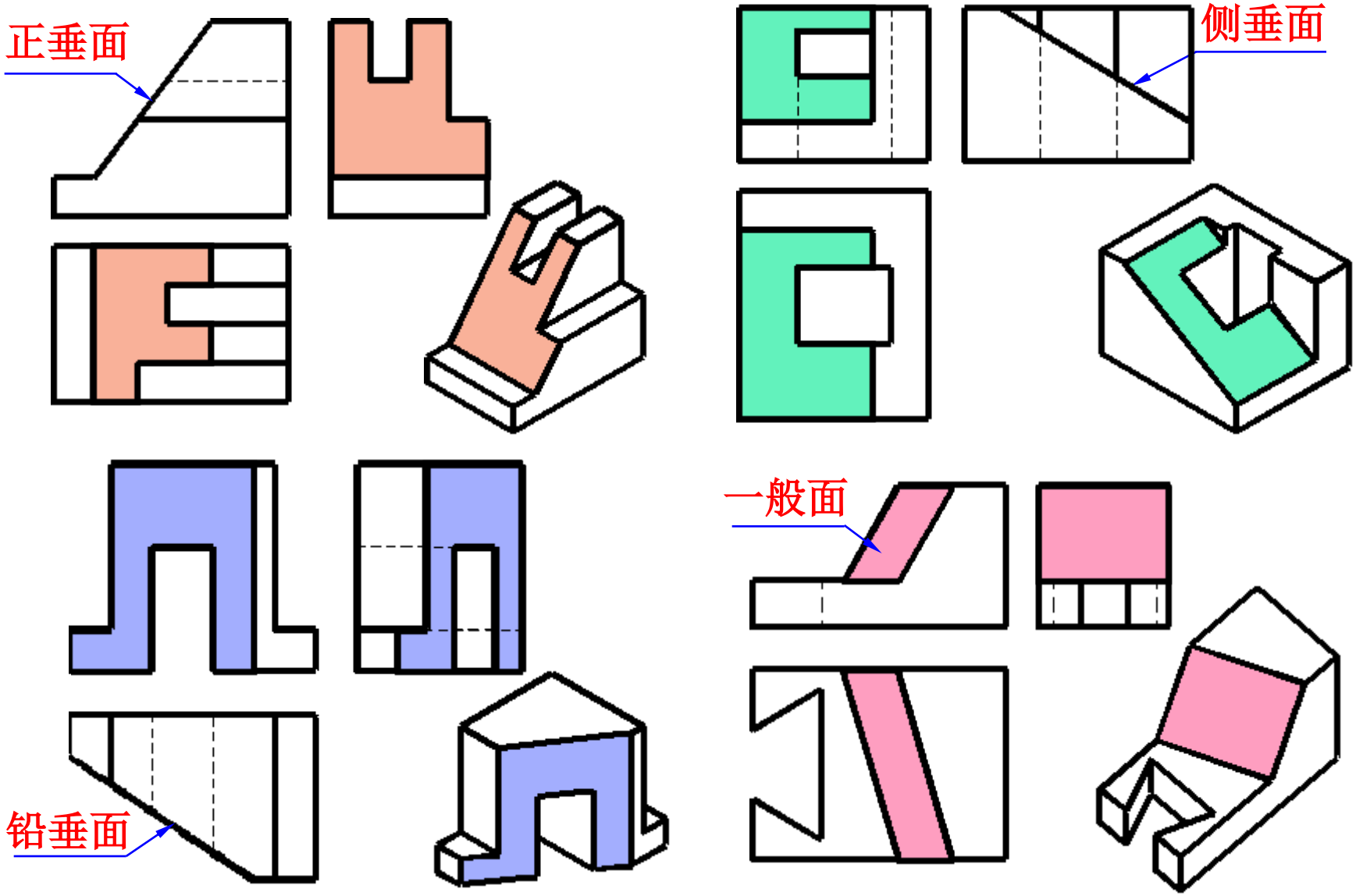
铅垂线的投影



侧垂线的投影

§5-4 读组合体视图

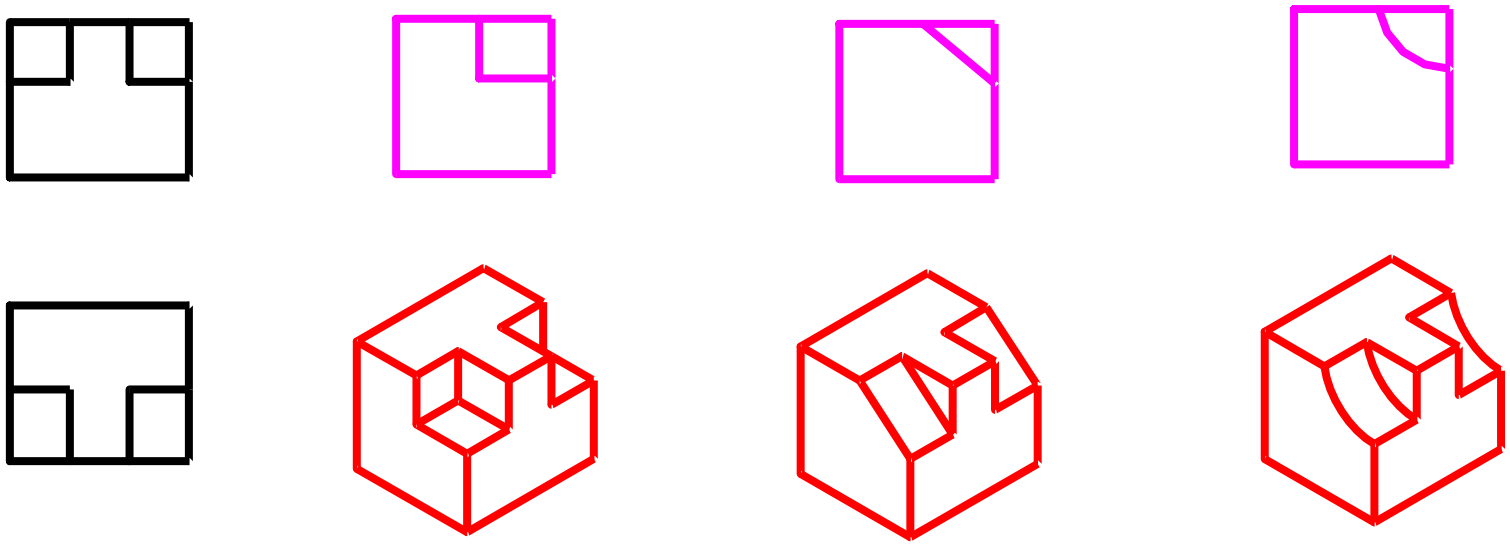
3、垂直面和一般面的投影具有类似性



§5-4 读组合体视图

一、读图的原则

1、要将几个视图联系起来看



物体的主、俯视图相同，但形状不同

§5-4 读组合体视图

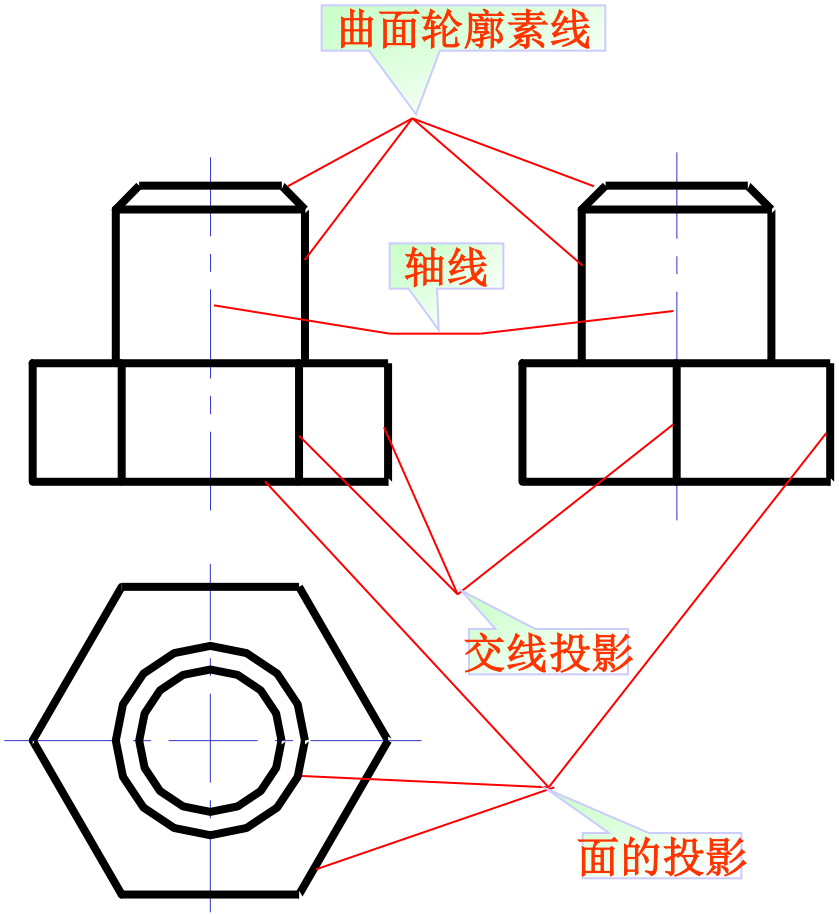
2、要弄清视图中图线的含义

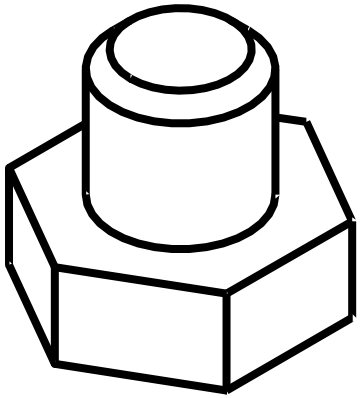
投影图中每一条线段可以是下列几何元素的投影：

(1)两表面的交线；

(2)特殊位置平面或曲面的积聚性投影；

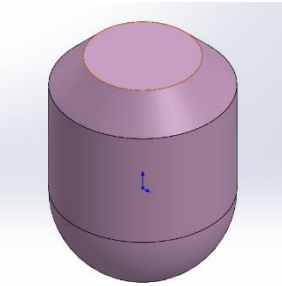
(3)回转曲面的投影轮廓。



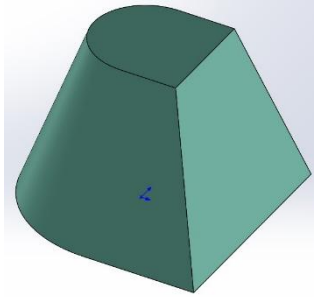


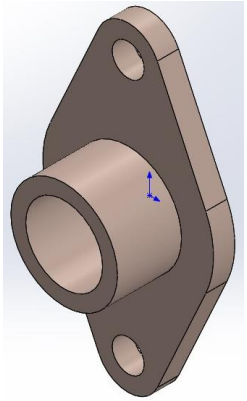
§5-4 读组合体视图

3、要弄清视图中“线框”的含义

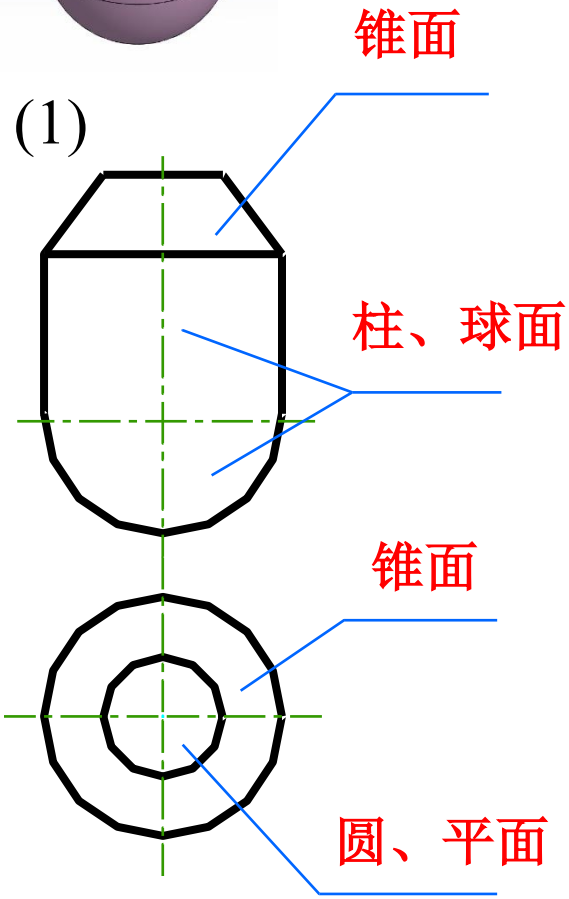


每一封闭线框可以是：
(1)平面的投影；
(2)曲面立体的投影；
(3)孔或槽的投影。

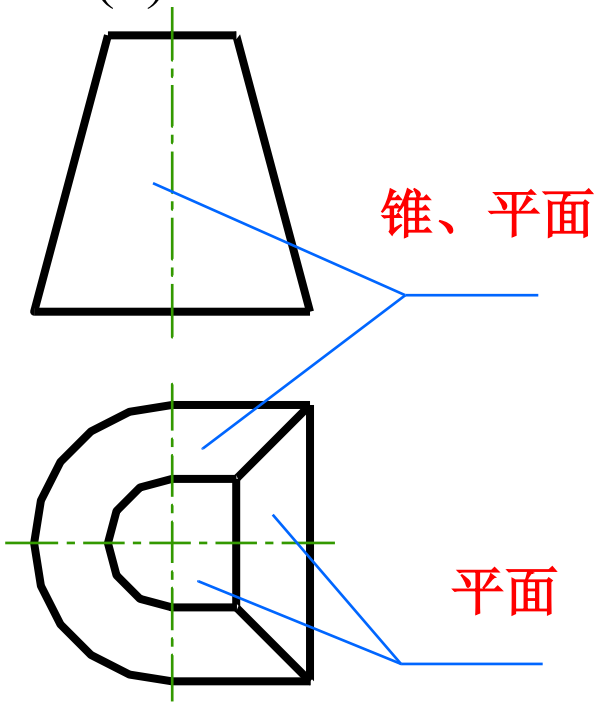




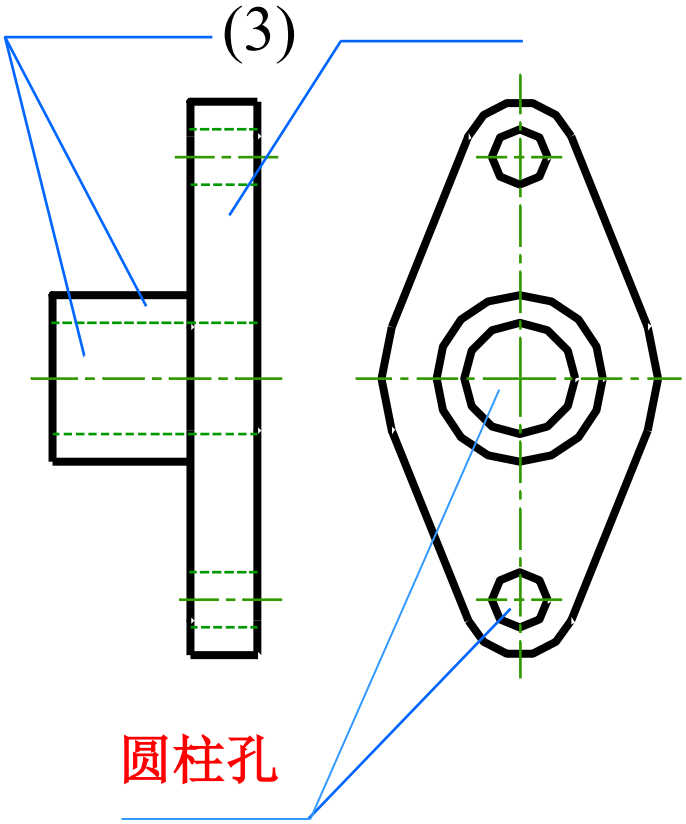
(1)



(2)

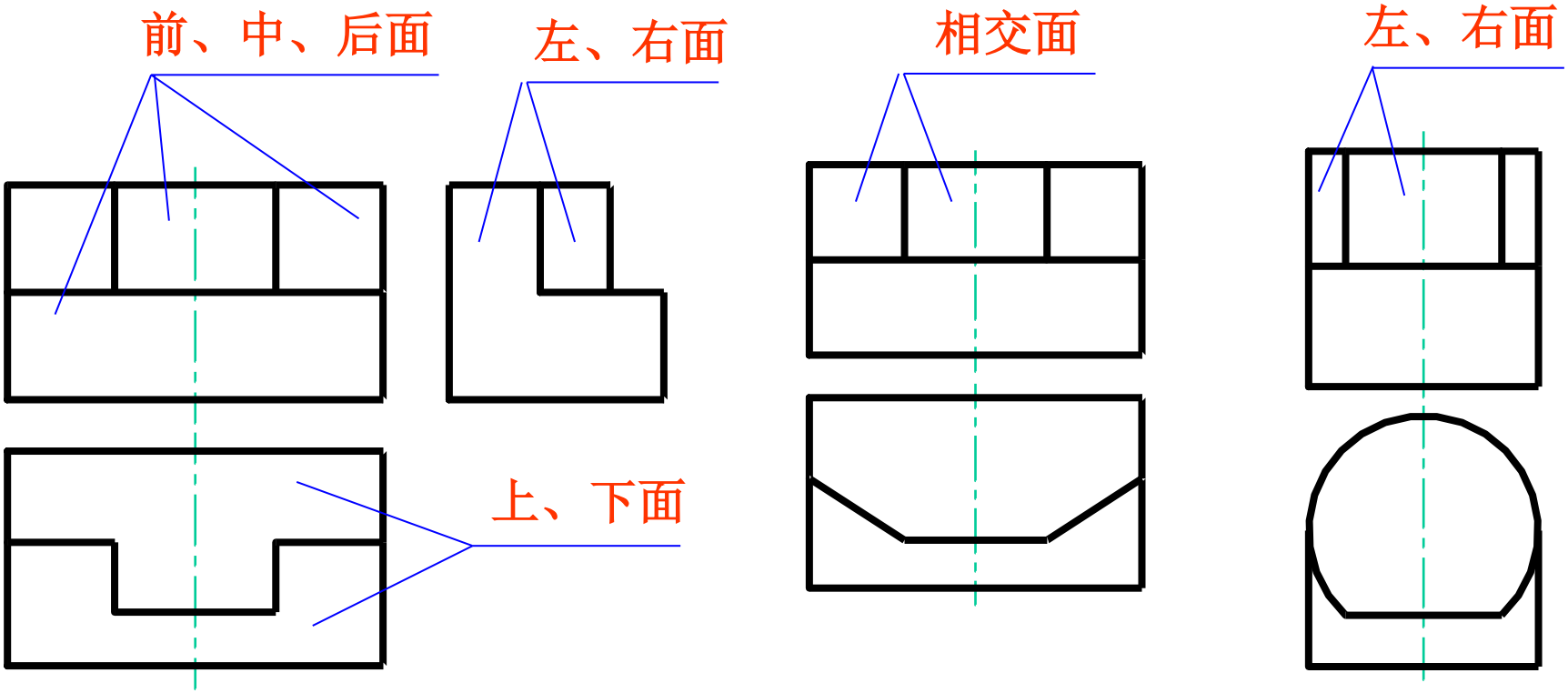


(3)



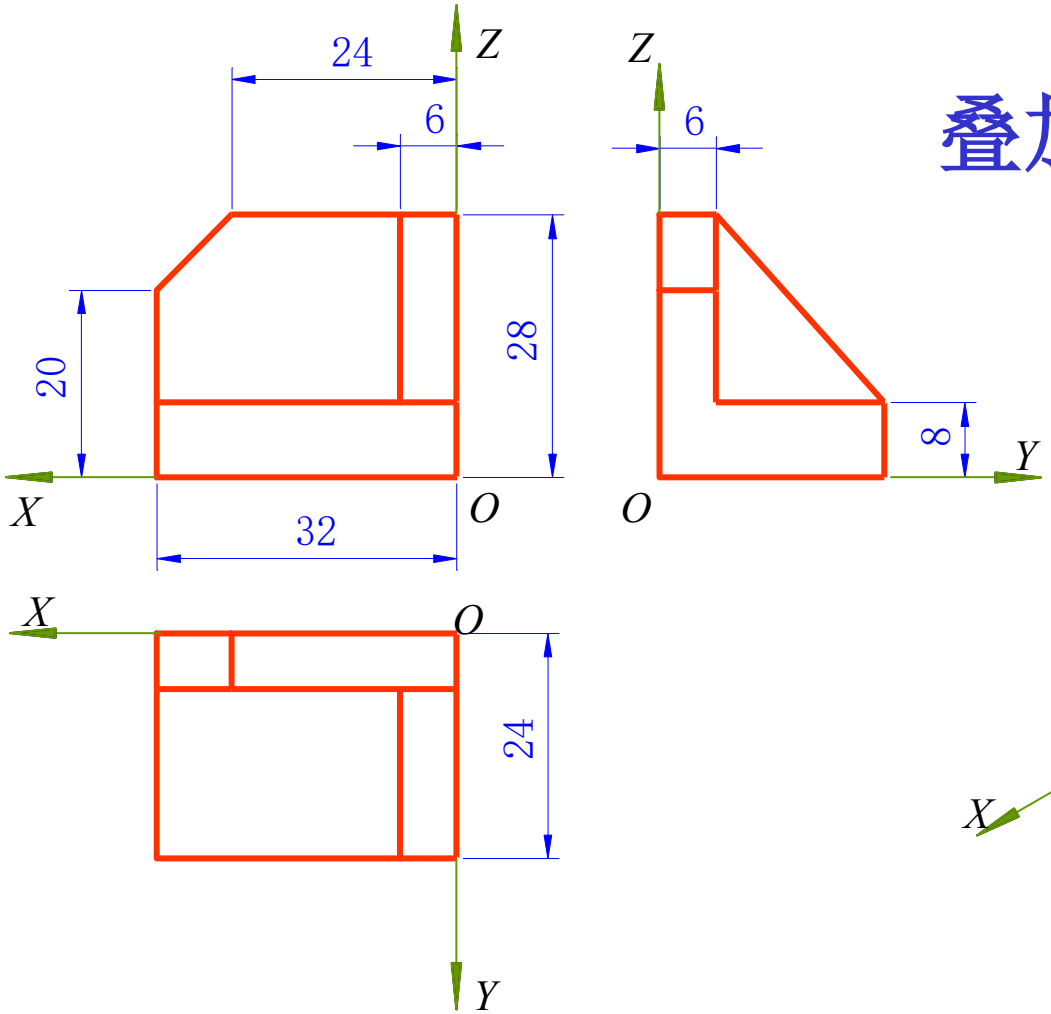
§5-4 读组合体视图

4、要判断出相邻表面间的相对位置

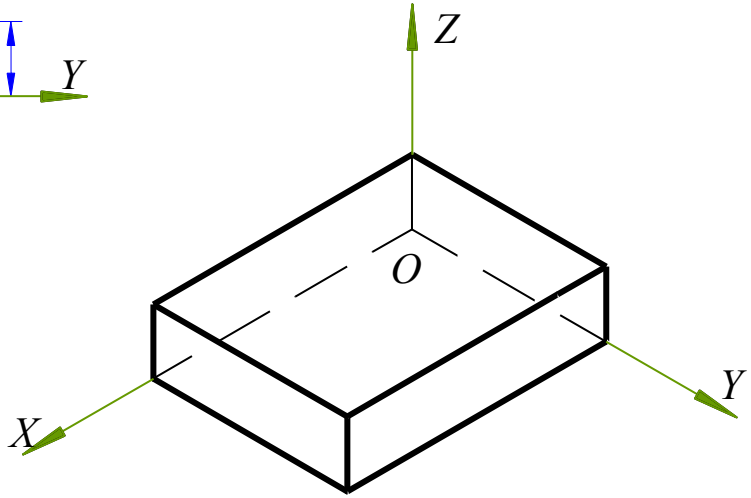


§5-4 读组合体视图

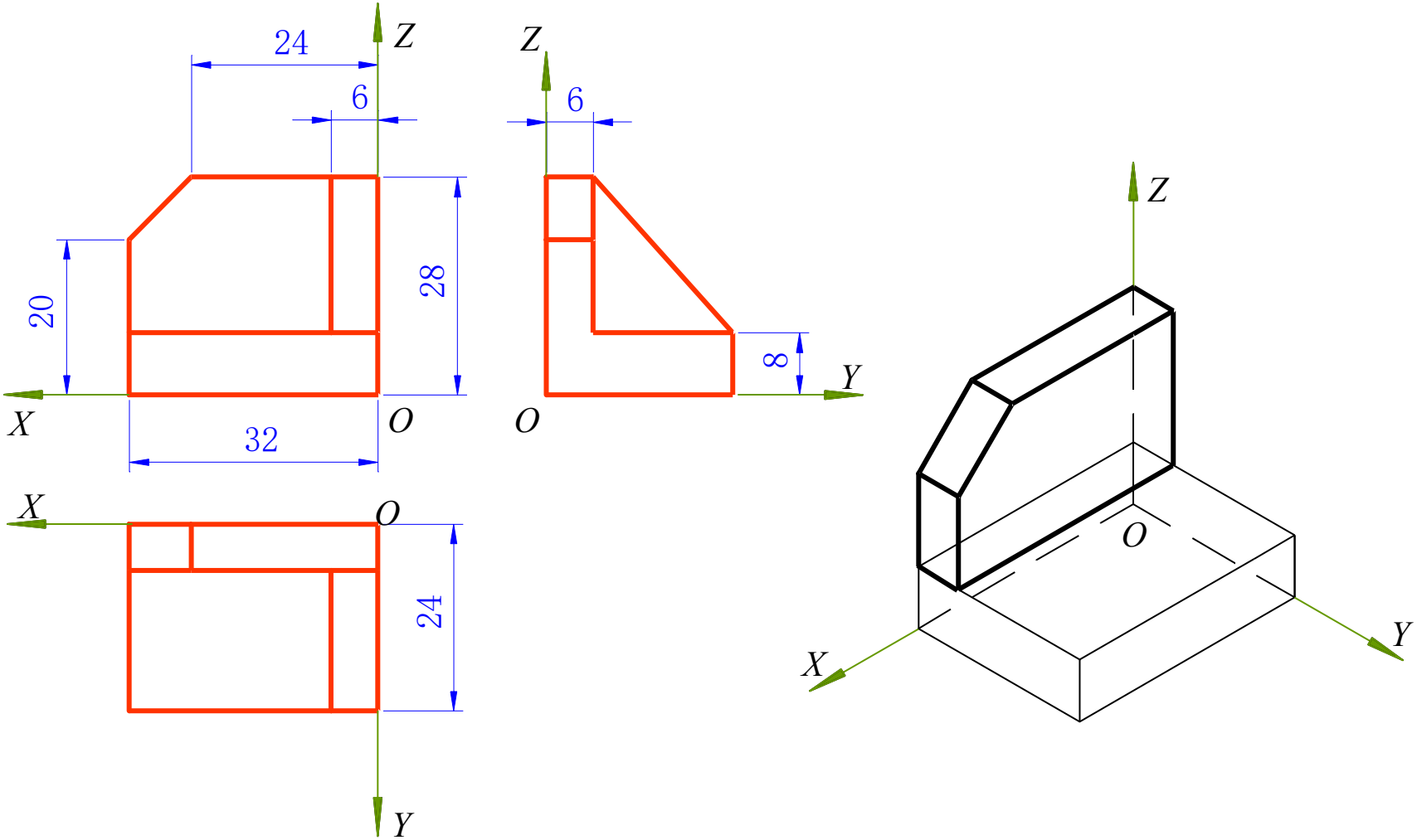
二、读图的方法（一）



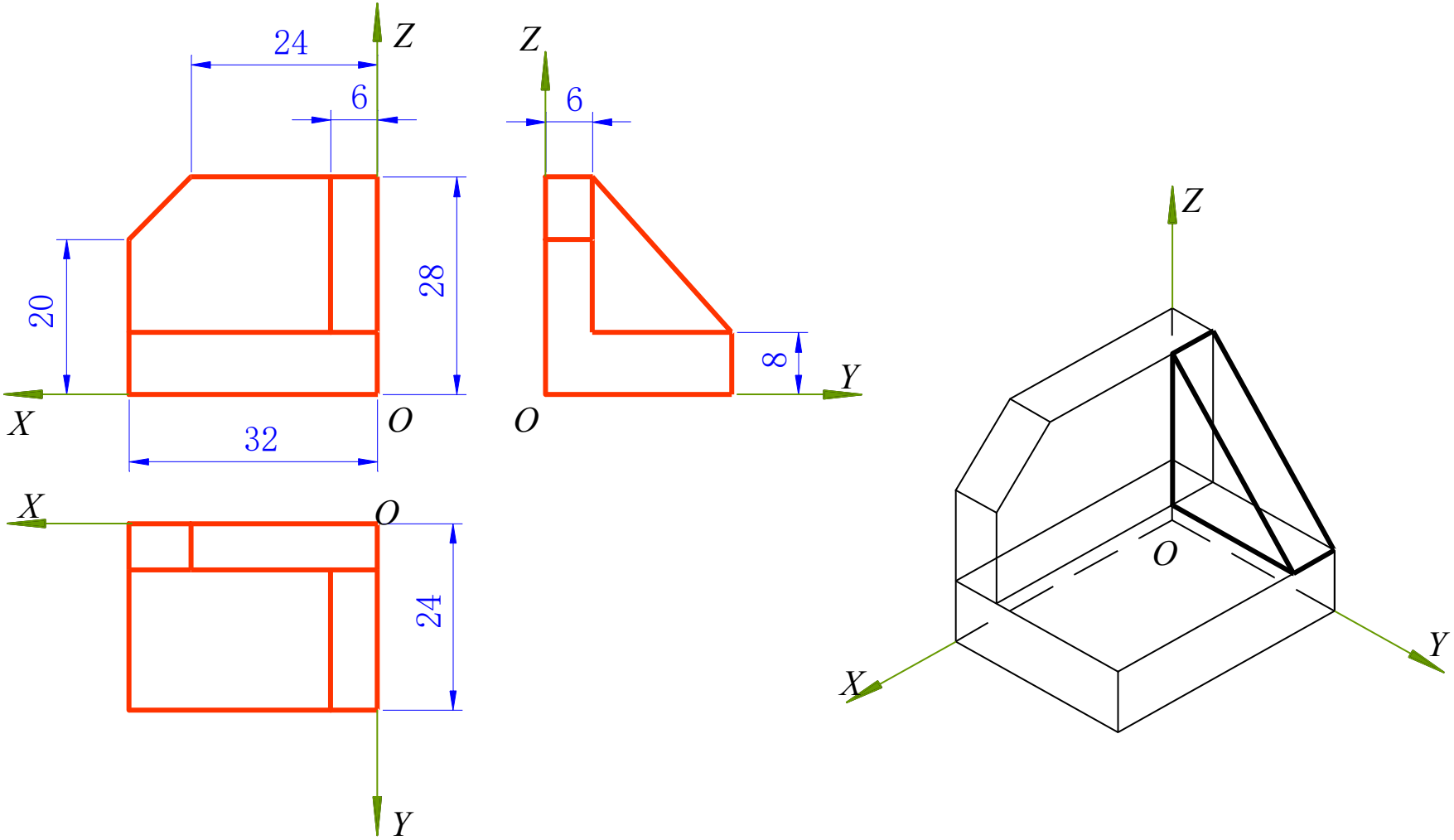
叠加法



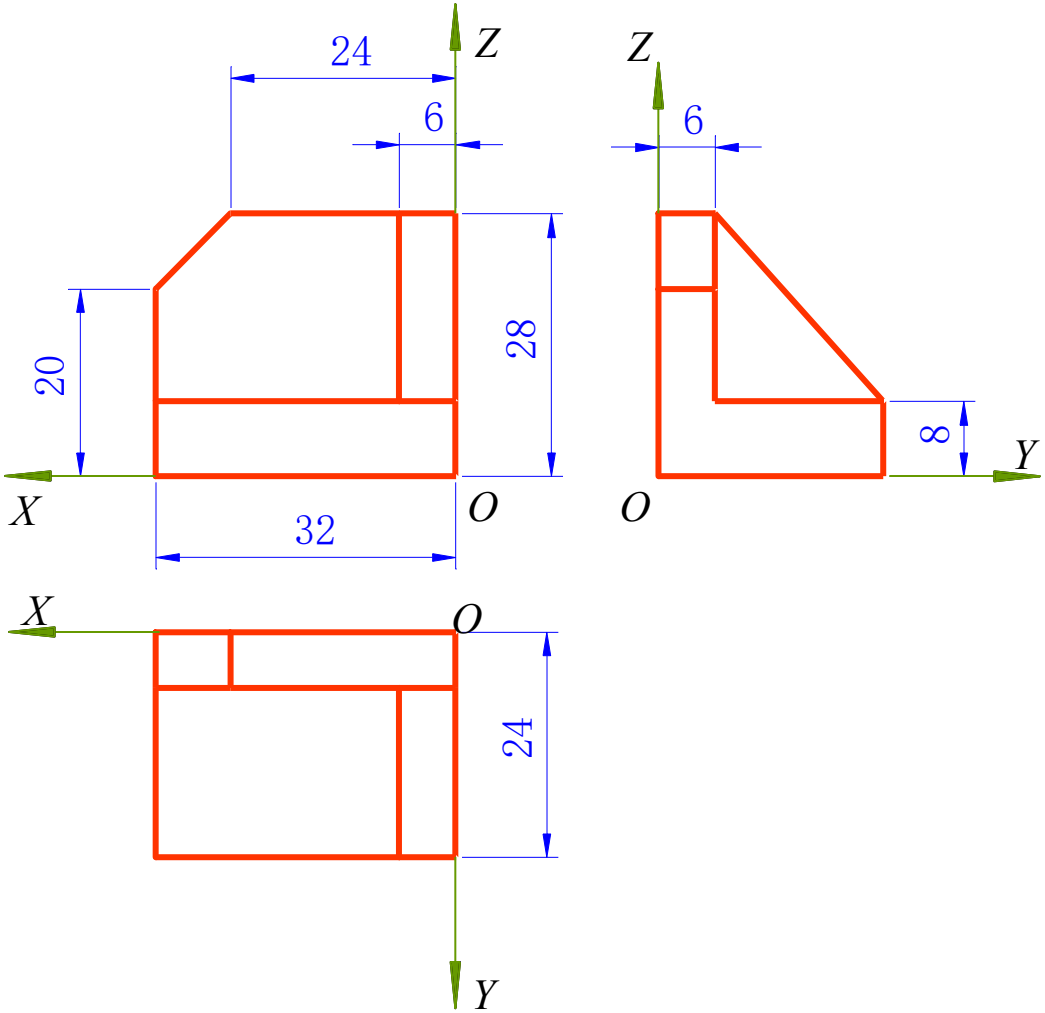
步骤一

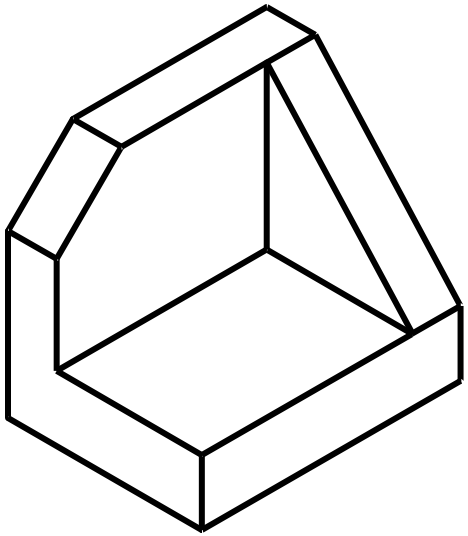


步骤二



步驟三





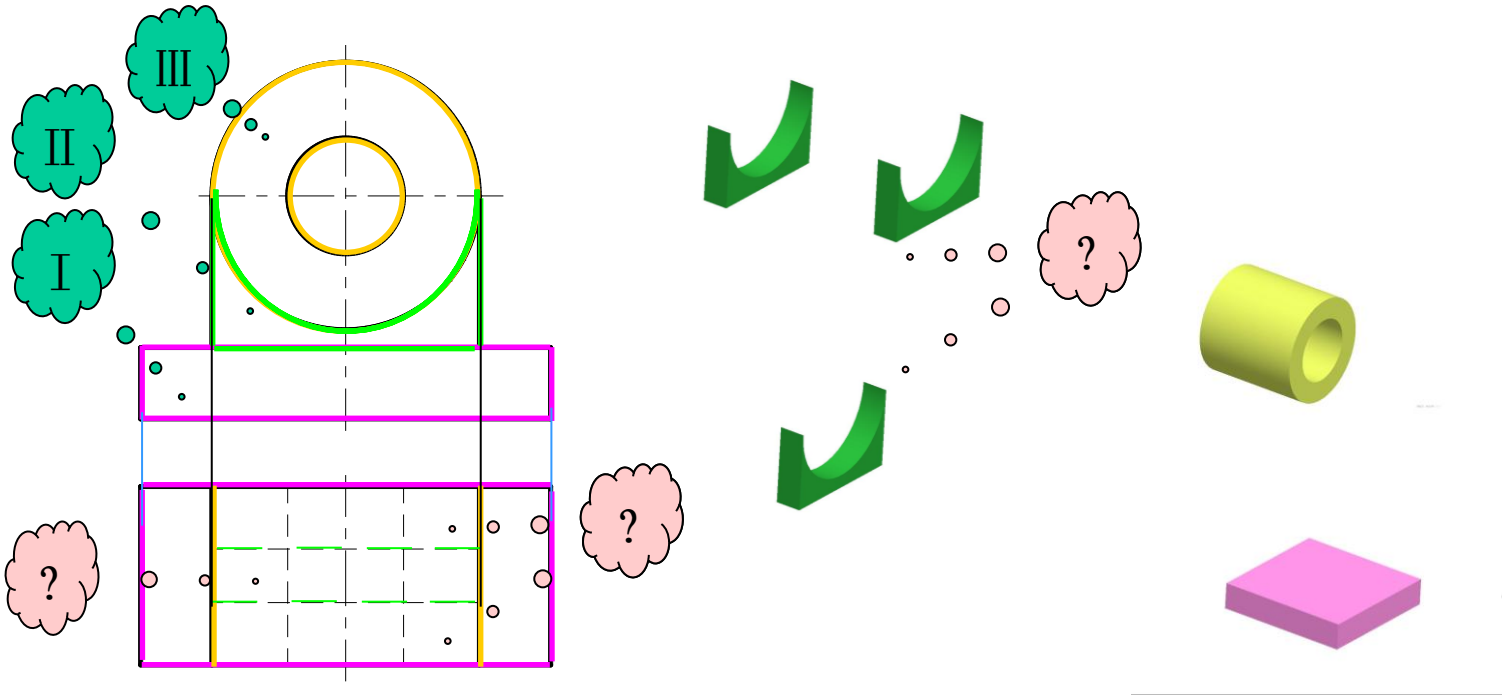
完成

§5-4 读组合体视图

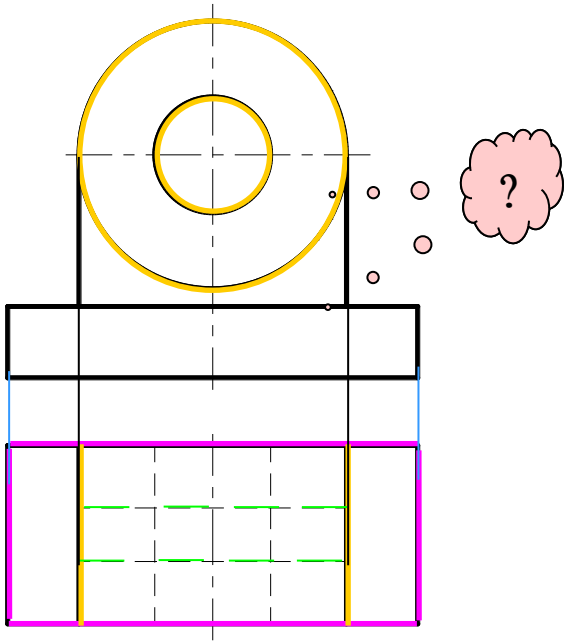
读图举例：例1

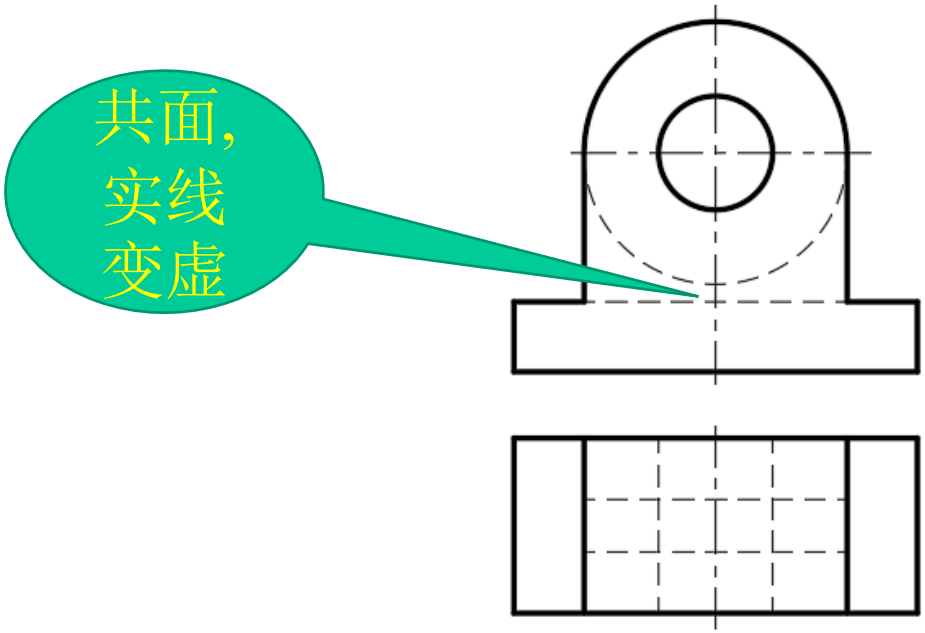
1.划线框，分部分；
2.对投影，想基本形体，定相对位置；
3.深入分析，弄懂细节；
4.综合起来，想出整体。

将组合体分解为： 1、底板 2、支撑板 3、圆柱



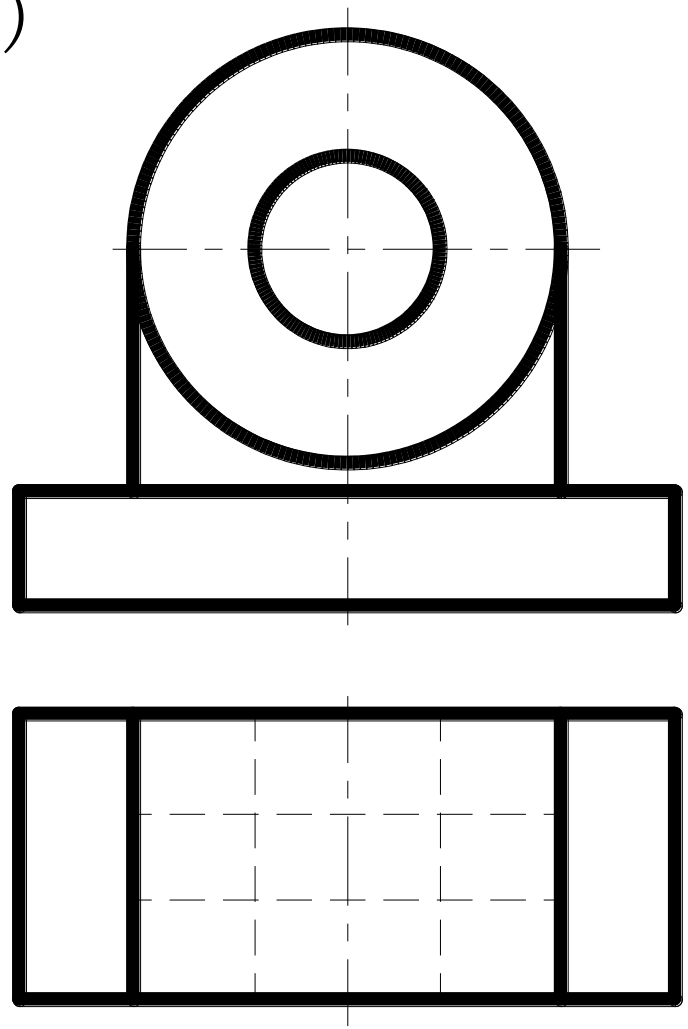
§5-4 读组合体视图

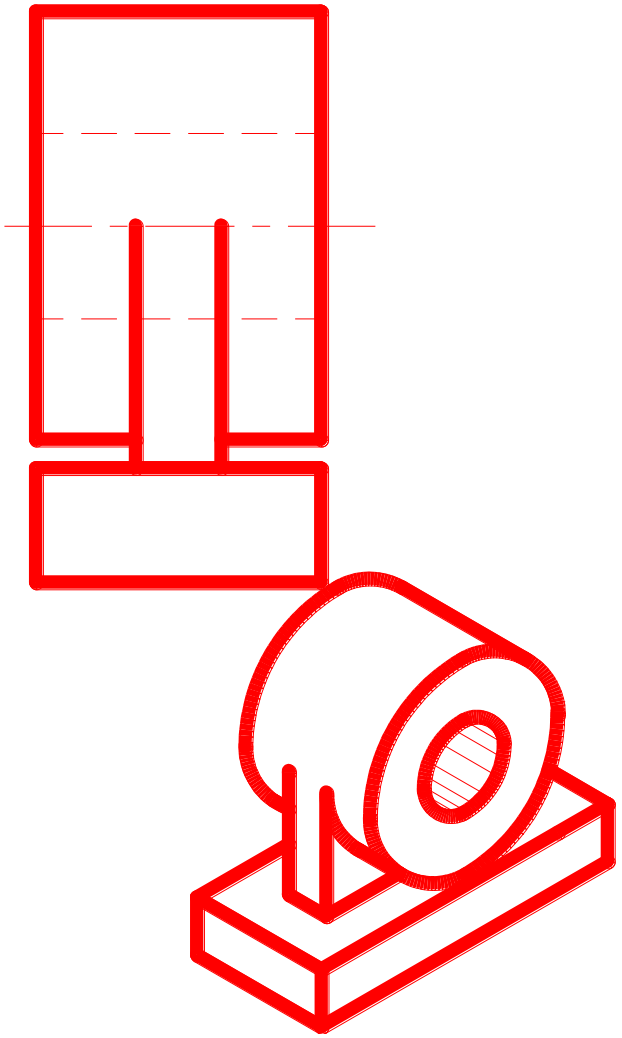




§5-4 读组合体视图

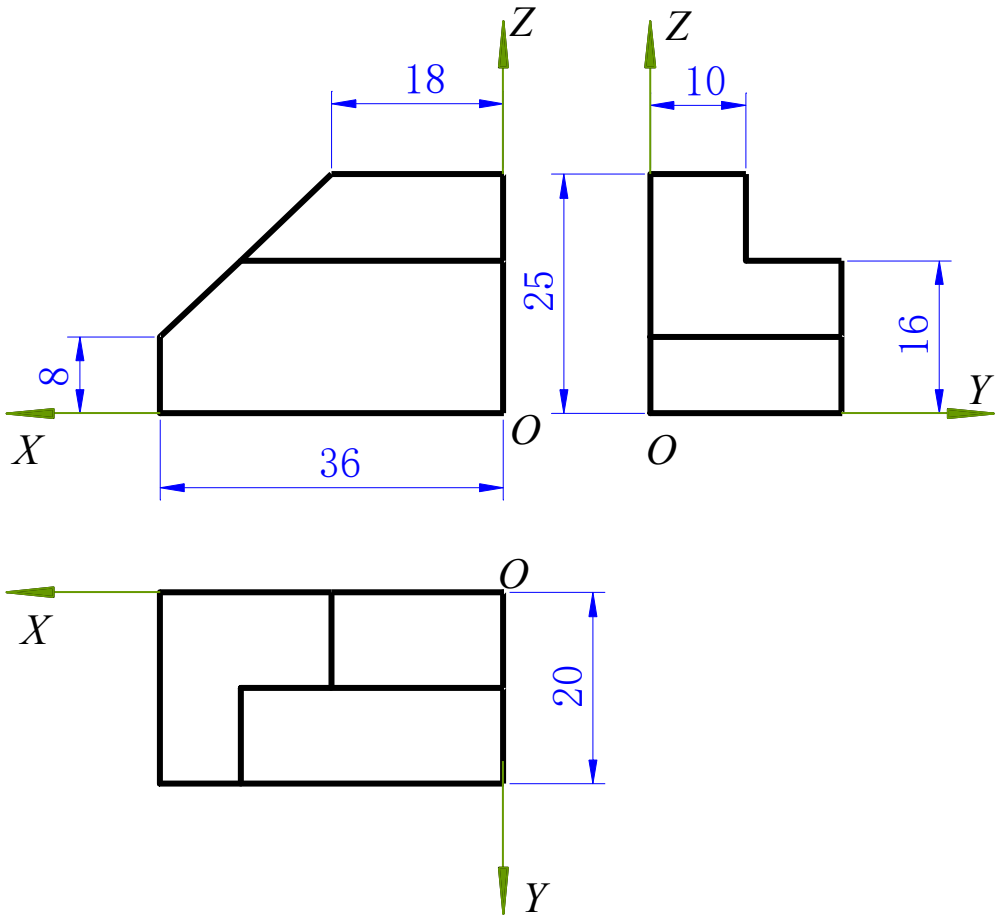
(2)

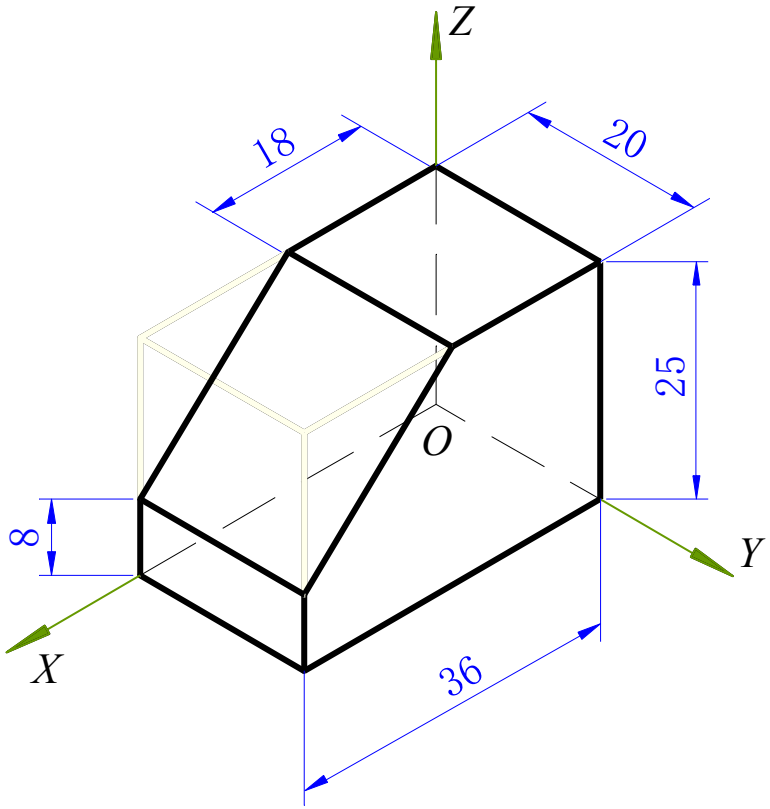




§5-4 读组合体视图

二、读图的方法（二）

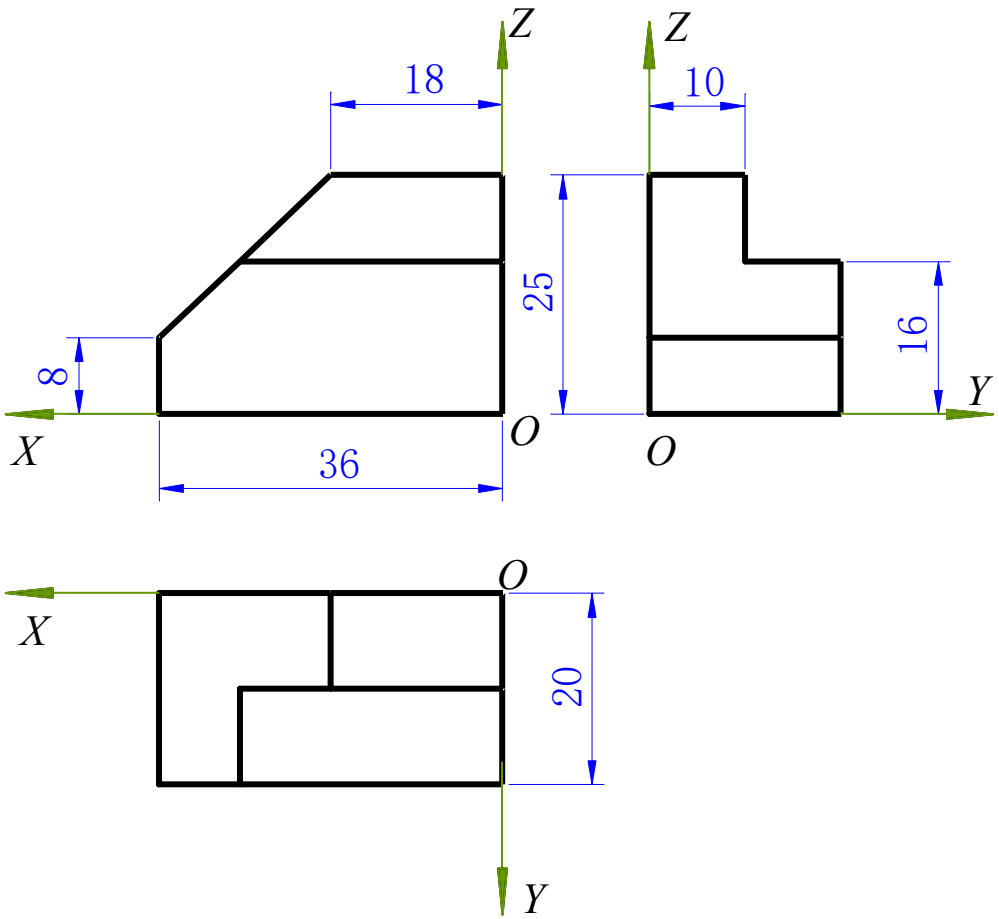


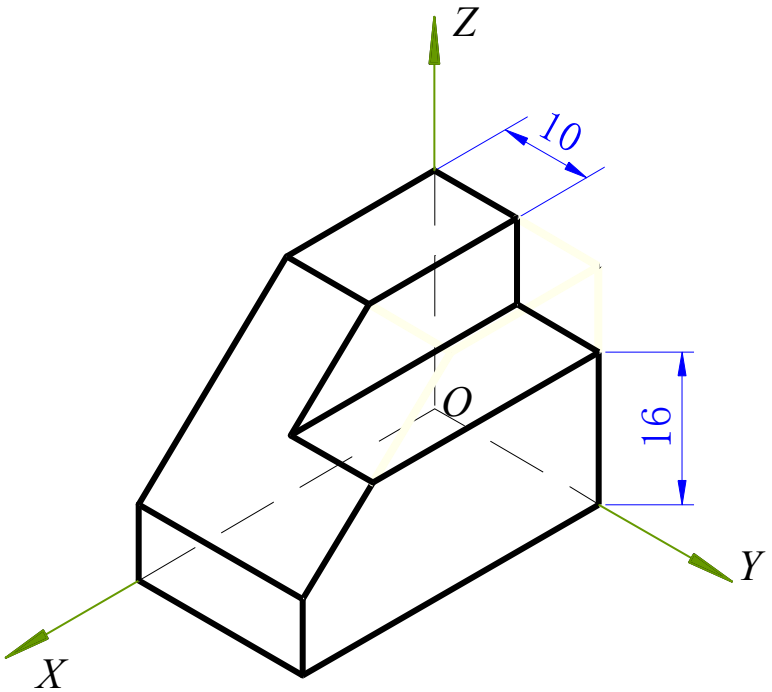


步骤一

切割法

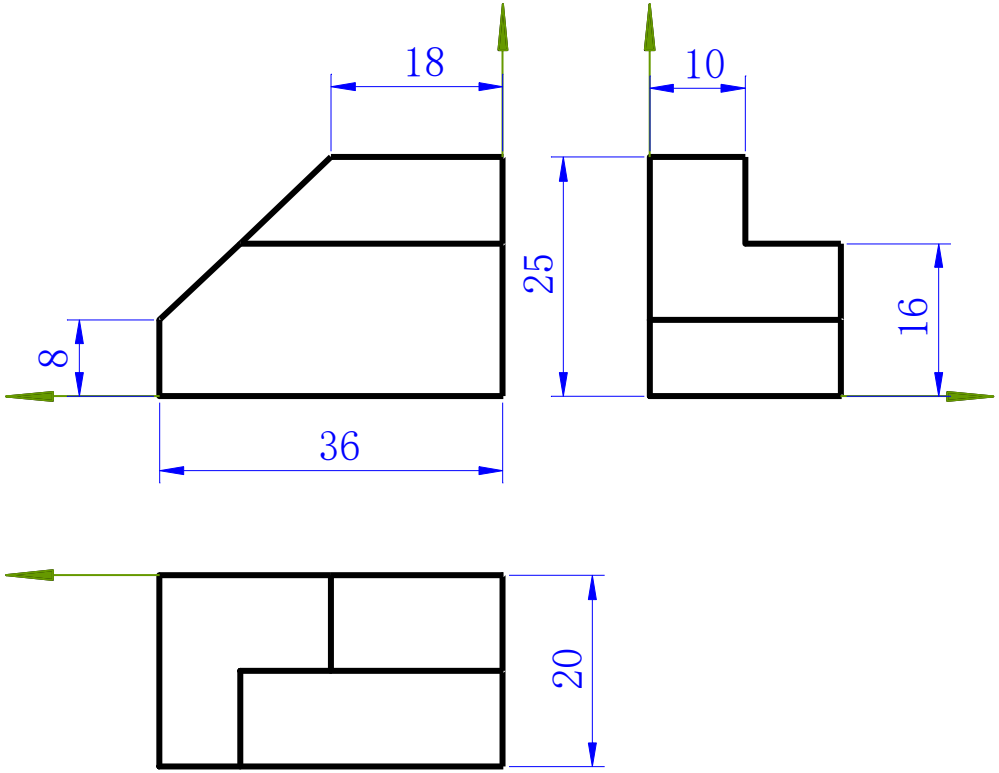
§5-4 读组合体视图

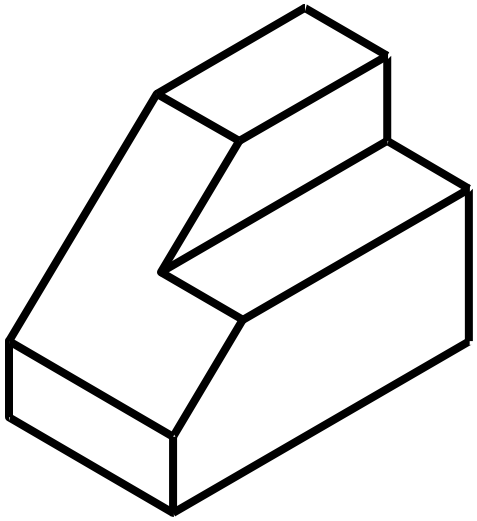




步骤二

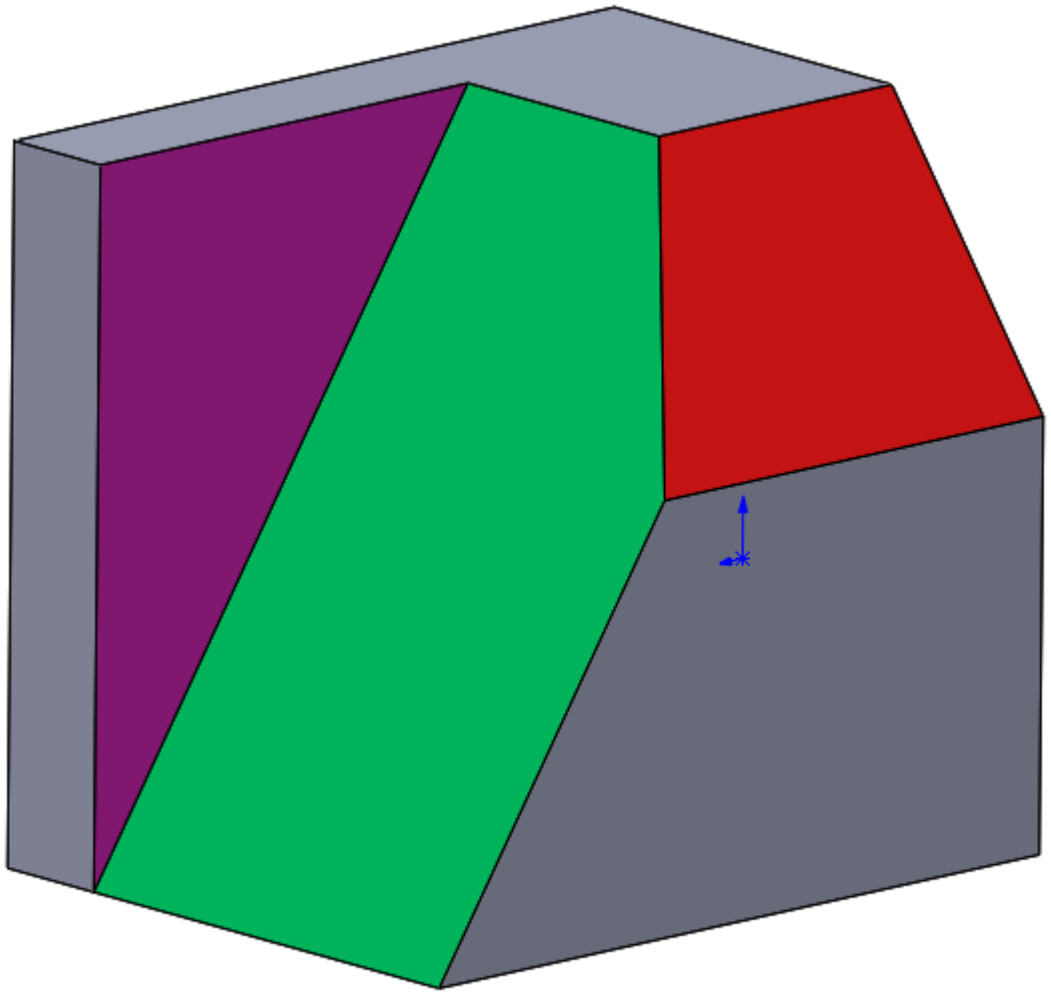
§5-4 读组合体视图





完成

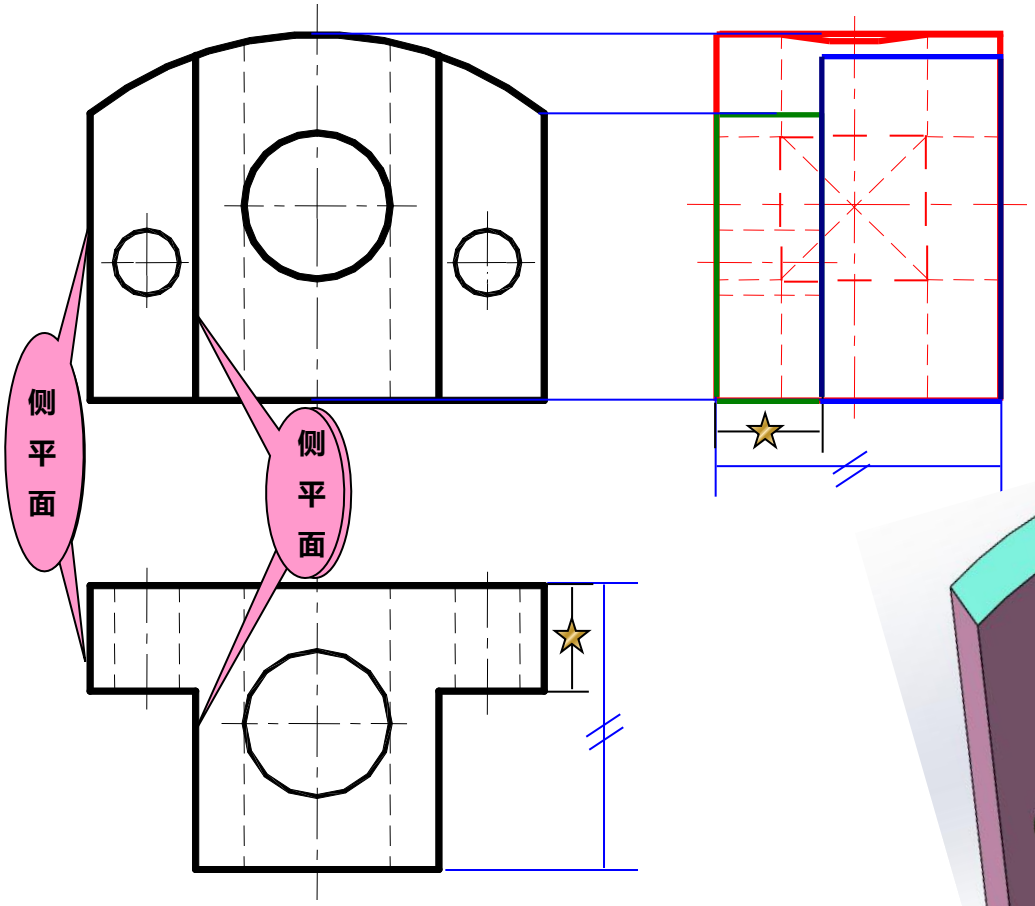
§5-4 读组合体视图

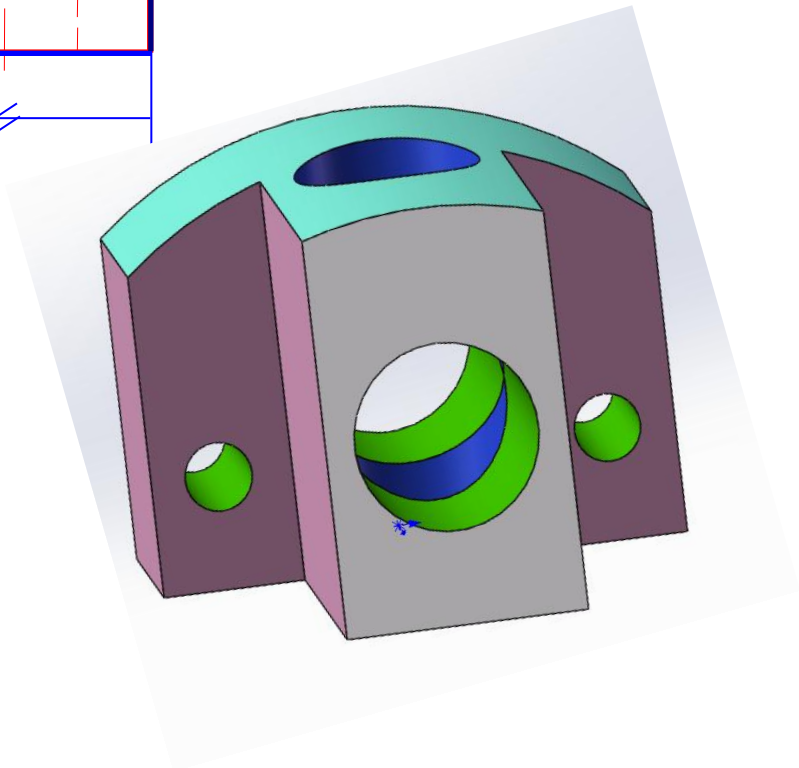


§5-4 读组合体视图

读图举例:

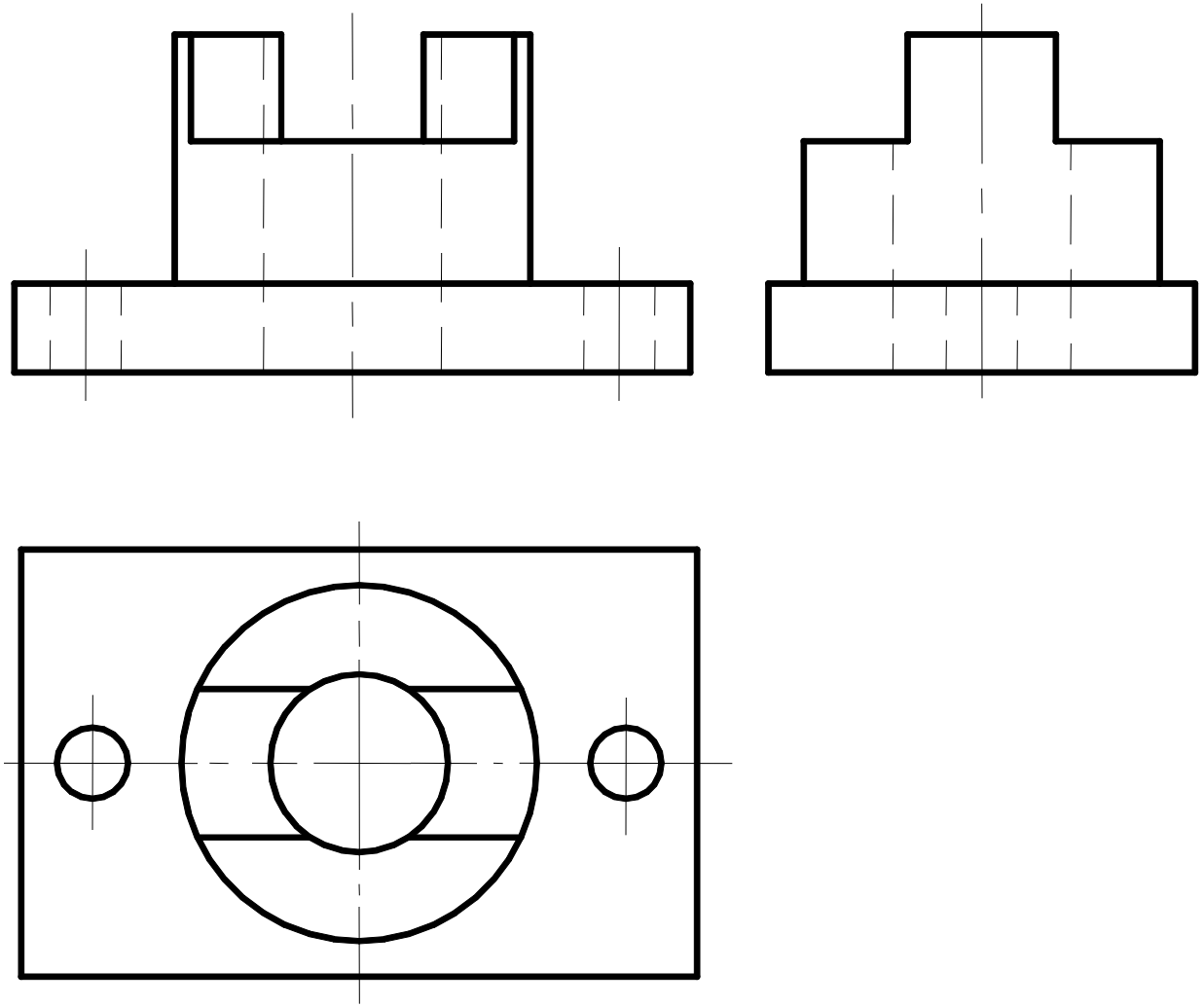
例2: 根据组合体的两个视图补画第三视图。





§5-4 读组合体视图

例1

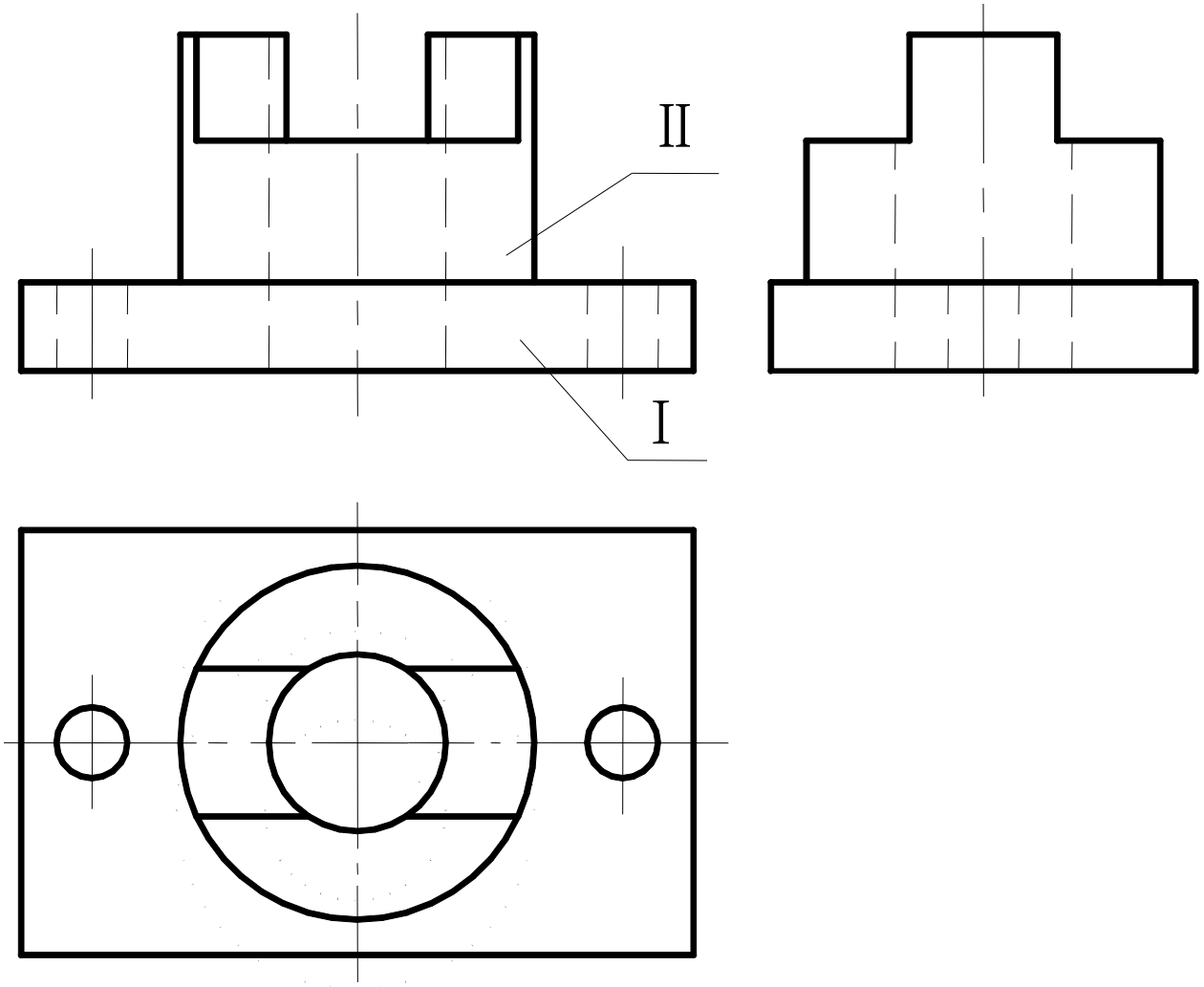


§5-4 读组合体视图

1、根据投影规律进行形体分析

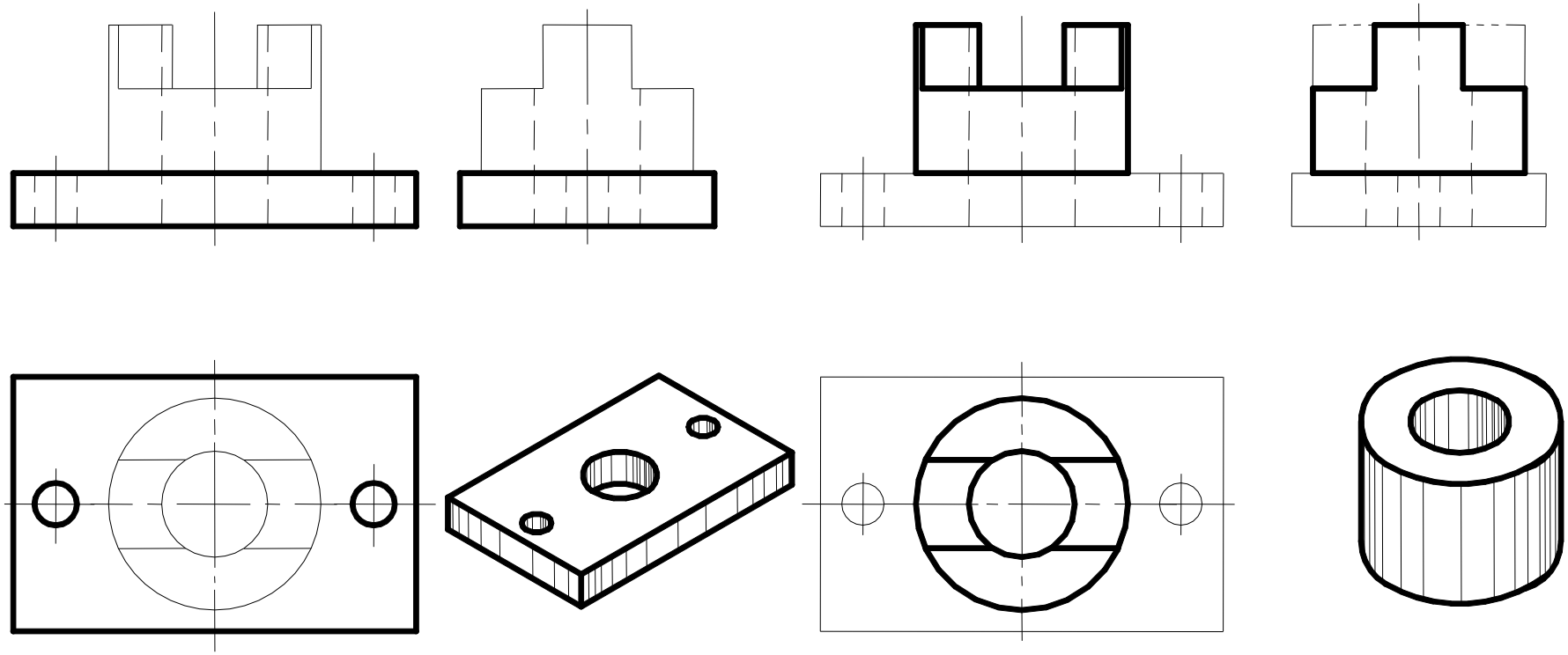
I、长方体

II、圆柱体



§5-4 读组合体视图

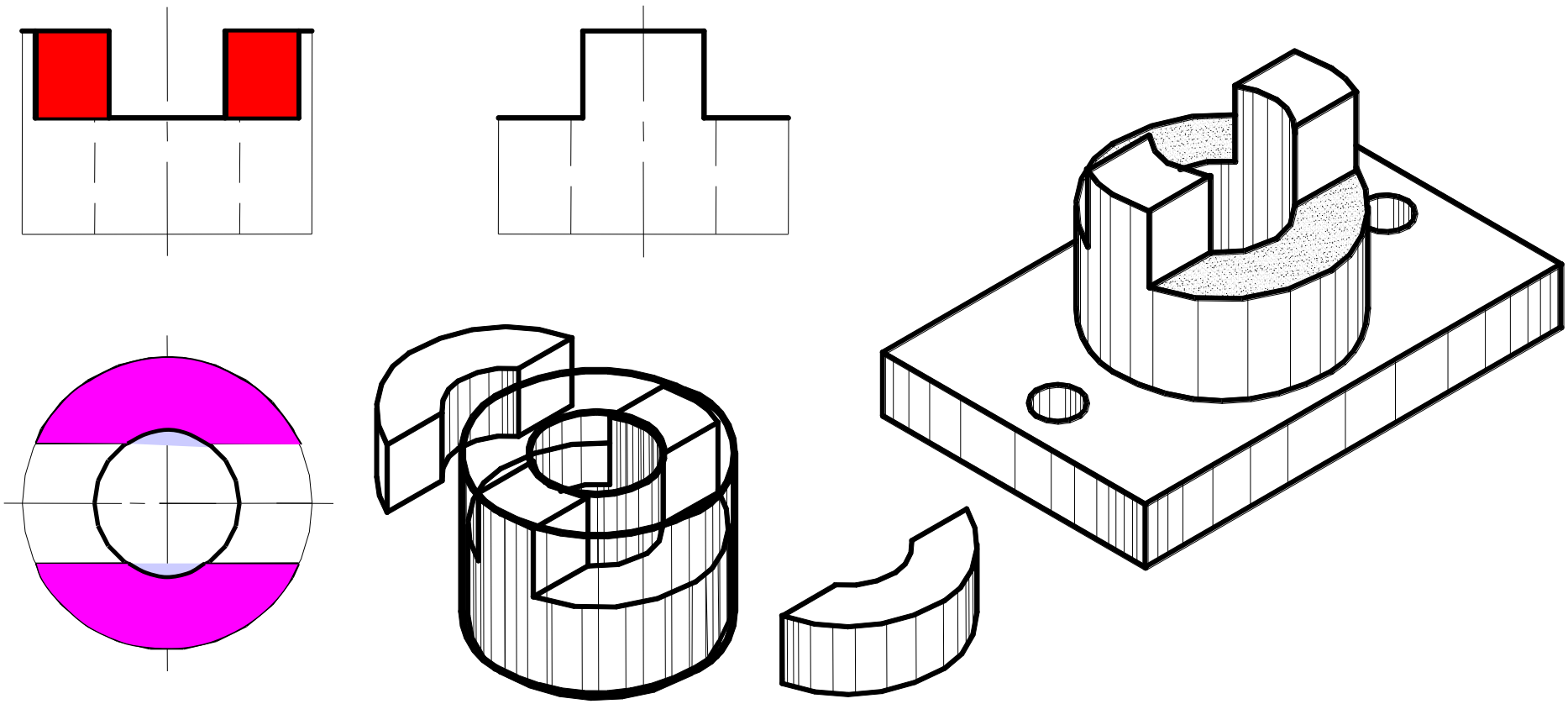
2、分基本形体，定相对位置



§5-4 读组合体视图

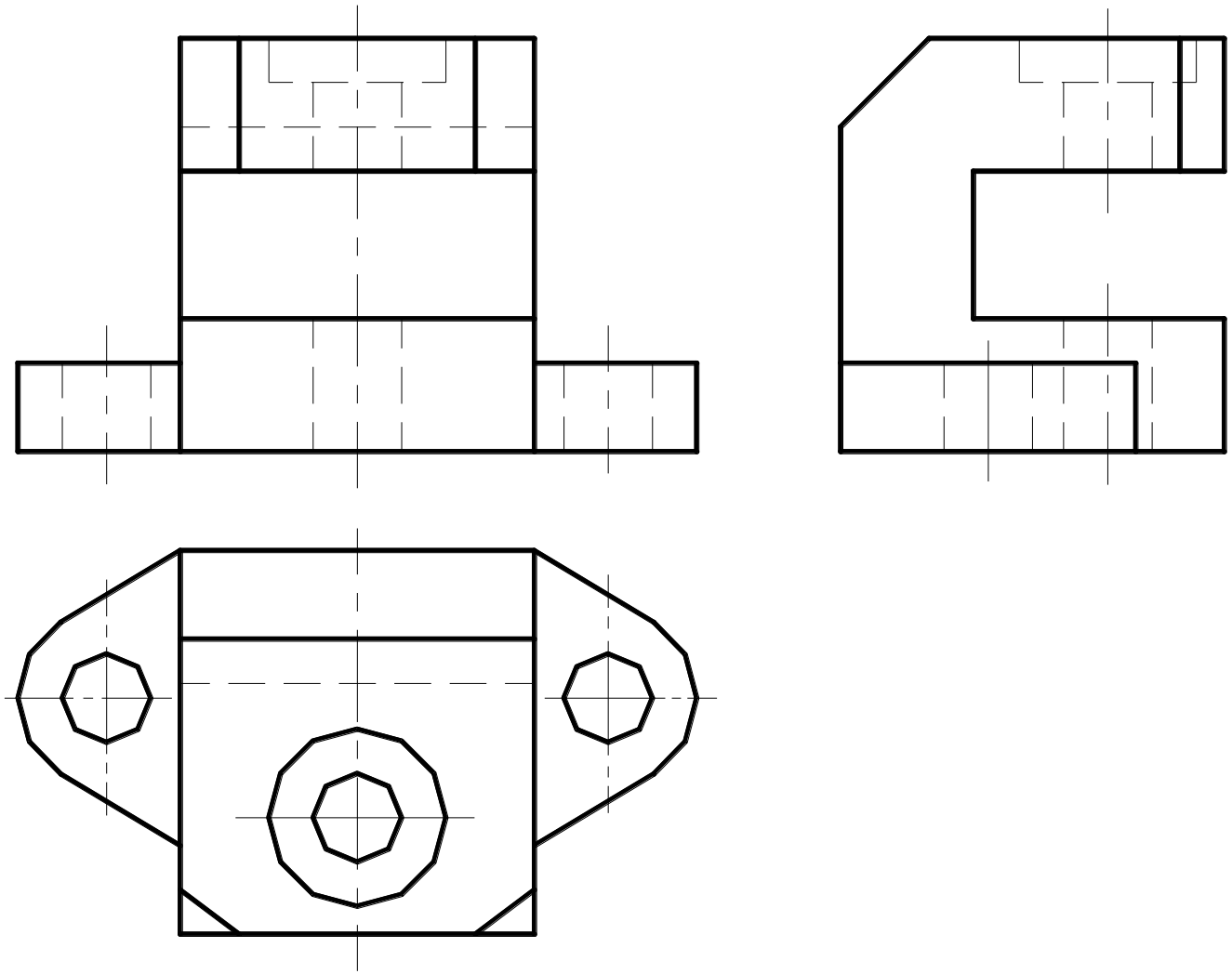
3、深入分析，弄懂细节

4、综合起来，想出整体`



§5-4 读组合体视图

例2、读图组合体三视图

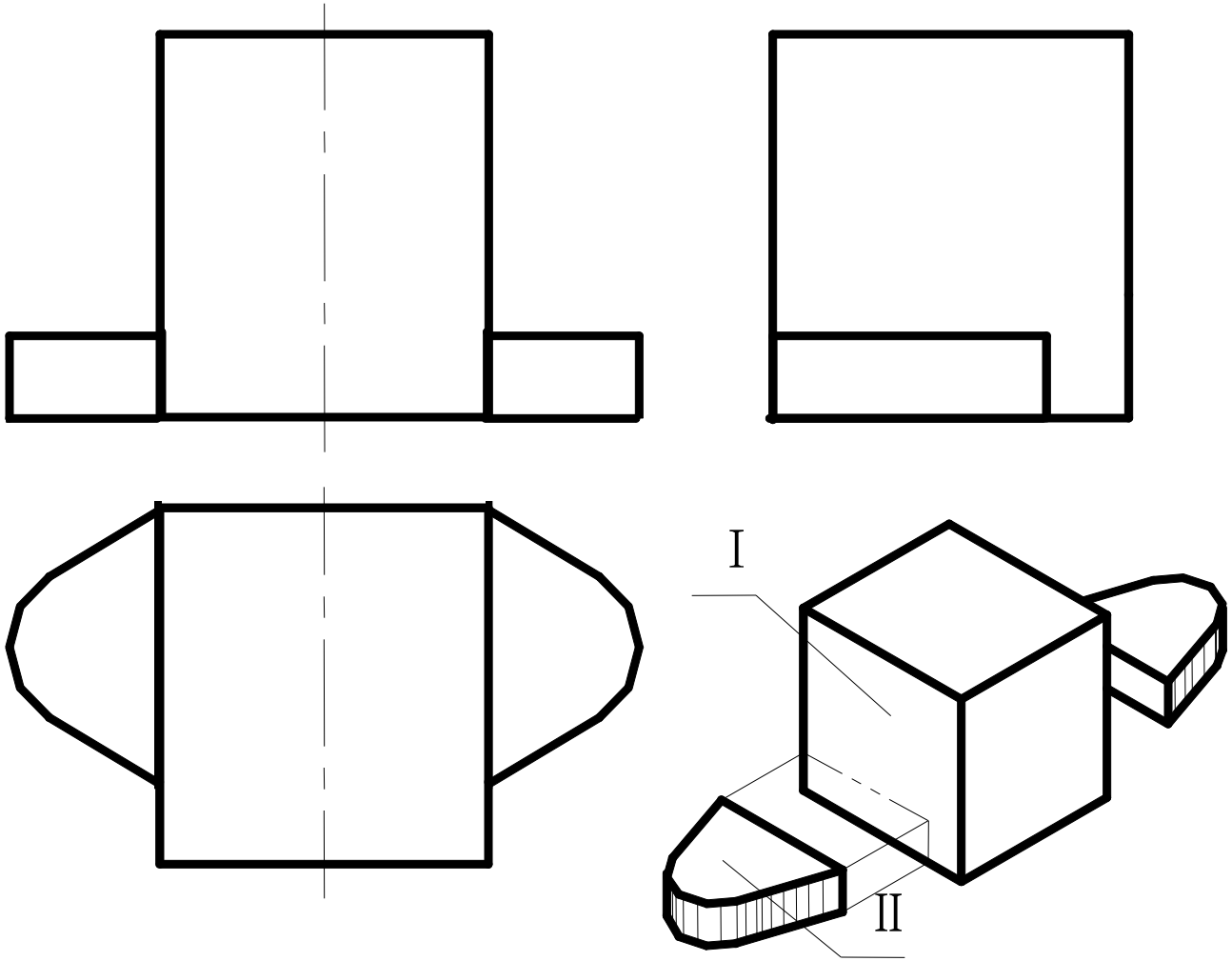


§5-4 读组合体视图

1)、将组合体分解成两部分

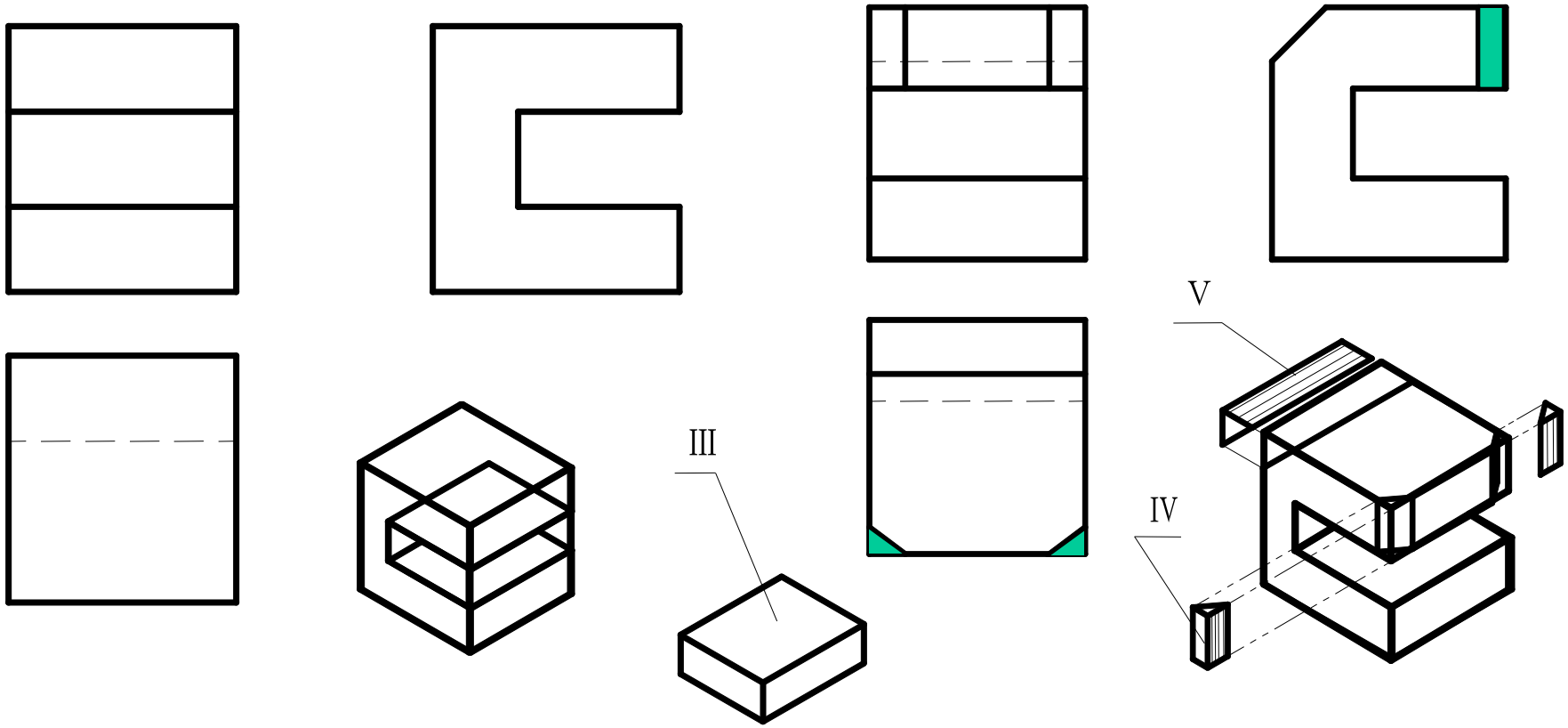
I、长方体

II、带圆弧的柱体



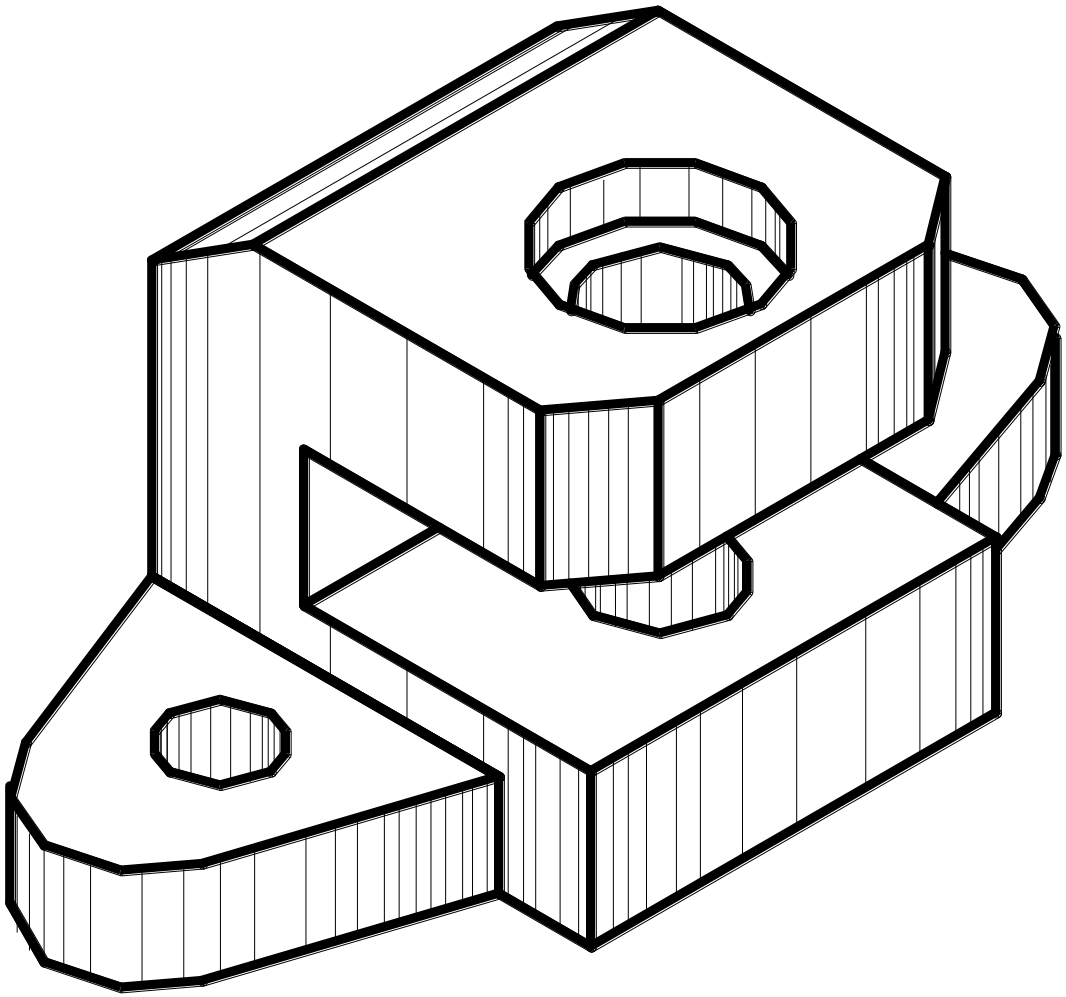
§5-4 读组合体视图

2)、长方体截切分析



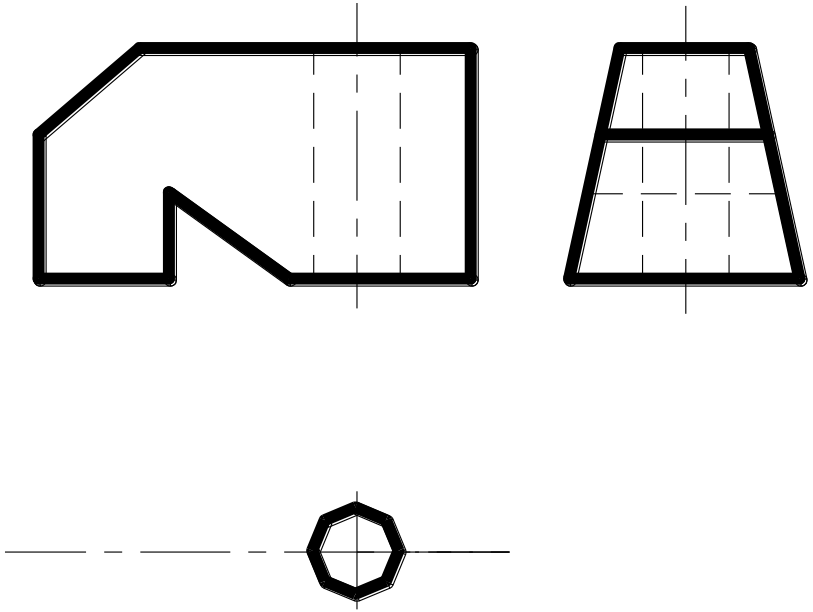
§5-4 读组合体视图

3)、想象出组合体的形状



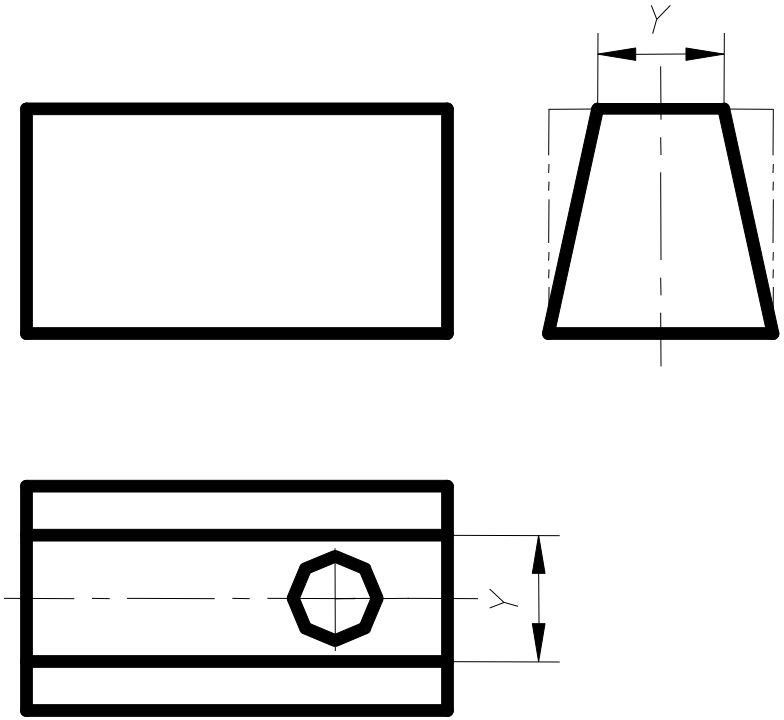
§5-4 读组合体视图

例3、读(a)图所示的组合体主、左视图，并补画俯视图



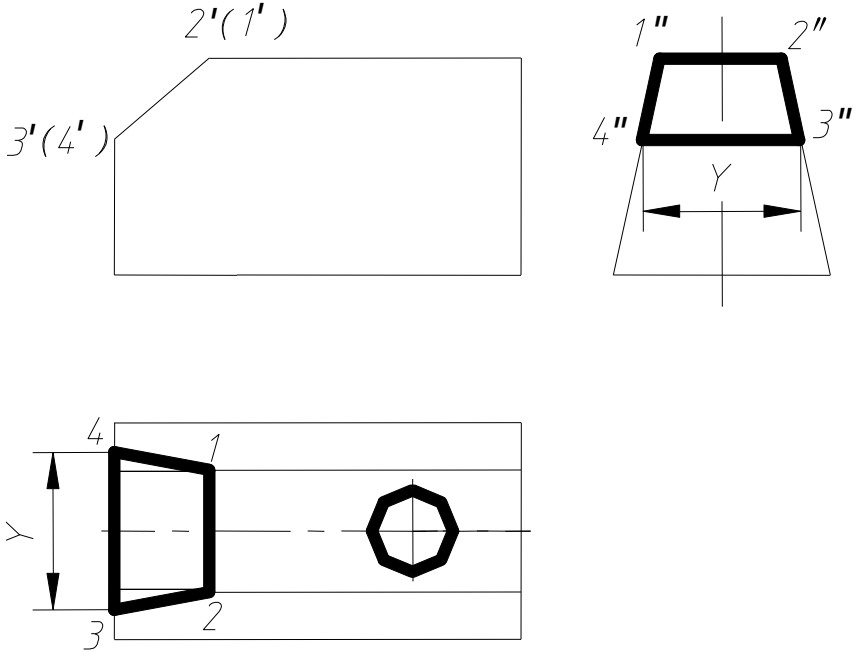
(a)

(1)补画长方体被侧垂面截切后的投影

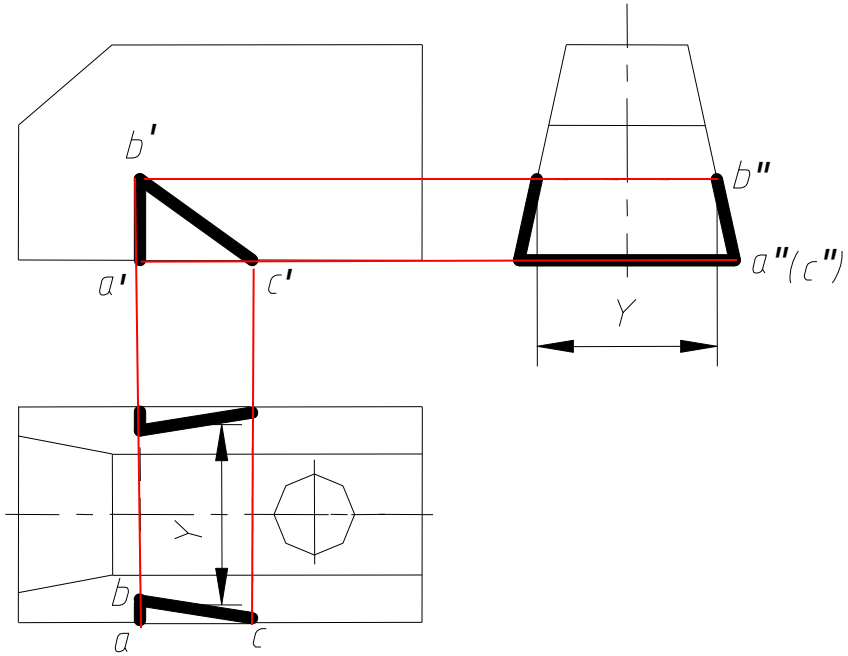


§5-4 读组合体视图

(2)补画长方体被正垂面截切后的投影

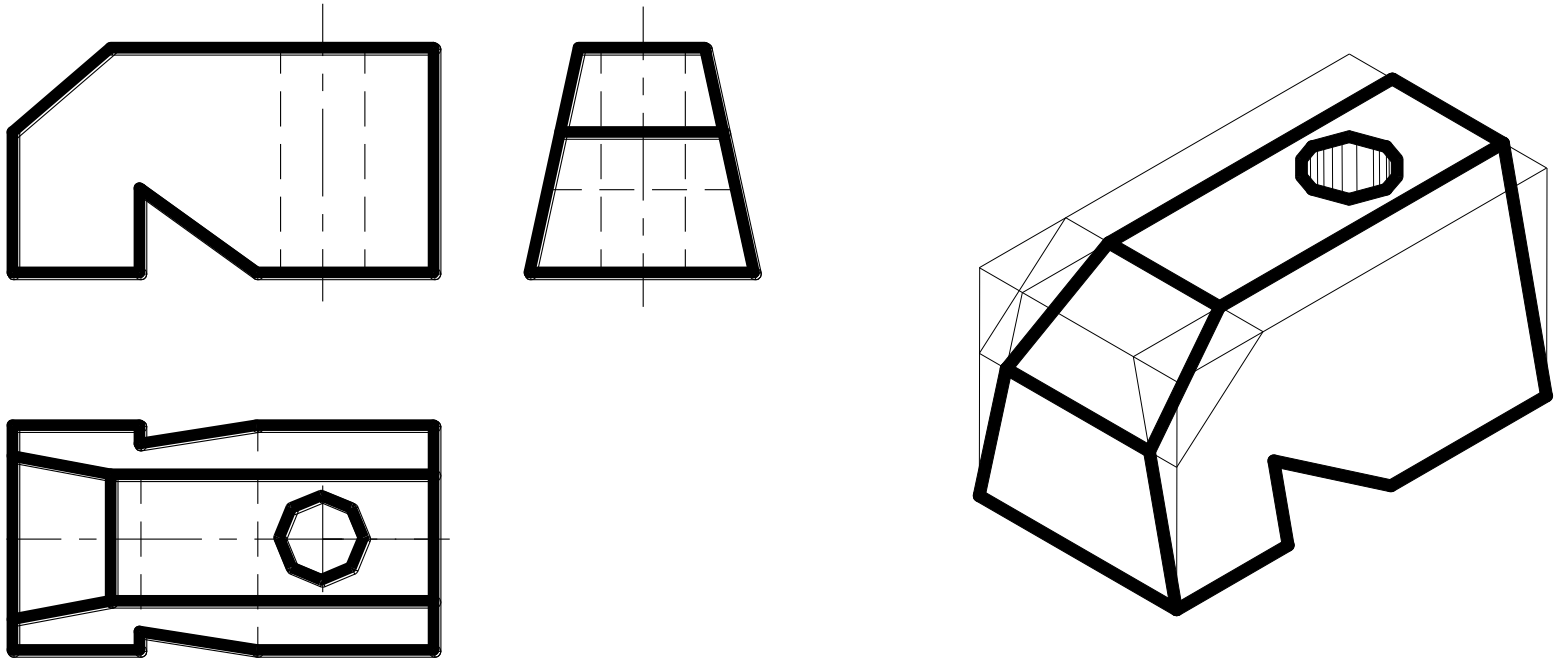


(3)补画侧平面、正垂面截切 后的投影



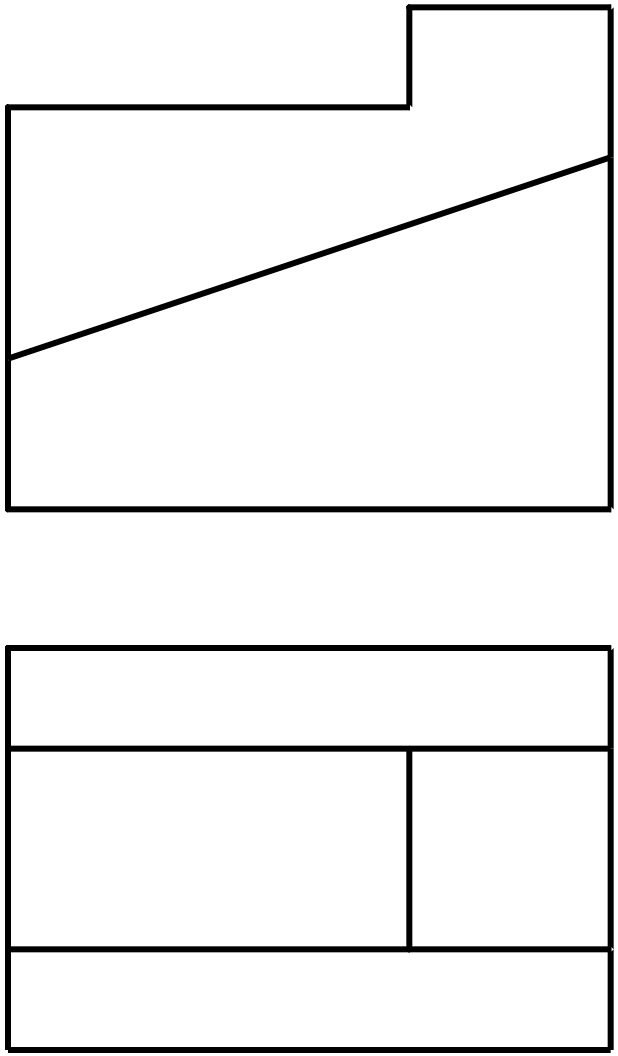
§5-4 读组合体视图

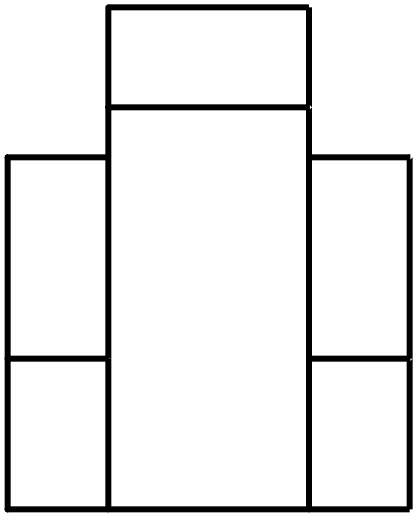
(4)检查加深，完成组合体投影图

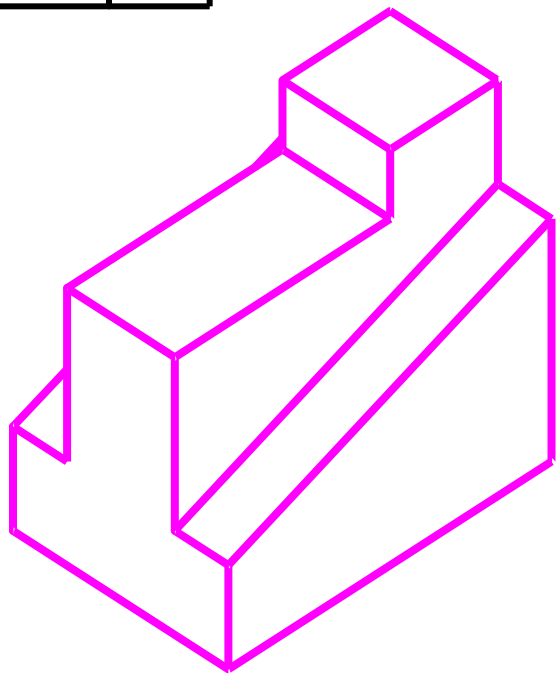


§5-4 读组合体视图

例4 由立体的轴测图画三视图

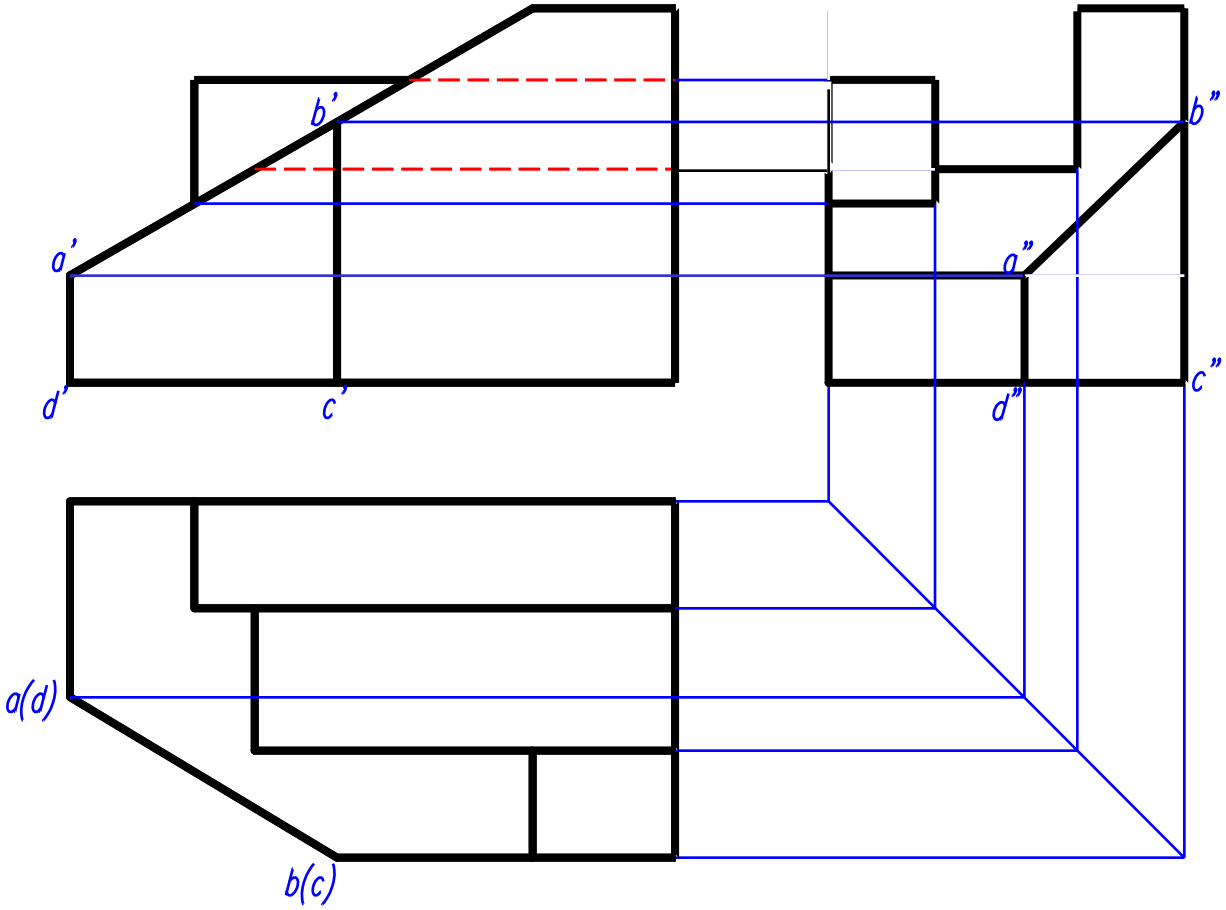




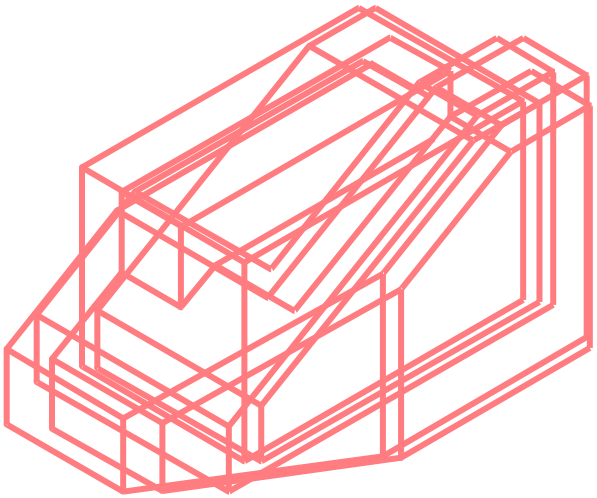


§5-4 读组合体视图

例5、由主、俯视图，画侧视图。

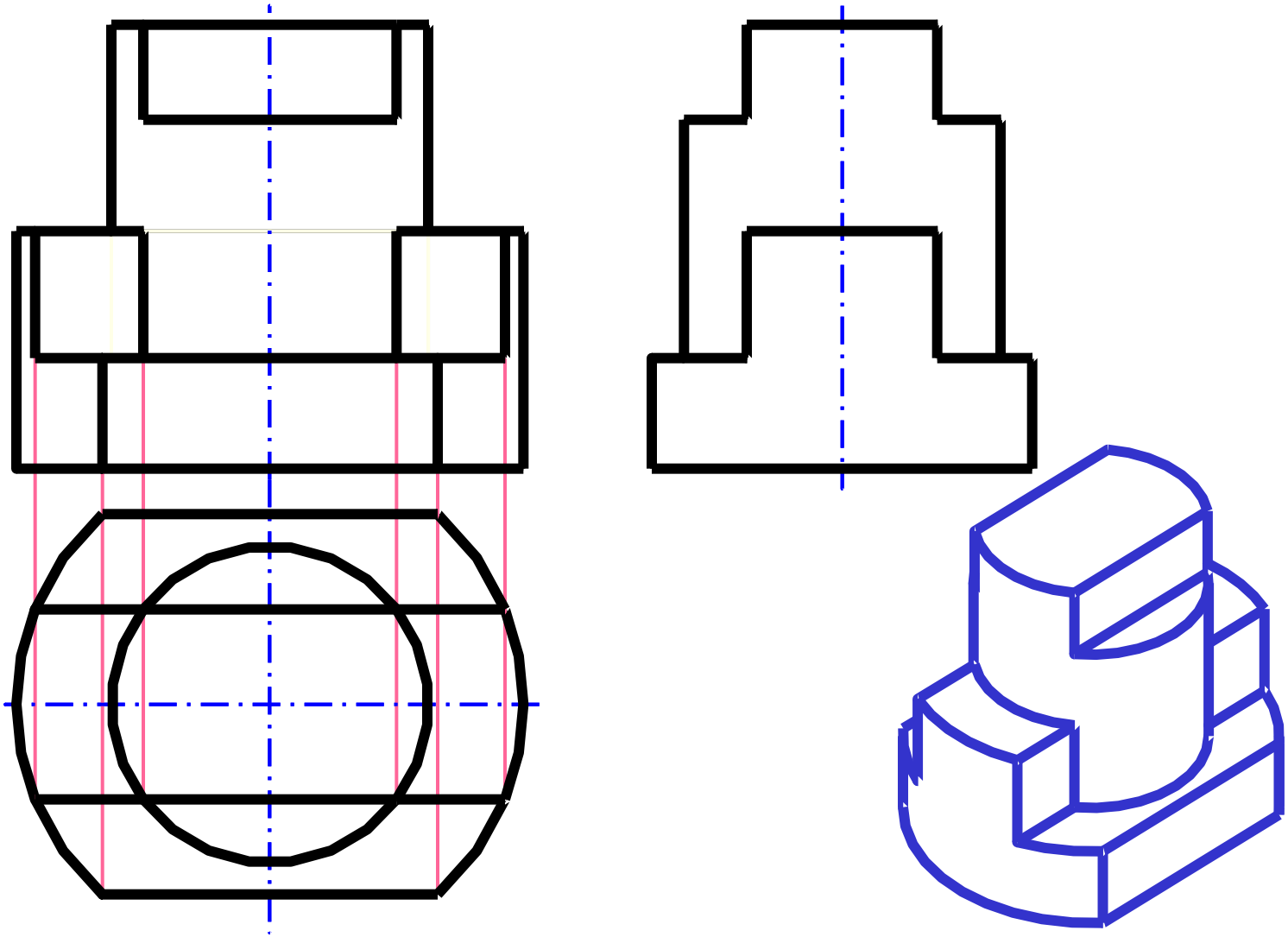


分析



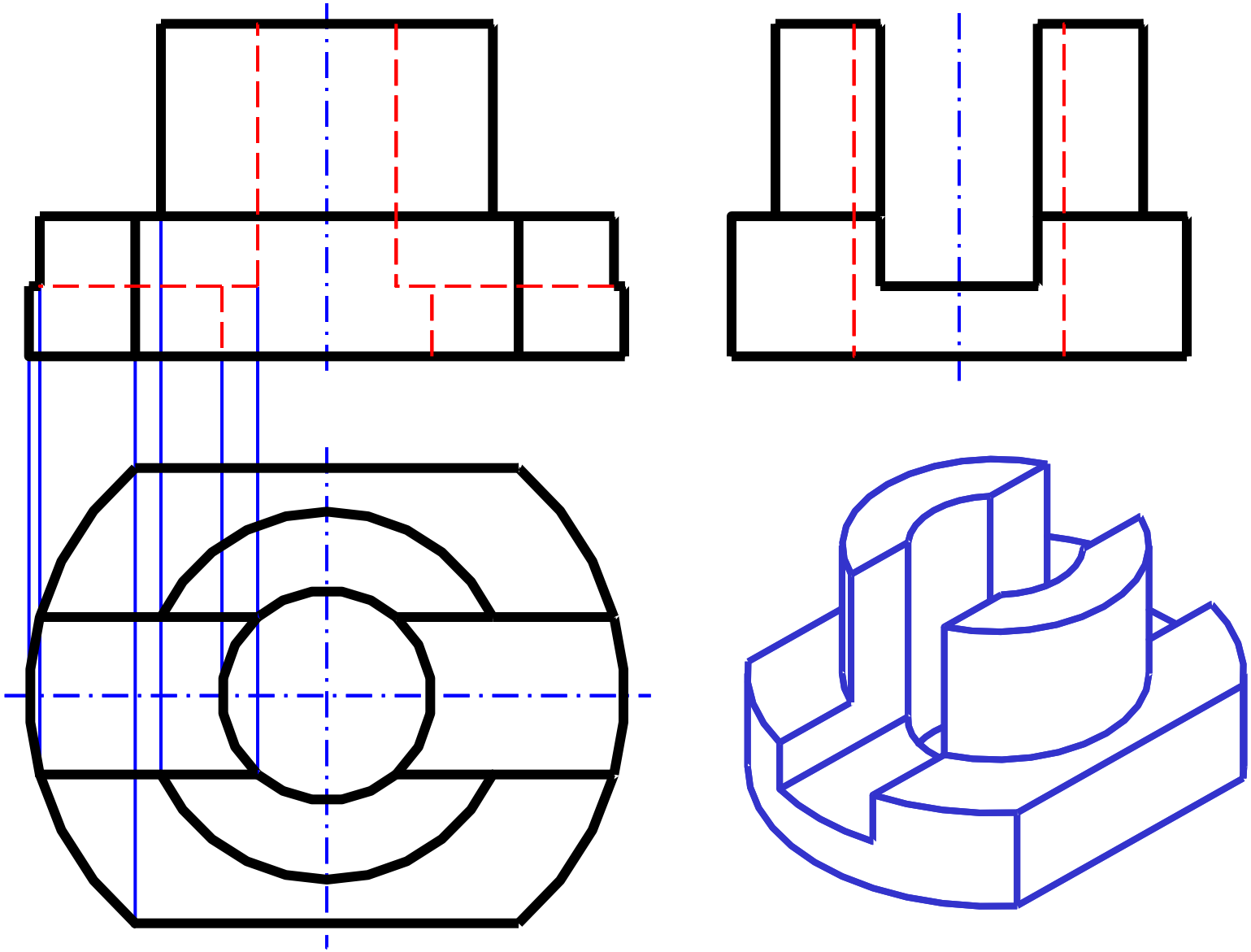
§5-4 读组合体视图

例6、由俯、侧视图，画主视图。



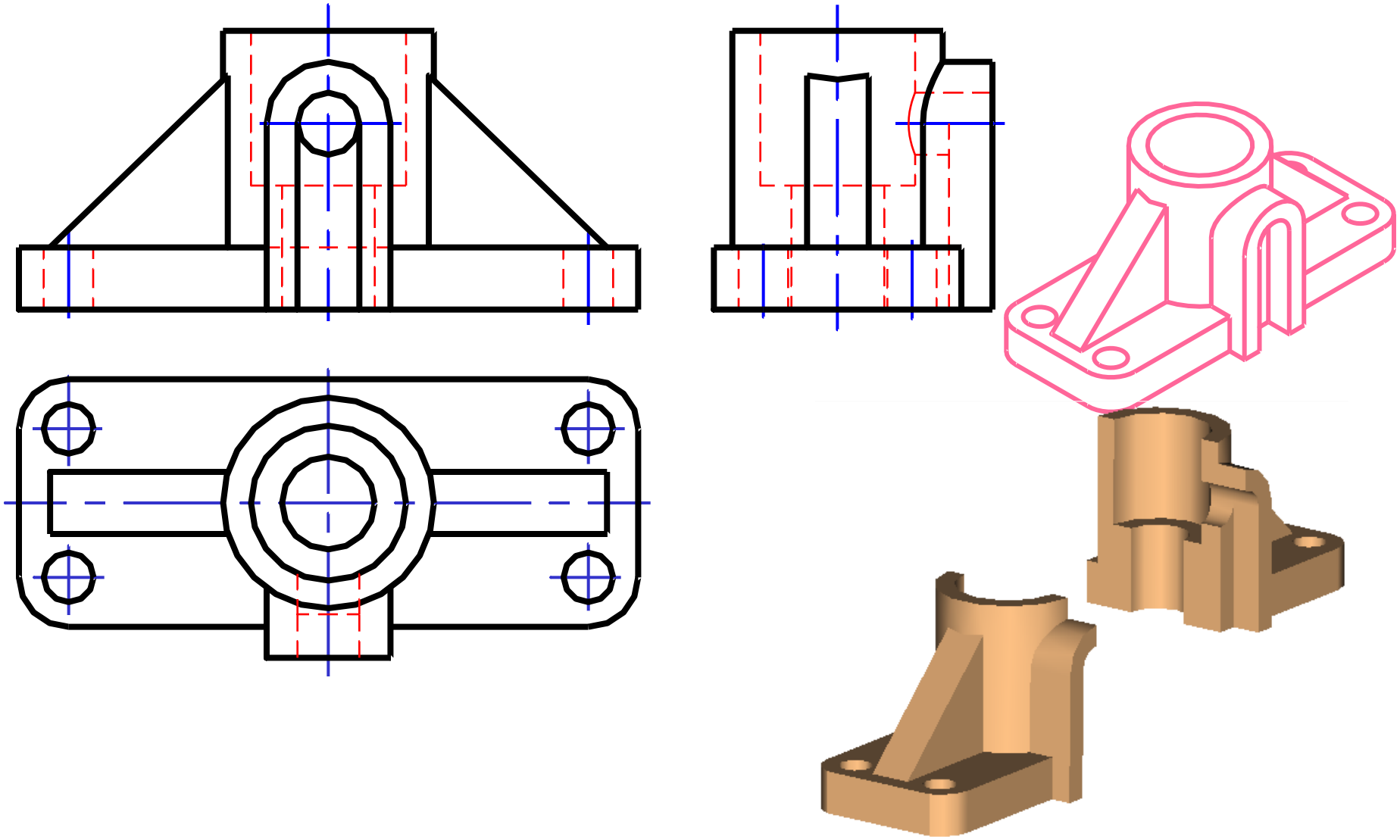
§5-4 读组合体视图

例7、由俯、侧视图，画主视图。



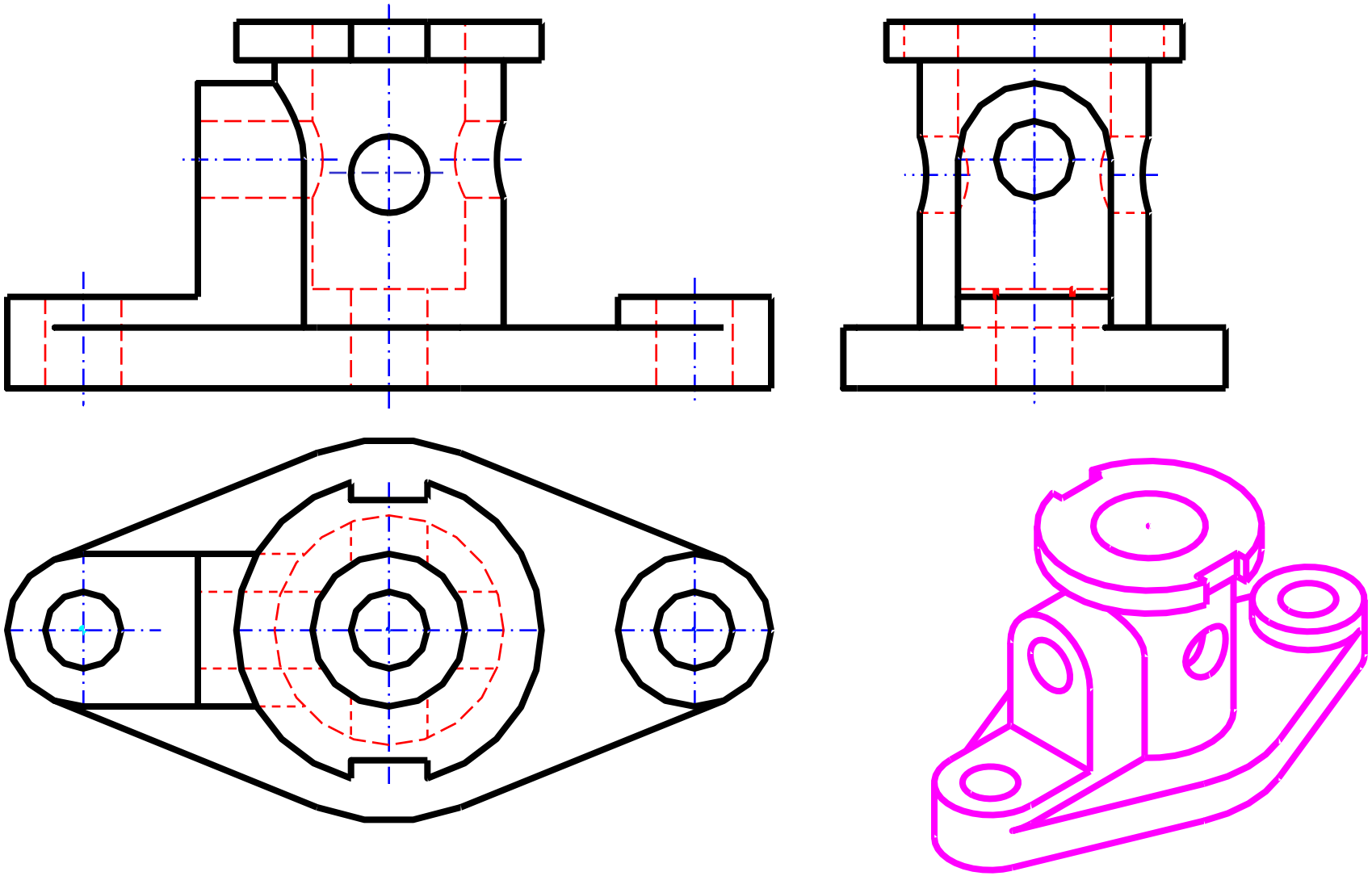
§5-4 读组合体视图

例8、由三视图想象出物体的形状



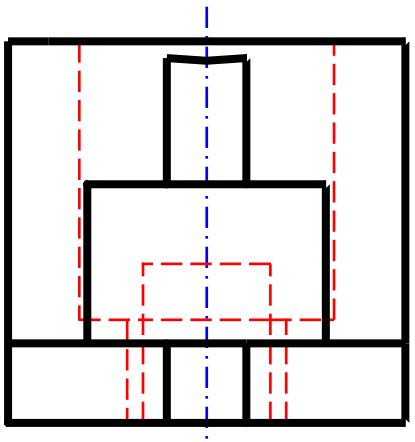
§5-4 读组合体视图

例9、 由三视图想象出物体的形状

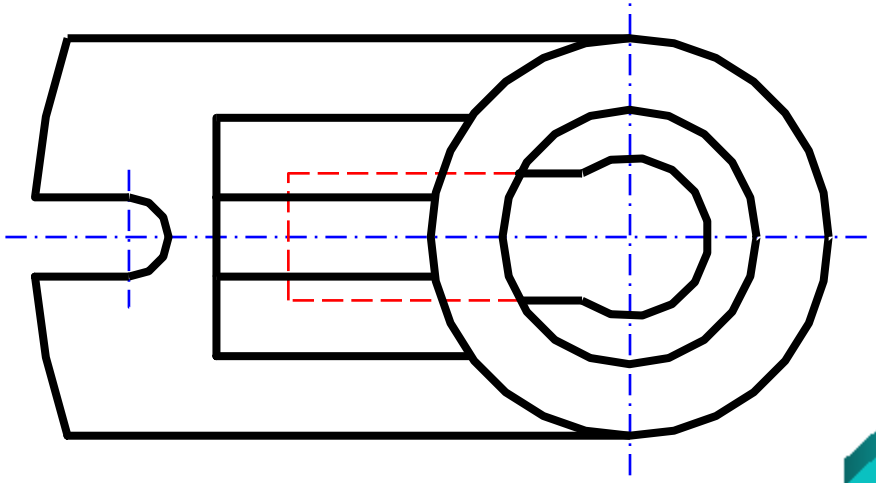


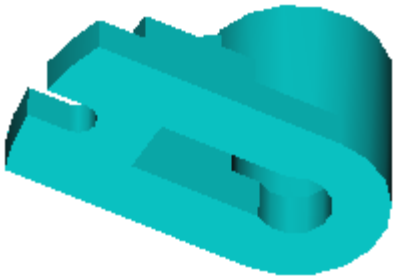
§5-4 读组合体视图

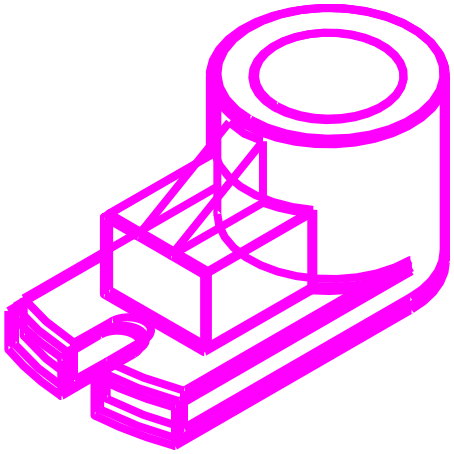
例10、 由俯、侧视图，画主视图。



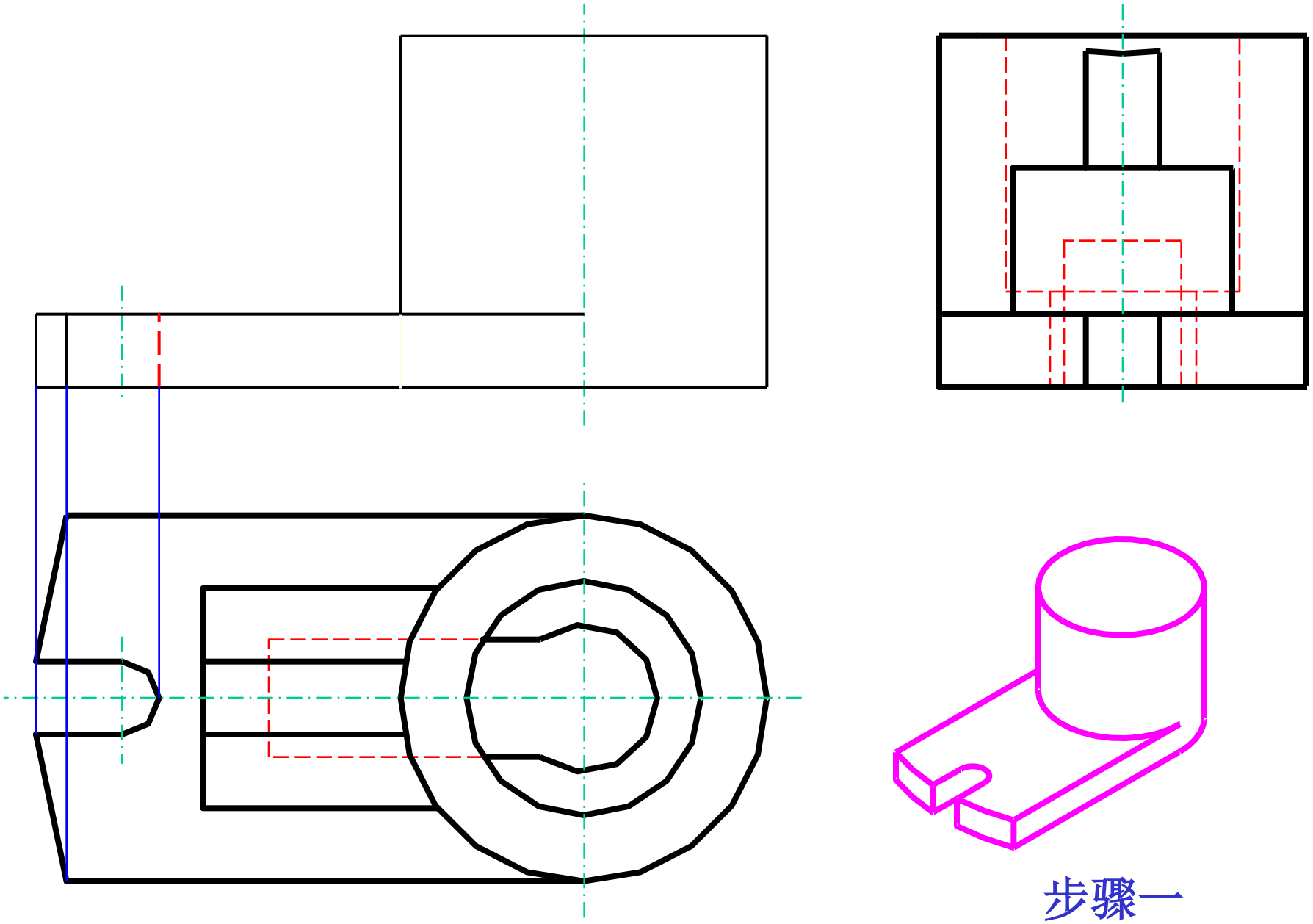
分析



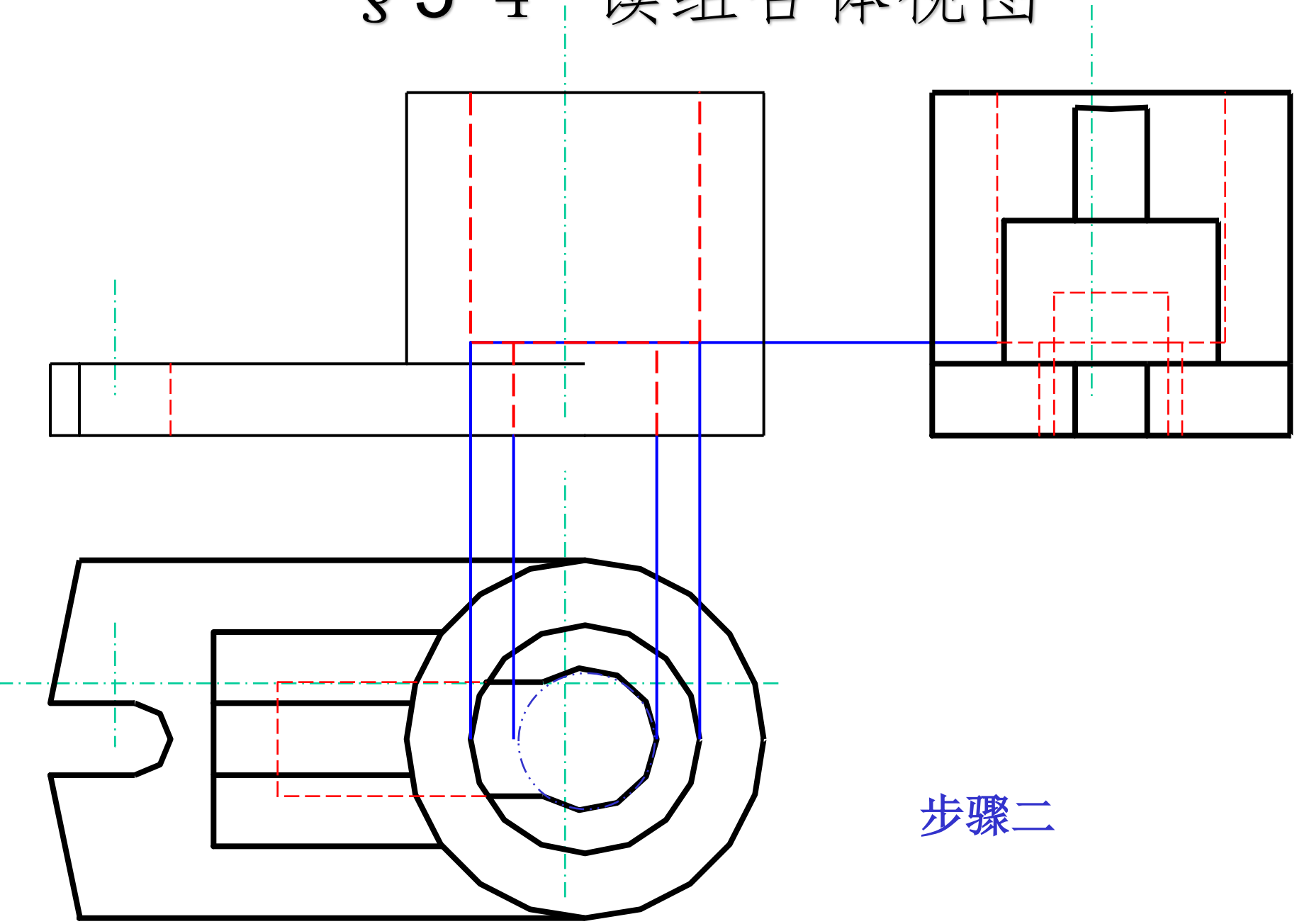




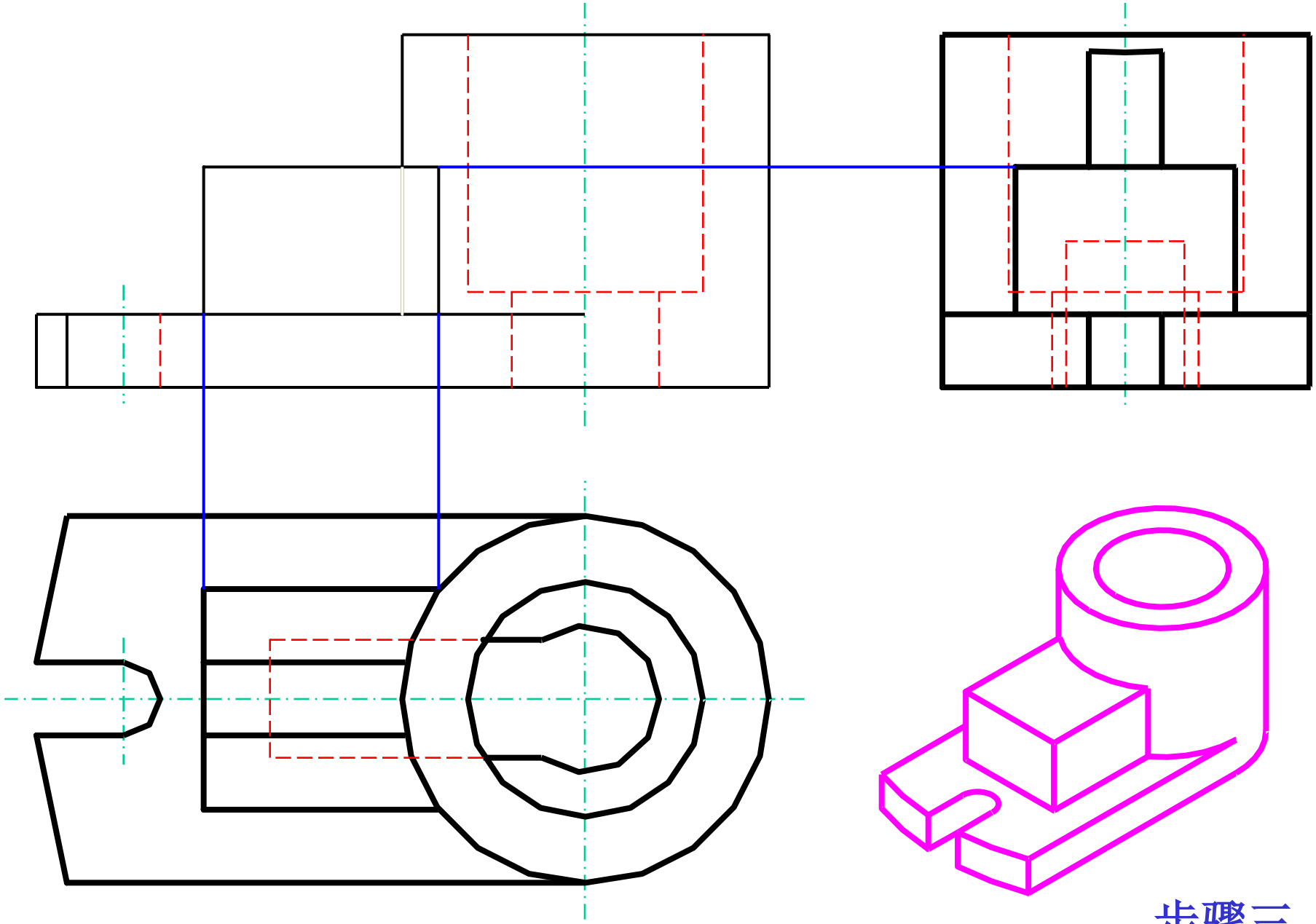
§5-4 读组合体视图



§5-4 读组合体视图

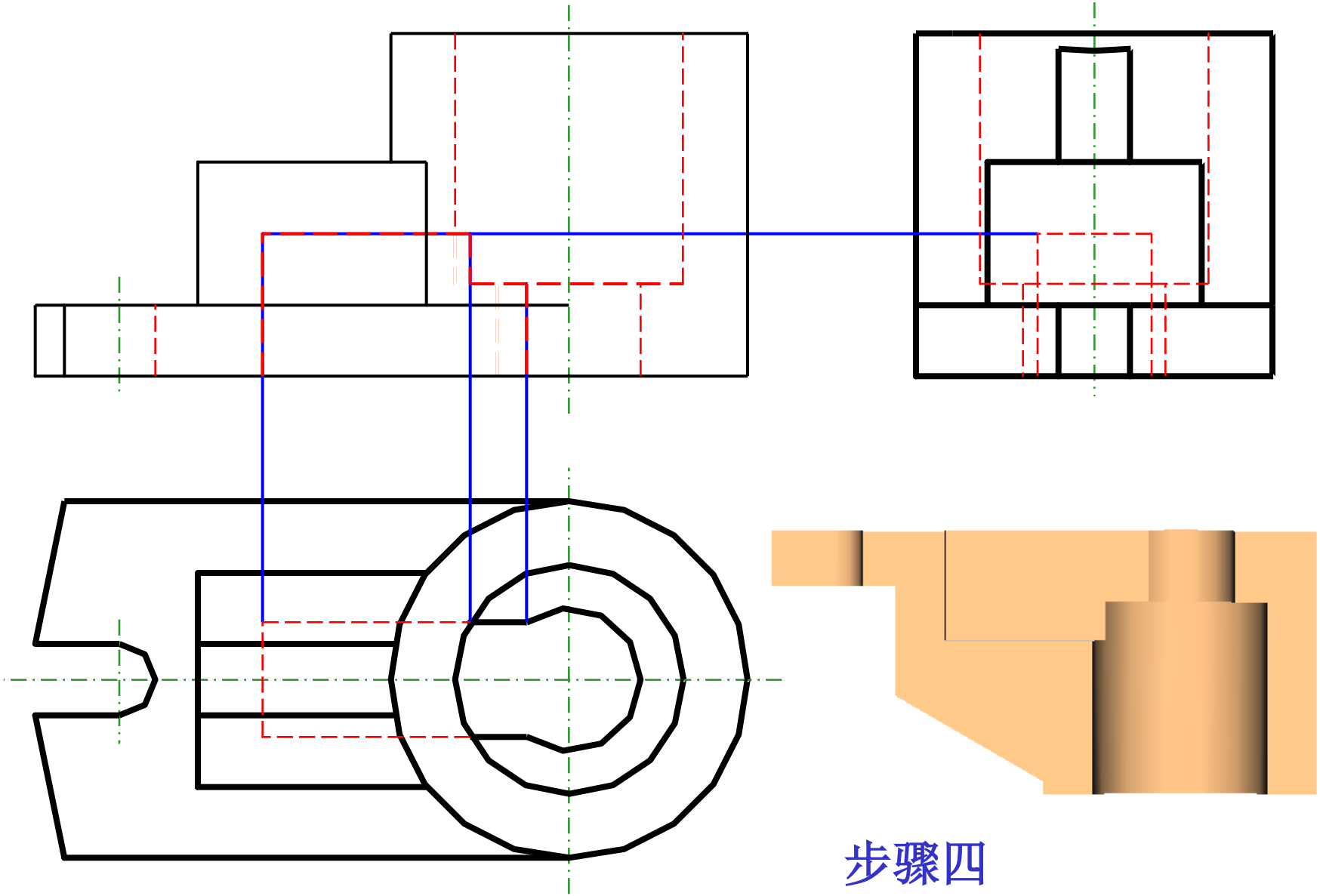


§5-4 读组合体视图



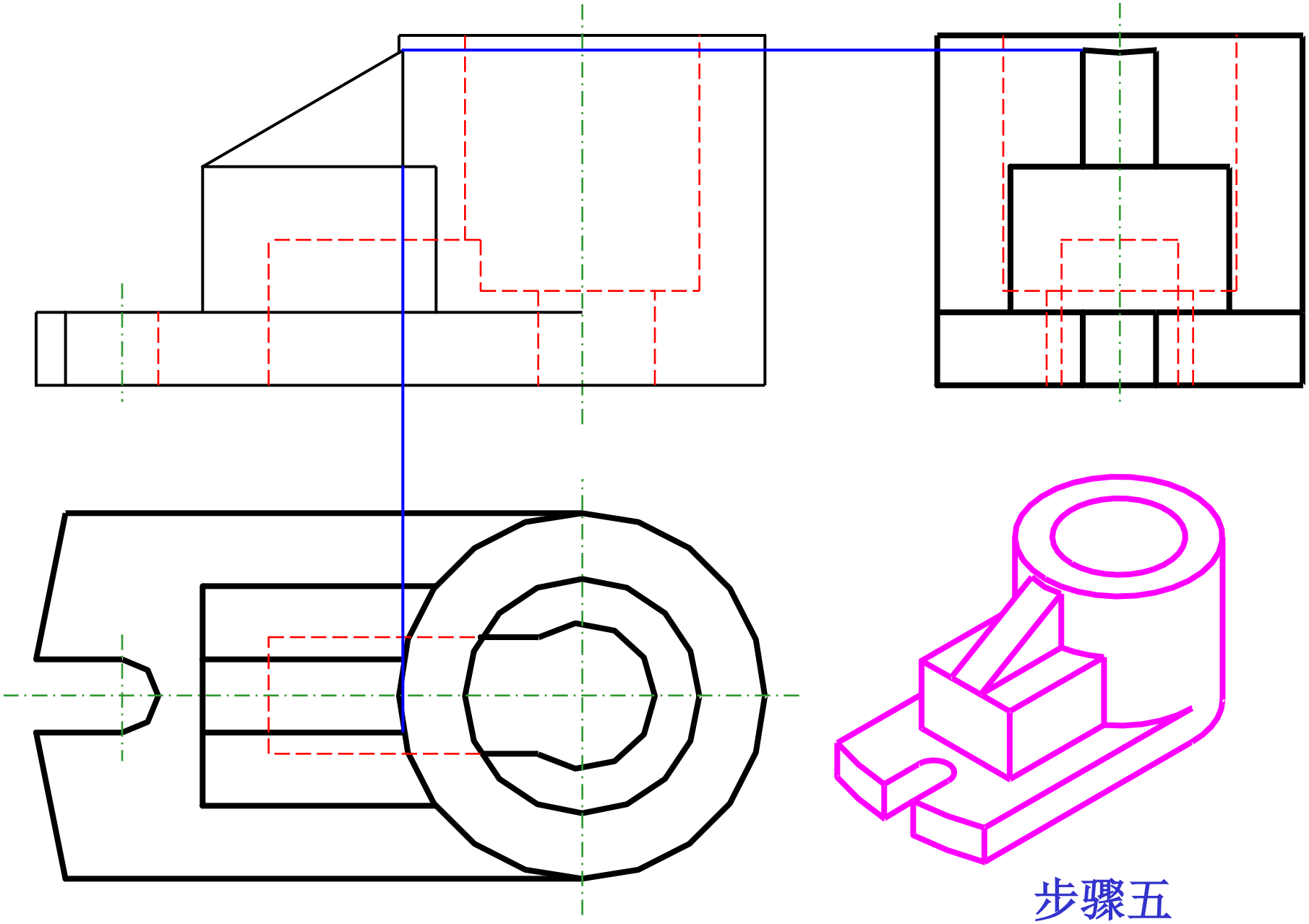
步骤三

§5-4 读组合体视图

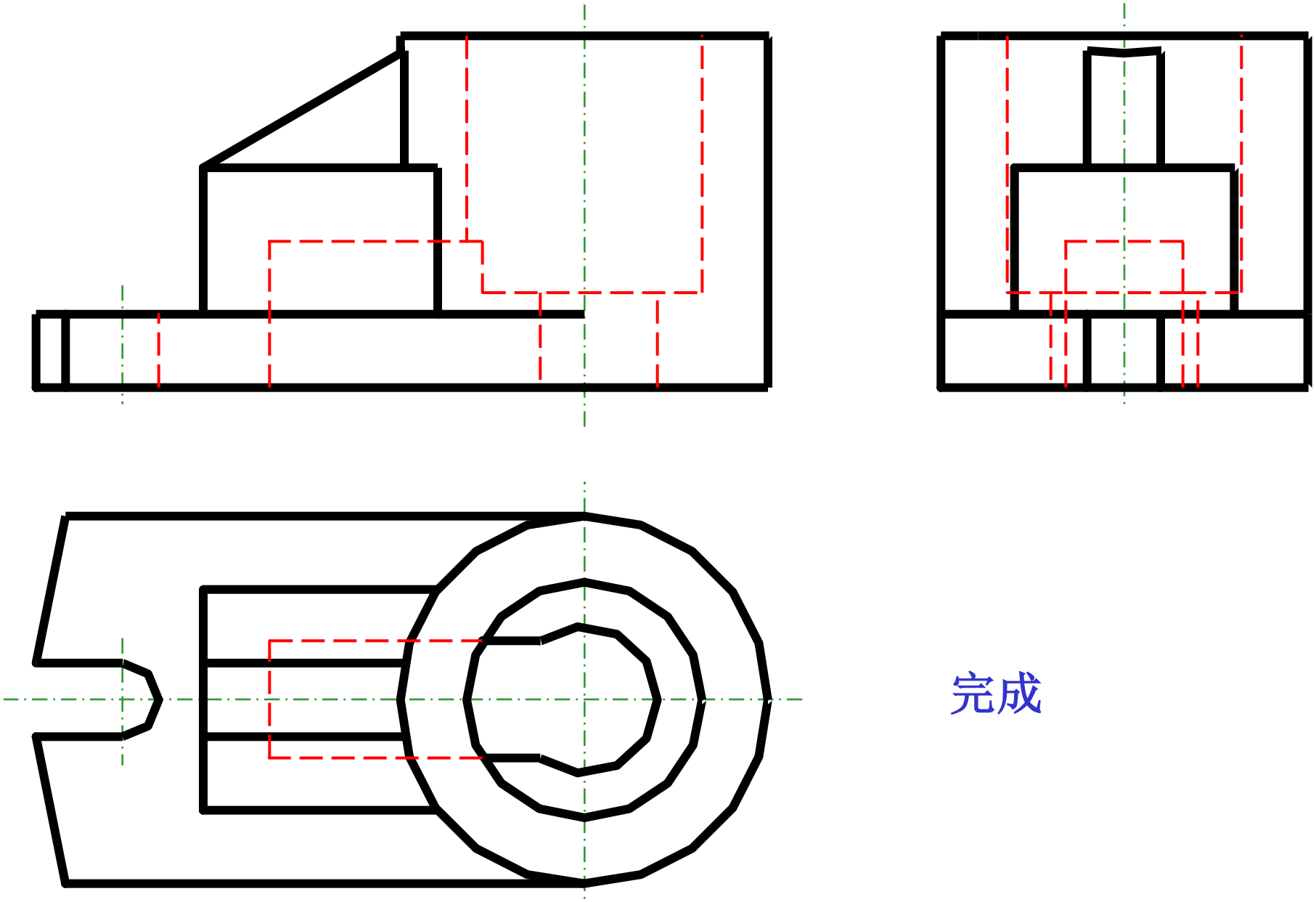


步骤四

§5-4 读组合体视图



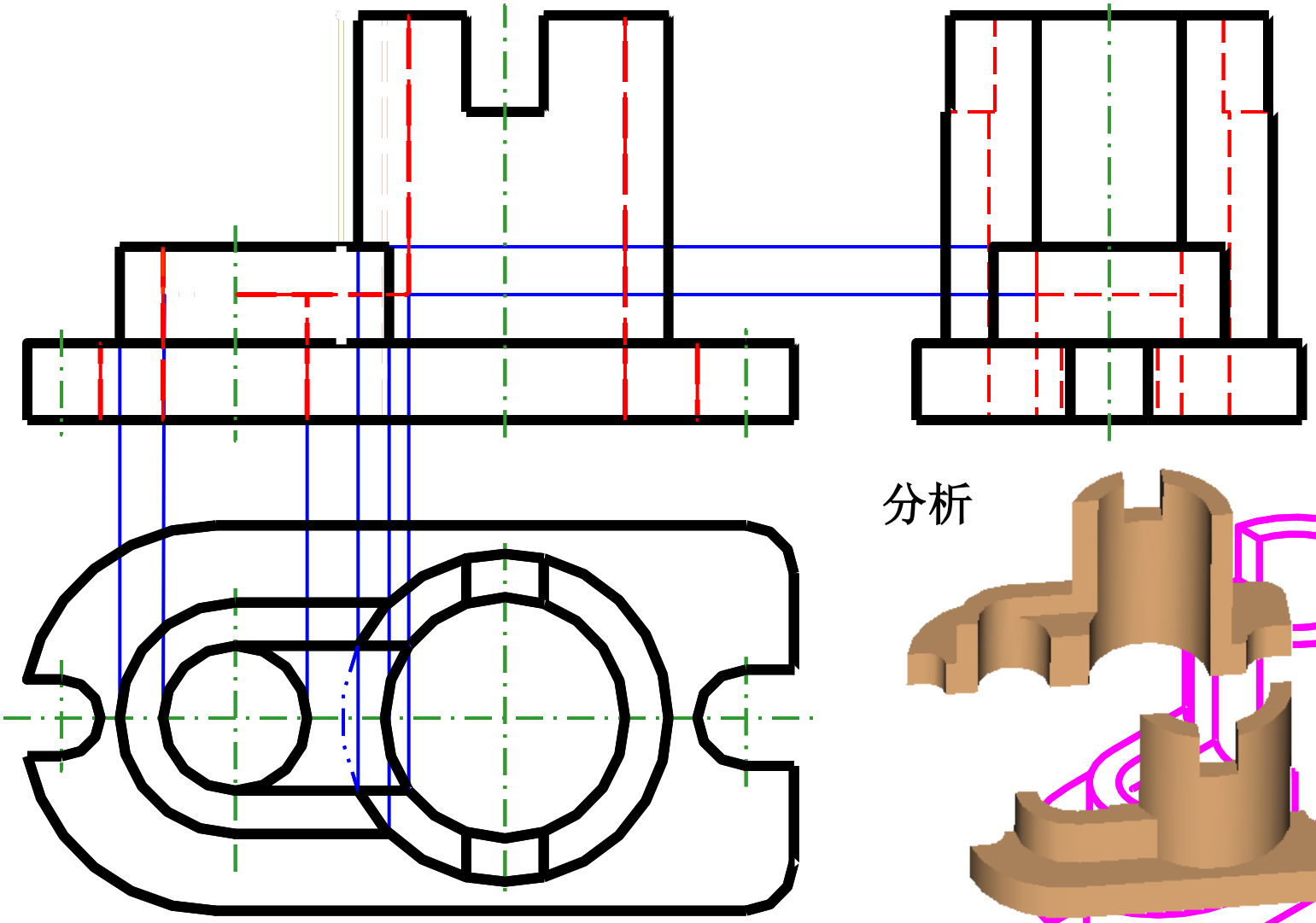
§5-4 读组合体视图



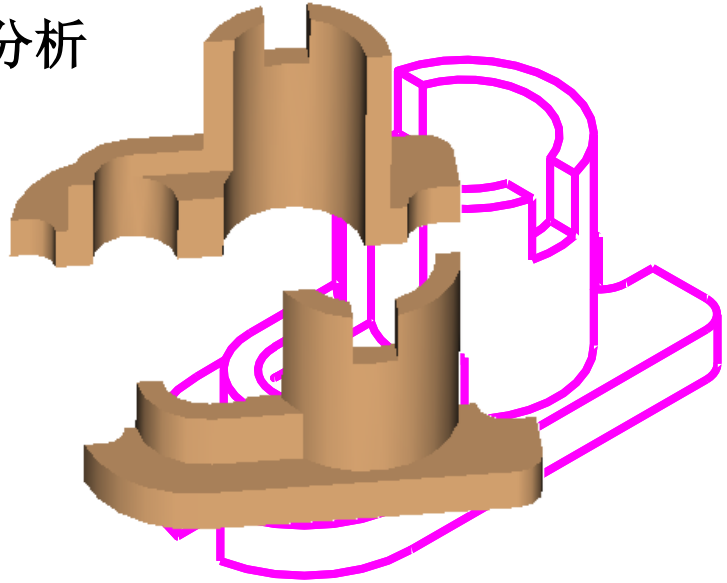
完成

§5-4 读组合体视图

例11、由俯、侧视图，画主视图。

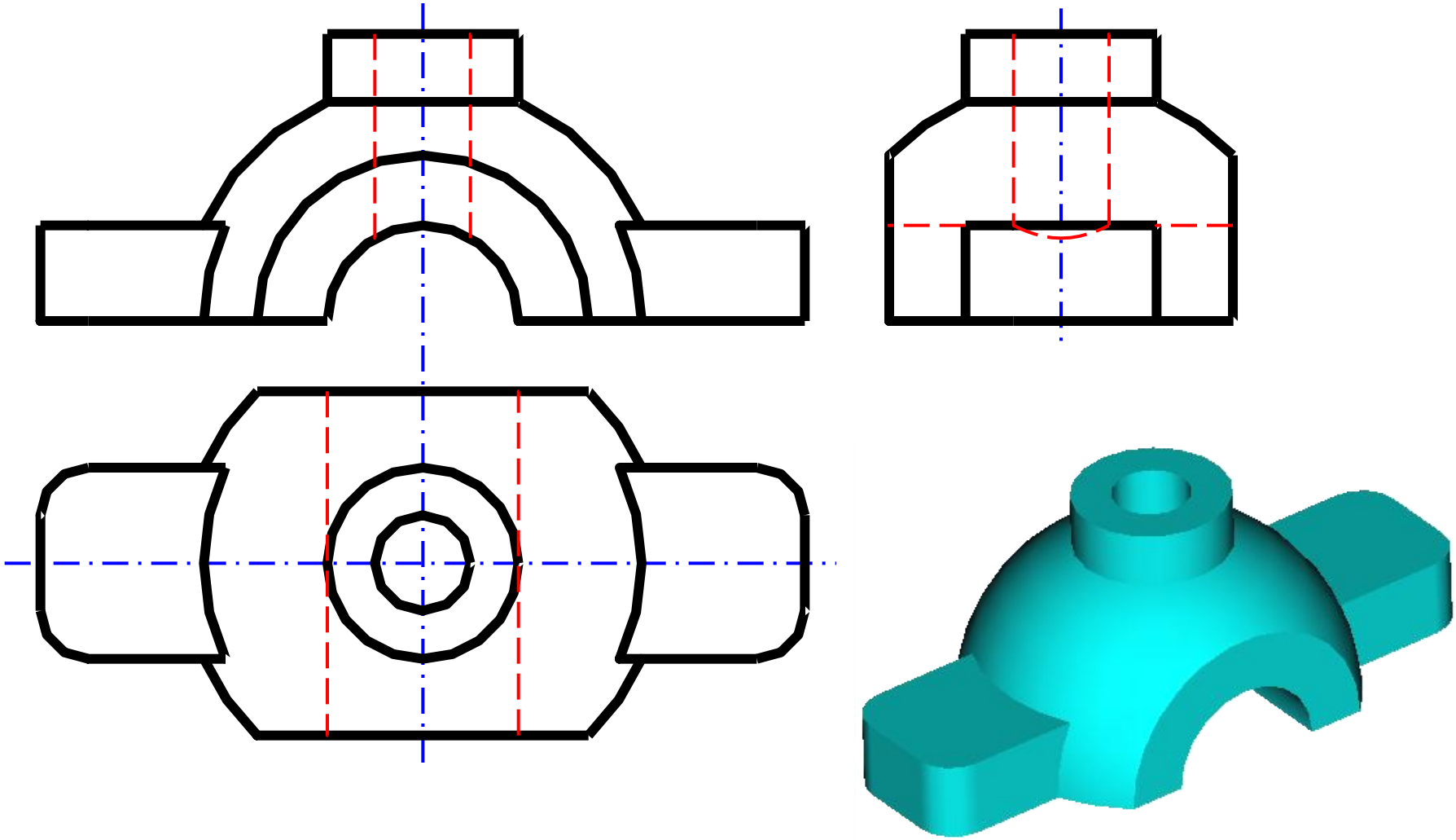


分析



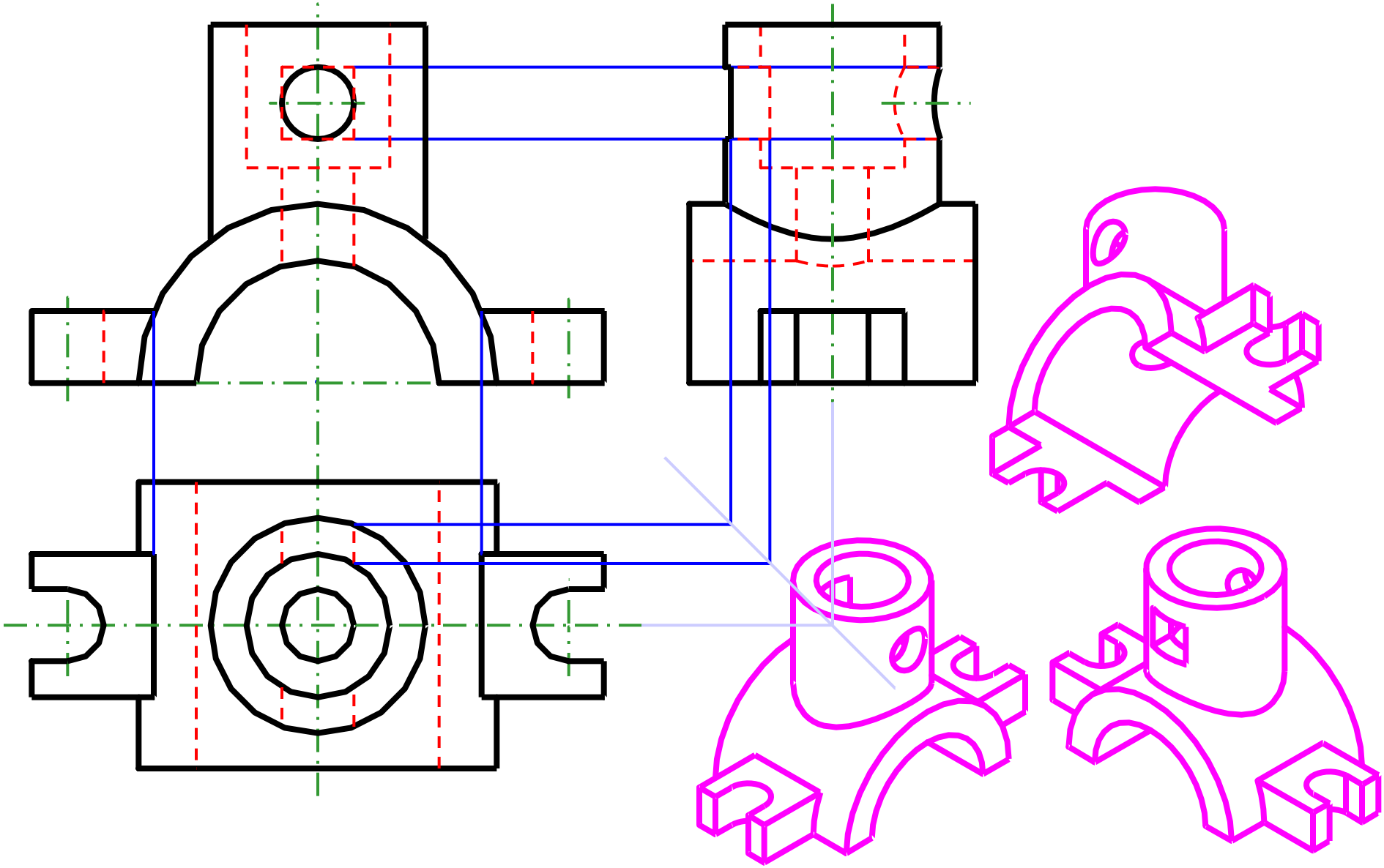
§5-4 读组合体视图

例12、补画视图中的漏线



§5-4 读组合体视图

例13、补画视图中的漏线



§5-5 构型设计

一、概述

根据不同的结构要求，将某些基本几何体按照一定的组合形式组合起来，构成一个新形体，并用三视图表示出来的过程。称为组合体构型设计。

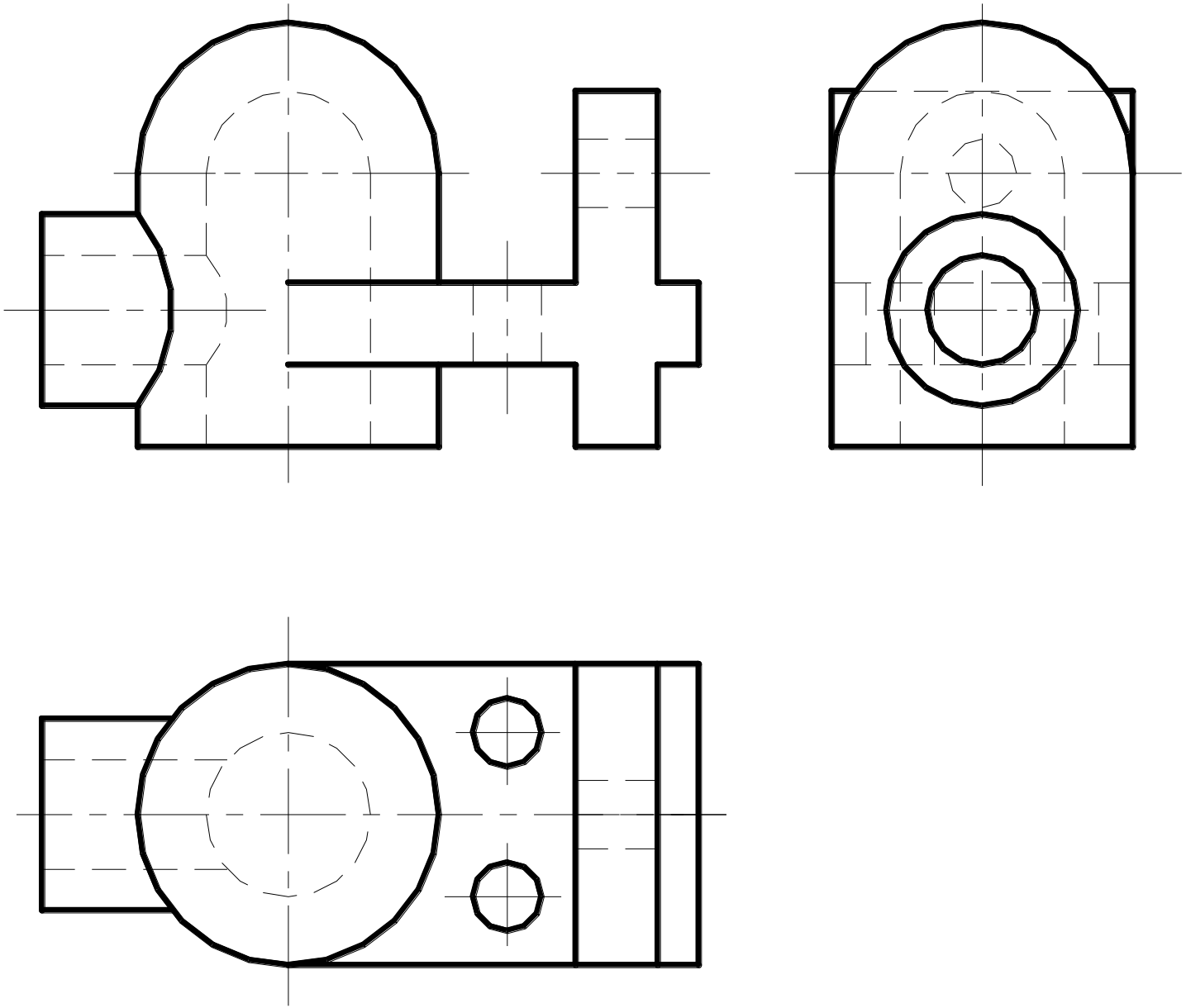
二、构型设计的基本过程

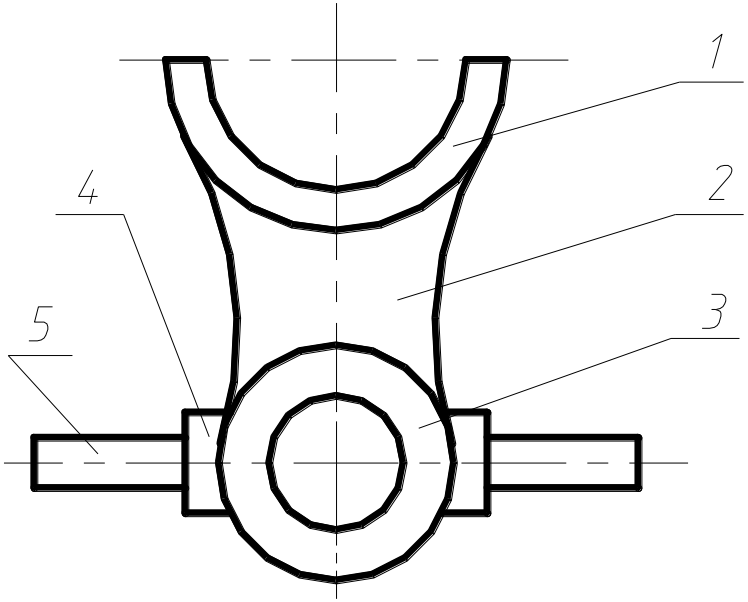
1、准备阶段——积累素材

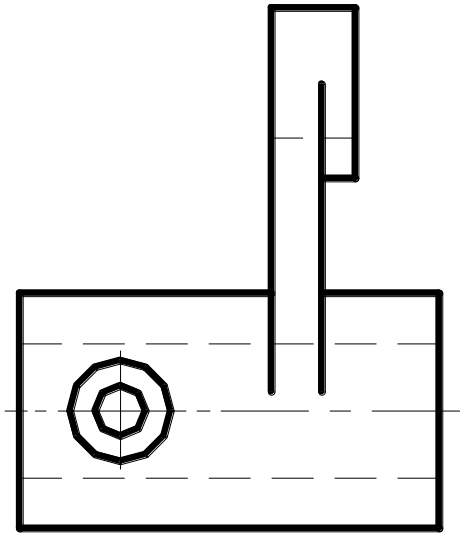
（1）分析实物 — 在构型设计前应当观看较多的组合体模型。仔细研究每个模型的构成和形体特点，能进行大致分类。

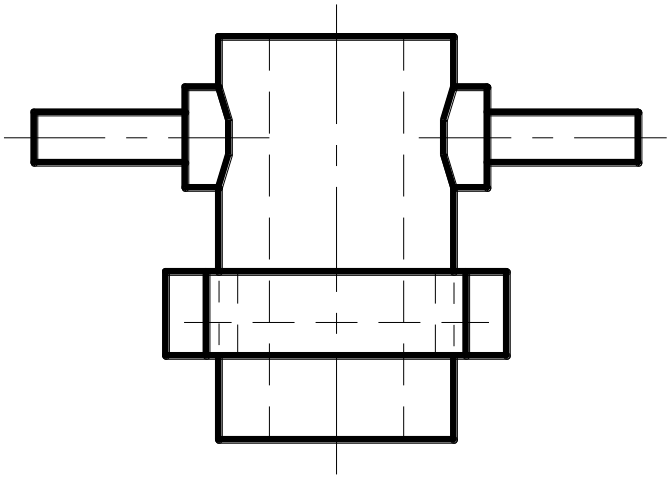
（2）分析构型设计图例 — 为开阔视野和思路，在观察、分析、记忆实 物的基础上，可选择一些新颖、独特、造型美观、重点突出的设计图进行具体分析。

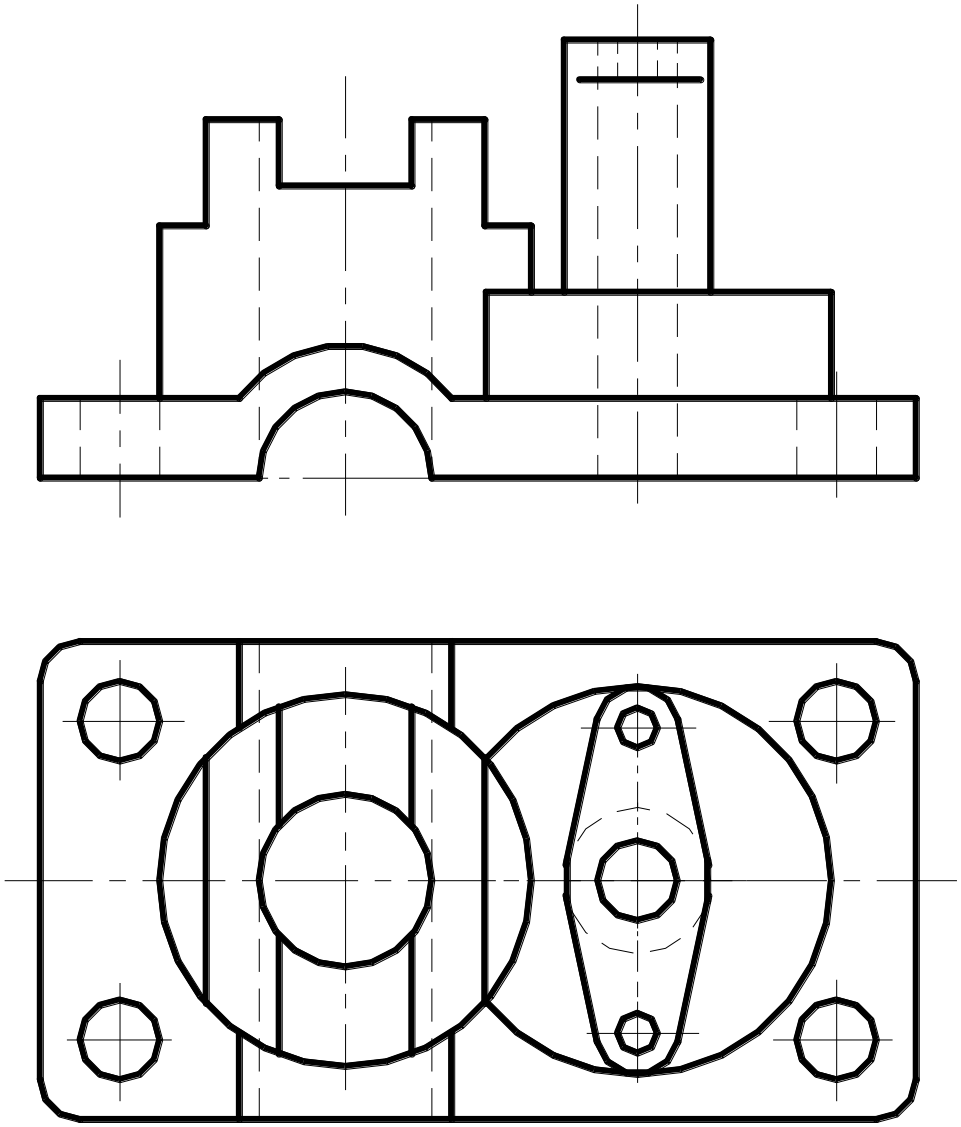
（3）观看构型设计优秀作业 – 精选数十幅构型设计优秀作业(布置展览橱窗)，供观看、分析、借鉴。下面是挑选出的几幅学生作业。

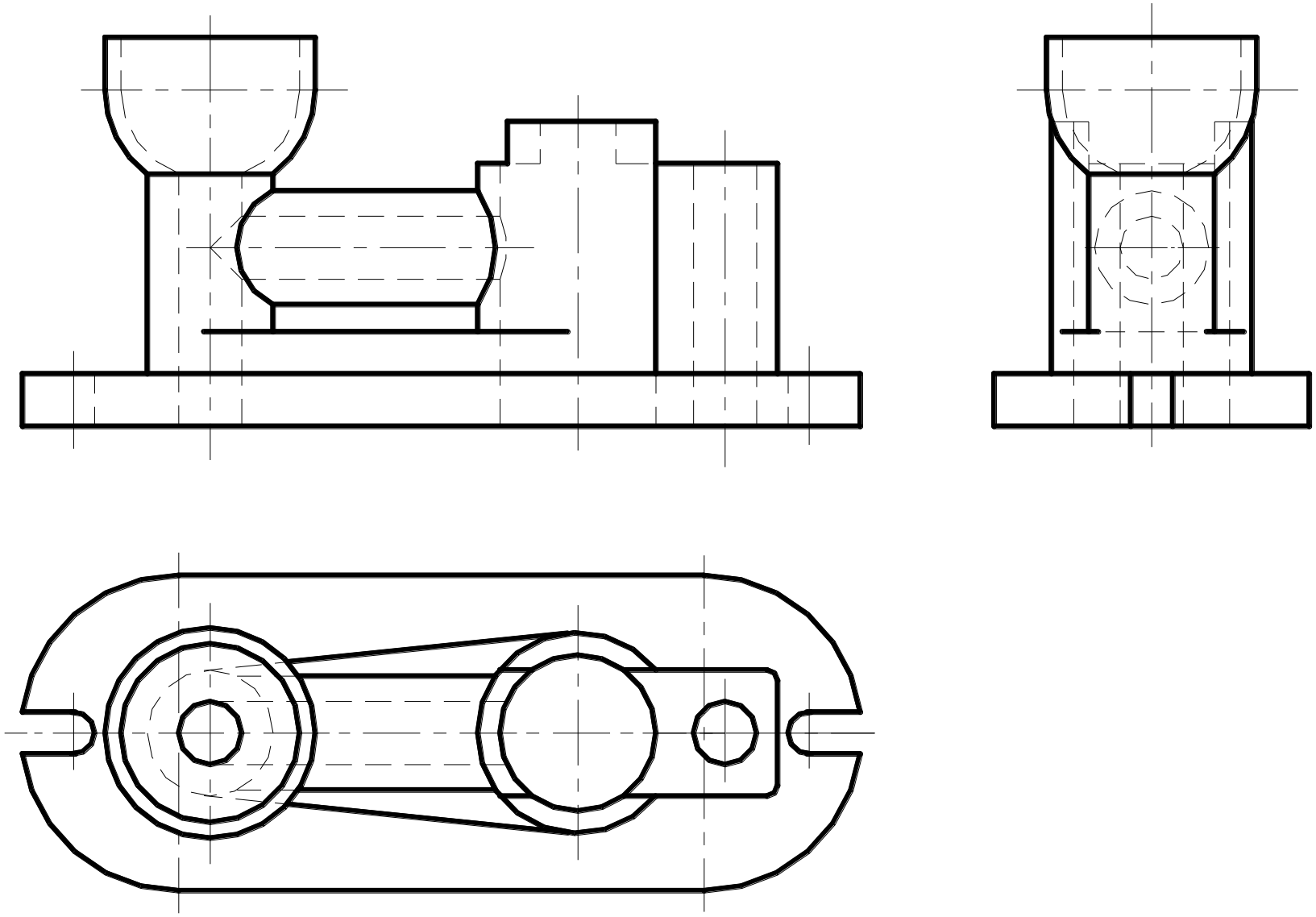



1
2
3
4
5





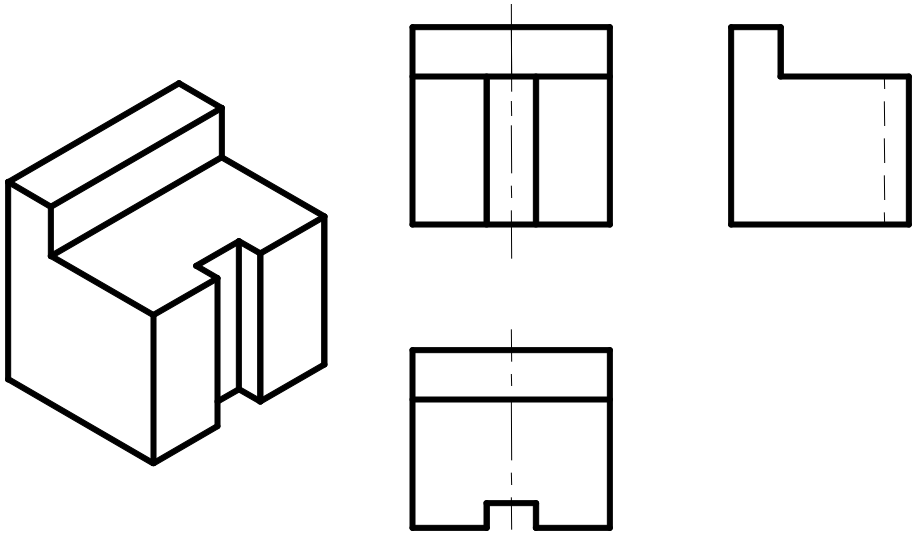


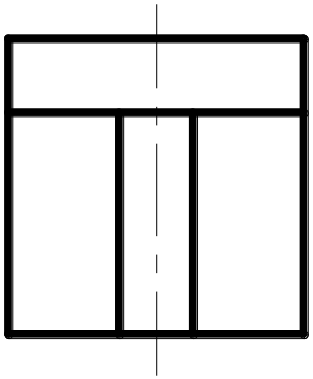


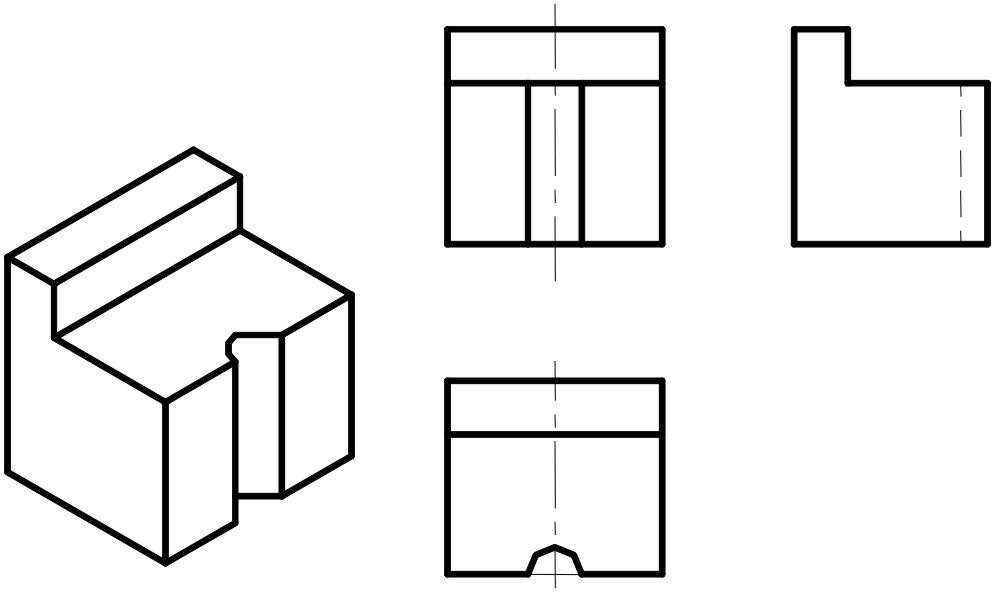
2、综合、想象、构思新形体

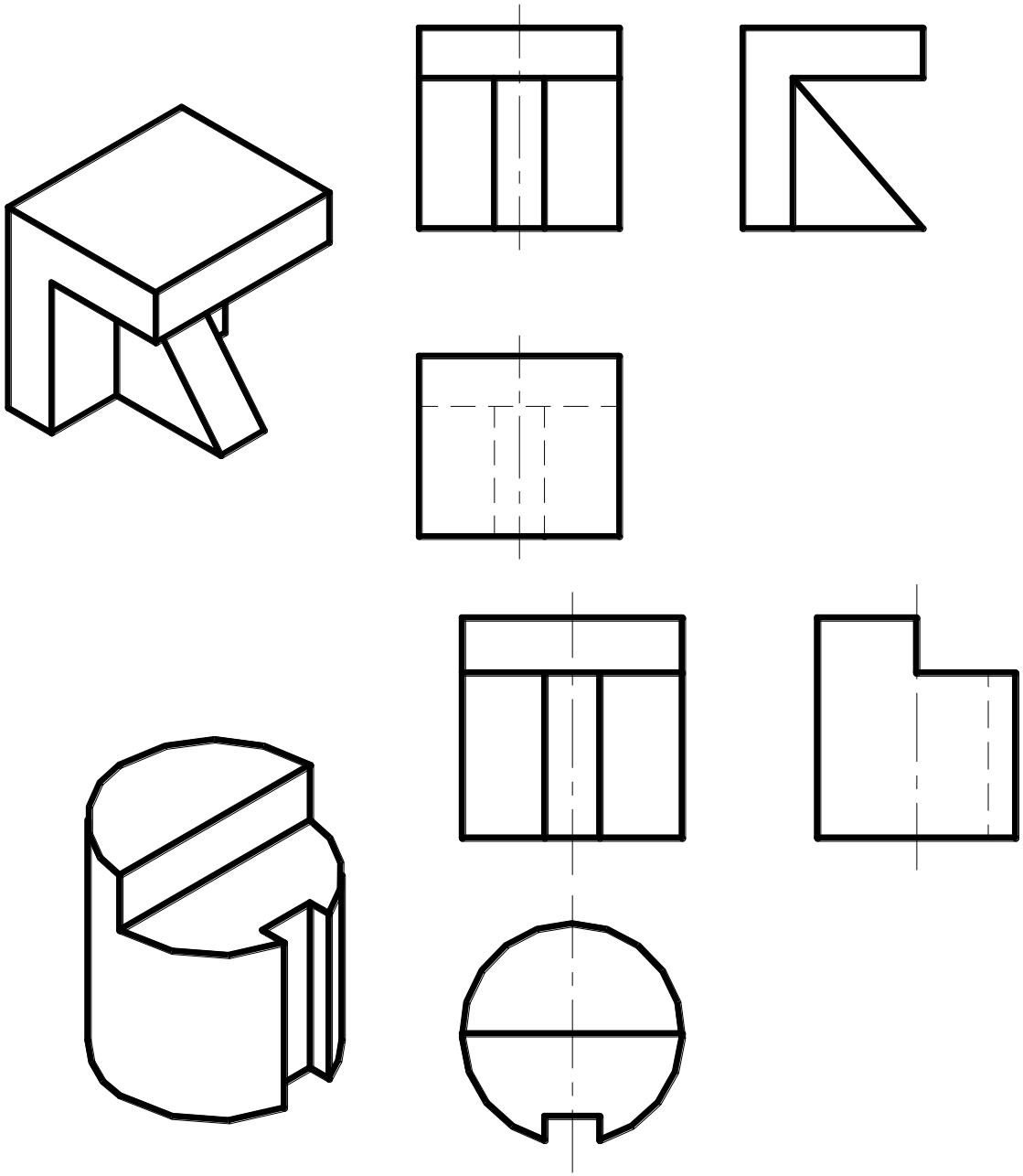
(1)由浅入深，进行一补二的练习。

(2)构思新形体。









3．画设计图

(1)画设计草图－先用草图纸画出新形体的三视图，称为设计草图。

(2)完成设计图－将修改后的草图按下述步骤画成正规设计图。

①选取比例，确定图幅。

②确定主视图。

③布置各视图，画出基准线。

④逐个画出各基本形体的三视图。

⑤完成组合体三视图的底稿。

⑥检查、描深，完成设计图。

三、构型设计举例

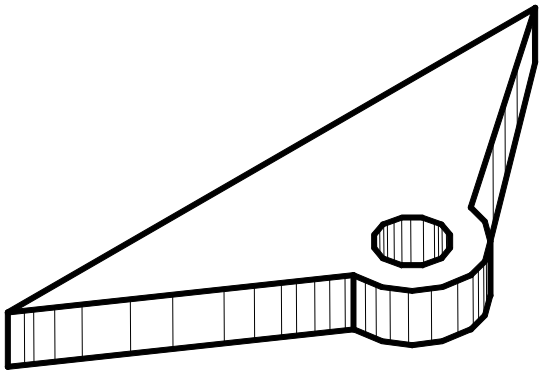
1、按下述规定条件设计组合体

由三个简单形体组合而成；组合形式为综合式；各基本形表面有相切、相交。

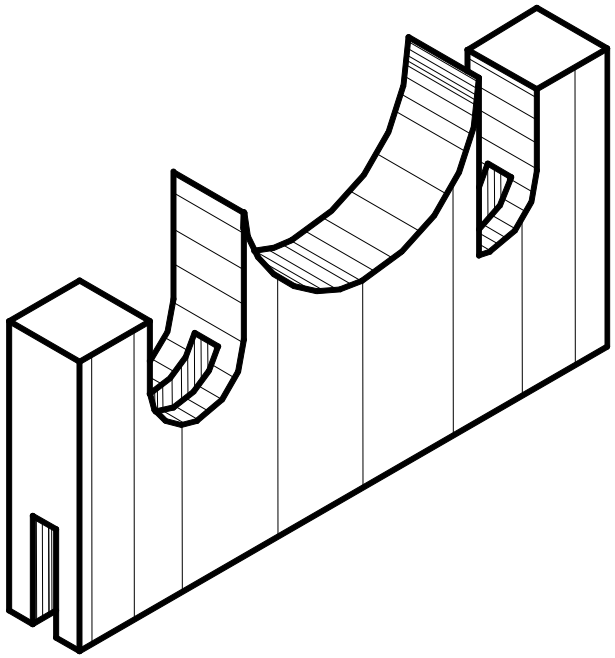
(1)选取素材　　取底板、支承板、圆柱体。

(2)综合、想象、构思出新形体。

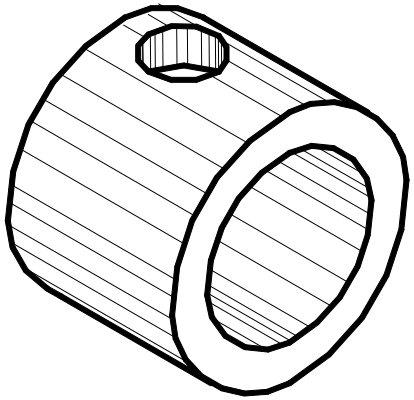
(3)按画图步骤画出组合体的三视图。



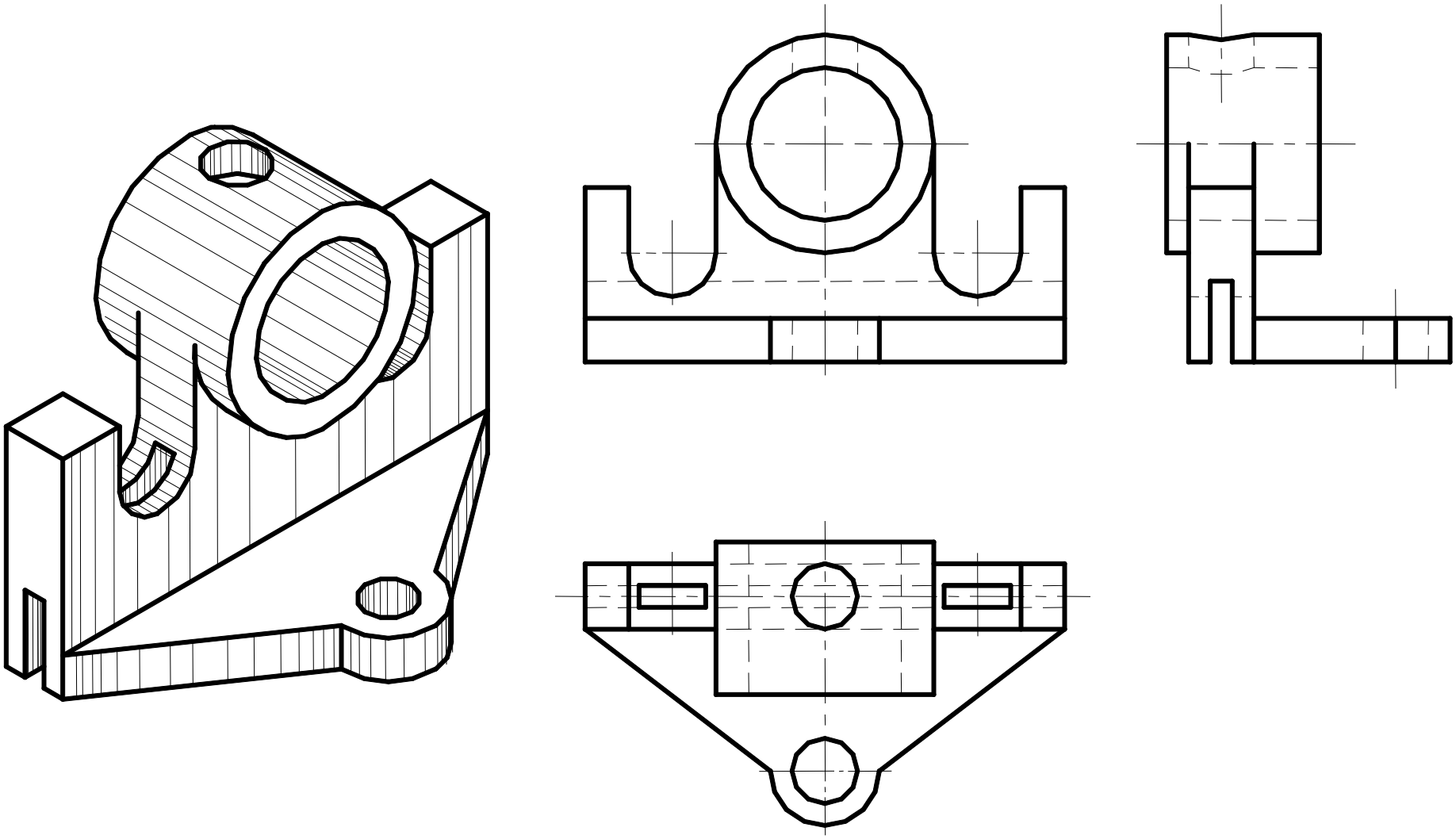
(a) 底板



(b) 支承板

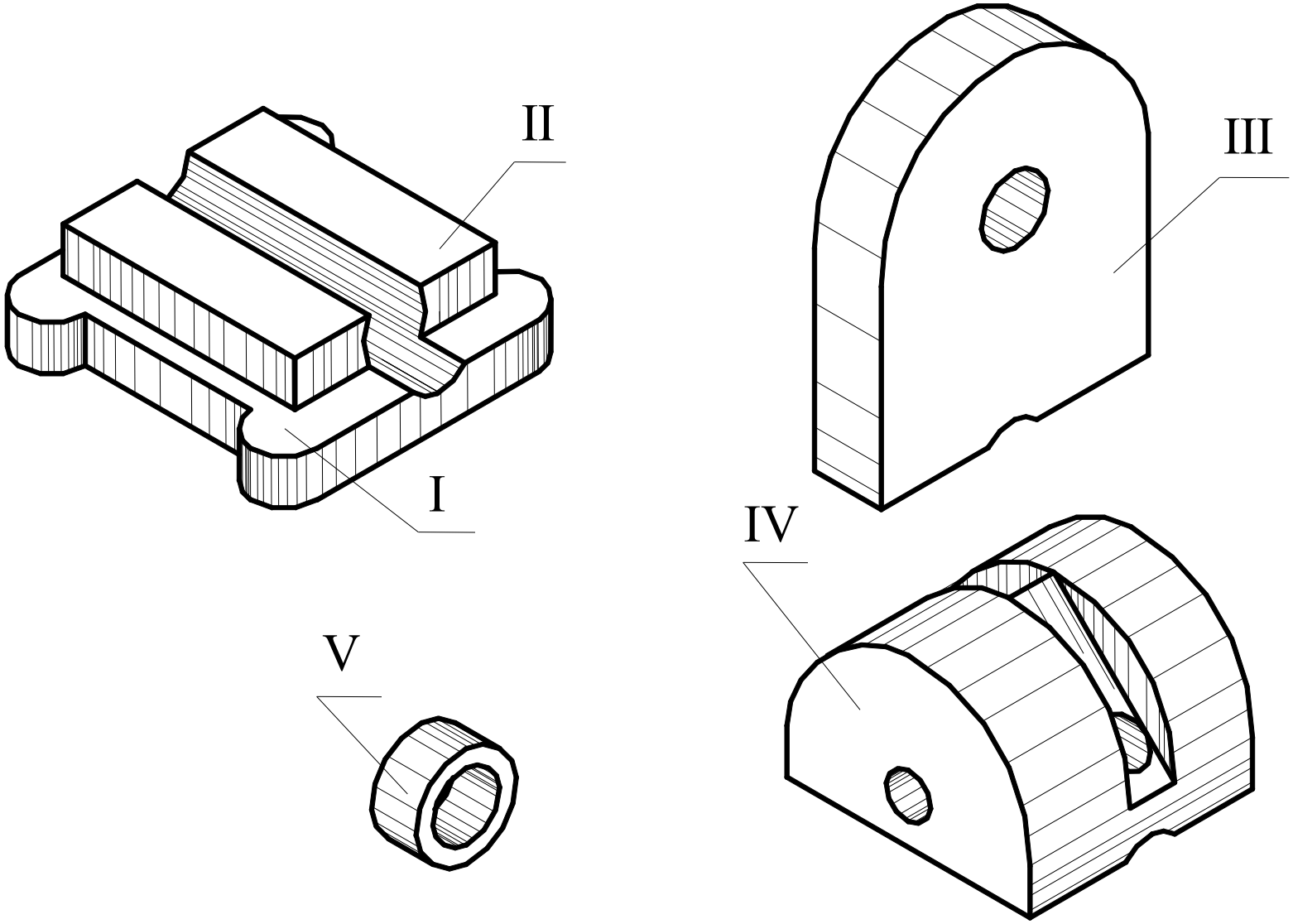


(c) 圆柱体



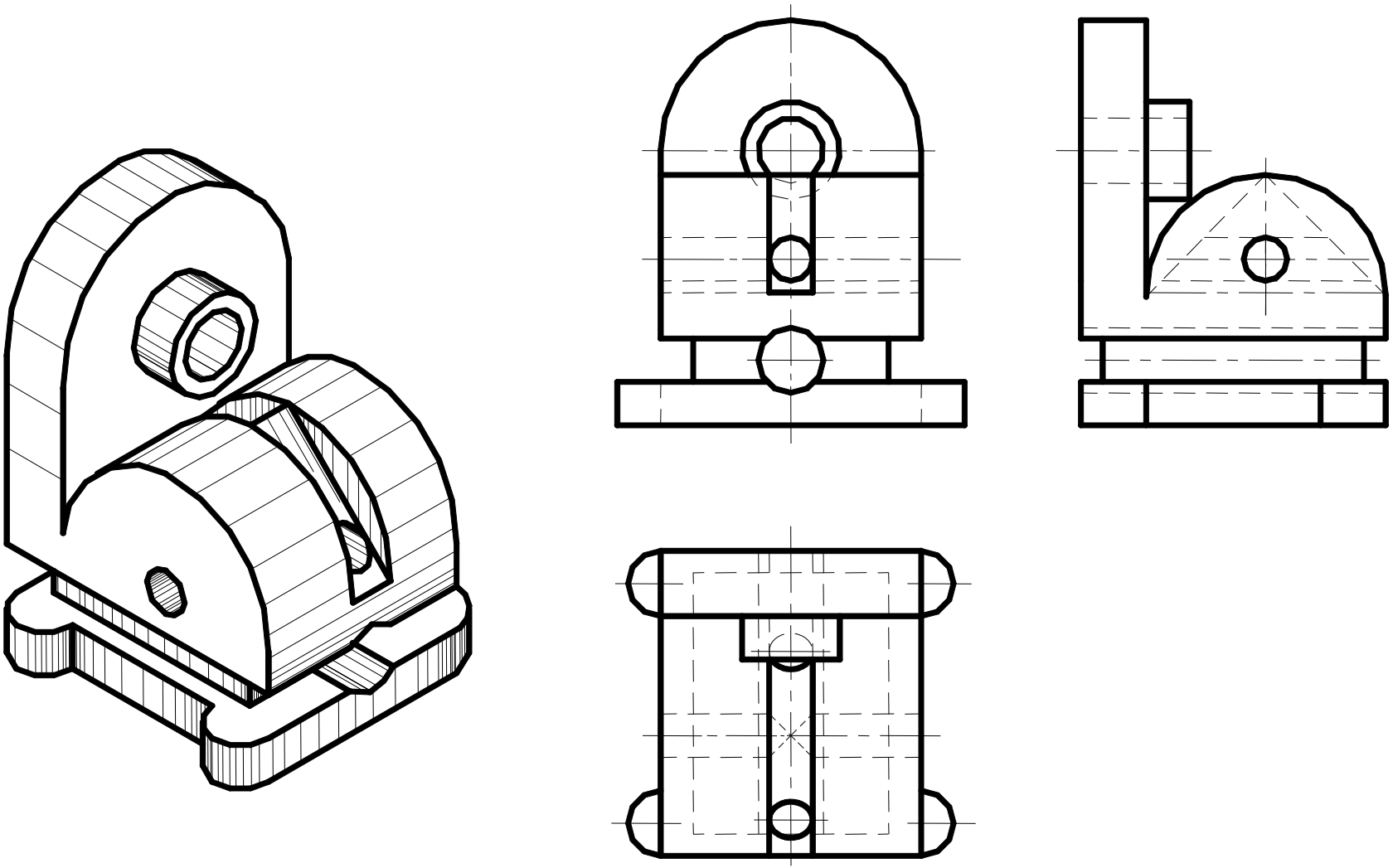
2、自由设计组合体

(1)选取素材　取底板、方板、半圆柱、圆柱体、立板。

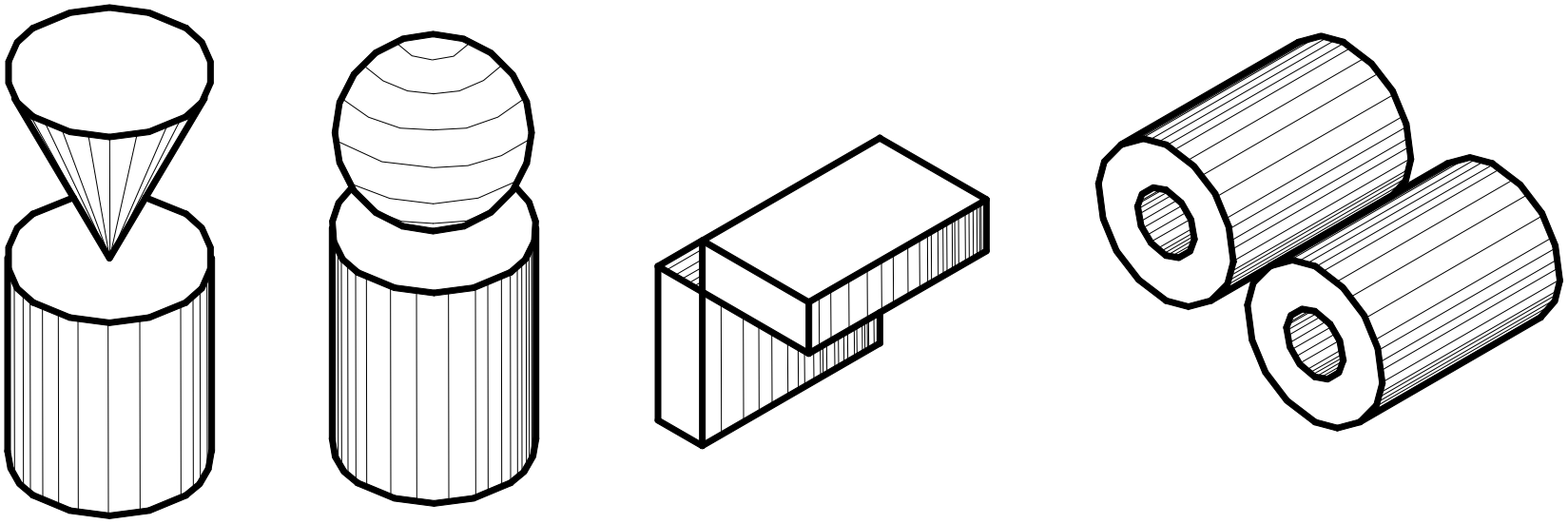


(2)综合、想象、构型。

(3)按步骤画出组合体的三视图。



四、注意事项

为使构型设计符合工程实际的要求，构型时应注意以下几点：

1．两体之间不能以点连接；
2．两体之间不能以线连接；
3．多观察实物或轴测图，增强组合体表象的储备；
4．掌握形体分析法，线面分析法及组合方式的联想方法；
5．重视构思表达的作业。